



Driftwood

Setup & Mod Guide

Minecraft 1.21.1 – NeoForge

Version 3.1.1

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Driftwood — Setup & Mod Guide

Wave -1 — Prerequisites

JDK 21

Tool	Role
Eclipse Temurin JDK 21 (LTS)	Java runtime and development kit for Minecraft 1.21.1

Install the **JDK** (not just JRE) — the JDK includes [javac](#) and [jlink](#) needed by NeoForge. Minimum version 21.0.0, any LTS build. On Windows, use the MSI installer and ensure `JAVA_HOME` is set to the install path (e.g., `C:\Program Files\Eclipse Adoptium\jdk-21.0.6.13-hotspot`). On Linux/macOS, use your package manager (`apt install temurin-21-jdk`, `brew install temurin21`).

Verify installation:

```
java -version
# Expected: openjdk version "21.0.x" ...
```

Performance Tuning

Recommended JVM arguments (for XMCL → instance settings → Java → JVM Arguments):

```
-XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:
+PerfDisableSharedMem -XX:SoftMaxHeapSize=12G -Xms8G -Xmx8G
```

System RAM	Recommended allocation
16 GB	<code>-Xms6G -Xmx10G -XX:SoftMaxHeapSize=8G</code>
32 GB	<code>-Xms12G -Xmx22G -XX:SoftMaxHeapSize=12G</code>
64 GB	<code>-Xms12G -Xmx32G -XX:SoftMaxHeapSize=12G</code>

Note: ZGC (Z Garbage Collector) with generational mode is the recommended GC for Java 21 with NeoForge. It provides consistently low latency (sub-millisecond pause times) and handles the large heap sizes common with modded Minecraft better than Shenandoah or G1GC. The Adoptium Temurin JDK 21 includes ZGC — no special Java build needed. `-XX:+AlwaysPreTouch` pre-initializes all memory pages at startup, preventing runtime page-fault stalls. `-XX:+PerfDisableSharedMem` disables synchronous `hsperfdata` file writes, removing a minor source of I/O-

induced latency. `-XX:SoftMaxHeapSize=12G` tells ZGC to target ~12G heap usage and GC more aggressively to stay near that level, while allowing bursts above it up to `-Xmx`. For this to take effect, `-Xmx` must be set higher than `-Xms`. The baseline uses equal values for simplicity; on a 64 GB machine you may want e.g. `-Xms12G -Xmx24G` instead to let `SoftMaxHeapSize` work.

X Minecraft Launcher (XMCL)

Tool	Role
XMCL	Mod manager and launcher for the modpack

XMCL manages mod and modpack installation, updates, and launching for this modpack. It supports both **CurseForge** and **Modrinth** as mod sources in one launcher, handles NeoForge installation automatically, manages Java versions per instance, and uses hard links to avoid duplicate mod storage. Open source, free, with no premium tier.

Setup:

1. Download and install **XMCL** from [xmcl.app](#) for your OS.
 - Windows: App Installer, AppX, or Zip (x64)
 - macOS: DMG (Intel or Apple Silicon)
 - Linux: Deb, RPM, AppImage, Flathub
 - Or via winget: `winget install CI010.XMinecraftLauncher`
2. Open XMCL and follow the [NeoForge](#) instructions below to create your instance.

Once the instance is running, install mods through the **Mods** tab from either CurseForge or Modrinth, or by dropping `.jar` files into the instance's `mods/` folder. XMCL uses hard links — mods are stored once globally and linked per instance, avoiding duplicates. All mod links in this document link to their CurseForge pages — XMCL can install from CurseForge directly via the Mods tab.

For texture packs, drop `.zip` files into the instance's **Resource Packs** tab or folder. For shaderpacks, drop `.zip` files into the instance's **Shader Packs** tab or folder. Access the instance folder from the instance settings.

NeoForge

Tool	Role
NeoForge	Mod loader for Minecraft 1.21.1

NeoForge is the mod loader powering this modpack. **XMCL installs NeoForge and Minecraft 1.21.1 automatically** when you create a NeoForge instance (see

XMCL above). Follow the XMCL steps — it handles the Minecraft version, mod loader, and Java runtime in one step.

If you prefer the official launcher:

1. Run **Minecraft 1.21.1** vanilla at least once from the official launcher to generate the game directory.
2. Download the **NeoForge installer** for 1.21.1 from neoforged.net.
3. Run it: `java -jar neoforge-1.21.1-<version>-installer.jar` → select “Install client”.
4. Launch the new **neoforge** profile once to generate the `mods/` directory.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.
 - Your Minecraft directory: Windows `%APPDATA%\minecraft`, Linux `~/.minecraft`, macOS `~/Library/Application Support/minecraft`

Important: Do not add any mod `.jar` files yet. Wave -1 is complete when the vanilla 1.21.1 client runs under NeoForge with no mods installed.

Skip this if you’ve played Minecraft before.

Controls (Default)

Action	Key
Move	W A S D
Jump	Space
Sneak	Shift
Sprint	Ctrl
Attack / Break	Left-click
Use / Place	Right-click
Inventory	E
Drop	Q
Hotbar slots	1–9

Your First 10 Minutes

1. **Punch a tree** — Hold left-click on a tree trunk until it breaks. Get 8+ logs.
2. **Make a crafting table** — Open inventory (E), place 1 log → 4 planks. Fill all 4 crafting slots with planks → crafting table.
3. **Craft tools** at the crafting table (3×3 grid):

Tool	Pattern
Wooden Pickaxe	3 planks top row, 2 sticks middle column
Wooden Axe	3 planks top-left, 2 sticks down
Wooden Sword	2 planks middle column, 1 stick below

1. **Gather stone** — Find grey stone, mine 8–12 blocks with your pickaxe.
2. **Build a shelter** — Dirt/wood walls + door (6 planks, 2 columns of 3) + torches (stick + coal/charcoal).
3. **Craft a furnace** — 8 cobblestone in a ring (center empty). Use it to smelt ore, cook meat, make charcoal.
4. **Survive the night** — Stay inside. Dig, expand, or sleep if you have a bed (3 wool + 3 planks).
5. **Day 2 priorities:** Find food, get iron (dig staircase-down until you see brown speckled blocks), find a village, make a bed.

Keybind to Set Immediately

Open **Options** → **Controls** → **Key Binds** and set these before anything else:

Action	Suggested Key	Why
Open Waystone Menu	<code>Ctrl+Shift+W</code>	Fast travel anywhere
Zoom	<code>C</code>	Sodium zoom
Toggle Dynamic Lights	<code>Ctrl+L</code>	Torch in off-hand = mobile light
Open Backpack	<code>B</code>	Sophisticated Backpacks
Open Fullscreen Map	<code>J</code>	JourneyMap

Wave 0 — Foundation

The essentials that make the game run well. Dependencies, performance, rendering, shaders, chunk loading, and texture packs. Install everything in this wave before launching.

Performance & Rendering

Mod	Role
Sodium	Rendering engine
Iris	Shader loader (v1.8.14, NeoForge native)
Lithium	Server-side game logic optimization
FerriteCore	Memory usage reduction
Distant Horizons	LOD rendering
ModernFix	All-in-one perf + bug fixes
ImmediatelyFast	Immediate-mode rendering
Entity Culling	Hide off-screen entities
Dynamic FPS	Background FPS reduction
NoisiumForked	Worldgen optimization (active fork)

ModernFix (configurable in-game via Mod List → ModernFix → Config, or `config/modernfix-mixins.properties`):

- `mixin.perf.dynamic_resources = true` — Load block/item models lazily instead of all at startup. **Biggest single launch-time speedup** in the pack. Reduces memory usage too. Disable if you see missing textures or broken models.

Shaderpack

Shaderpack	Description
Complementary Unbound	Realistic visual style, Potato→Ultra profiles, block-specific effects
Euphoria Patches	Optional add-on for Complementary Shaders — extends with handcrafted features, all disabled by default

Euphoria Patches is a shader add-on, not a standalone shaderpack. It patches Complementary Shaders (Reimagined or Unbound) with additional visual options. Install via its patcher mod (requires NeoForge, no additional mod dependencies). All features are disabled by default and can be individually enabled from the shader options menu.

Recommended Euphoria Patches settings to enable (Shader Options → Popular Settings tab, then apply the preset):

Performance impact is rated for the RTX 4080 SUPER (16GB VRAM, 4K, High profile) targeting 70-80 FPS. The base Complementary settings already leave headroom (no Entity Shadows, Shadow Res 2048, DH LOD 128-256 chunks), so Euphoria's overhead should be absorbed without dropping below the target. If FPS dips occur, disable the Heavy-impact features first.

Category	Setting	Value	Perf	Why
World	Colored Lighting	ON	Moderate	Dynamic block light colors — glowstone, redstone, sea lanterns
World	Seasons	ON	None	Leaves/grass shift colors with Serene Seasons calendar
World	Darkness Desaturation	ON	None	Colors desaturate in darkness for more atmospheric nights
World	Better Bedrock Noise	ON	None	Smoother, more natural bedrock layer pattern
Atmosphere	Better Clouds	ON	Heavy	Volumetric cloud improvements over base Complementary (High quality)
Atmosphere	No Rain Above Clouds	ON	None	Rain only falls below cloud layer (Iris-only)
Atmosphere	Aurora Colors	Custom	None	Fine-tune aurora hue/saturation for personal taste
Water	Better Water	ON	Heavy	Improved water reflections and wave animation
Effects	Lava Edge Effect	ON	Low	Glowing edge on lava blocks for depth (Iris-only)
Effects	End Portal Rays	ON	Low	Light rays emanate from End portal frames (Iris-only)
Effects	Epic Thunderstorm	ON	Moderate	More dramatic lightning visuals

Category	Setting	Value	Perf	Why
				(Iris-only); cost only during storms
Effects	Soul Sand Valley Overhaul	ON	Moderate	Richer fog and particle atmosphere in soul sand valleys
Effects	Dragon Death Effect	ON	Low	Spectacular visual during Ender Dragon death
Entities	Better Waving Vines/Sugar Cane	ON	None	Smoother plant waving animation (Iris-only)
Entities	Third-Person Interactive Foliage	ON	Low	Foliage reacts to player in third person (Iris-only)
Entities	Third-Person Soul Sand Eyes	ON	Low	Soul sand valley eyes track player in third person (Iris-only)
Mod Support	Better Modded SSBL	ON	Moderate	Auto-emissive modded light sources at correct light levels

Tip: The Popular Settings preset applies the most impactful options in one click. Browse individual categories afterward to fine-tune. Features marked “Iris-only” require Iris (not Oculus) — this pack uses Iris on NeoForge, so all features work.

Settings for NVIDIA RTX 4080 SUPER (16GB VRAM, 4K) — target: stable 70-80 FPS:

Setting	Value
Visual Style	Unbound
Profile	High
RP Support	Integrated PBR+
Shadow Resolution	2048
Shadow Distance	12 chunks
Real-Time Shadows	High
Light Shaft Quality	Medium

Setting	Value
Entity Shadows	OFF
Detail Quality	High
Cloud Quality	High
Water Reflection Quality	Medium (max)
Block Reflection Quality	Medium
Anti-Aliasing (FXAA)	ON
Edge Shadow (SSAO) Quality	Medium
Advanced Colored Lighting	8 chunks
Texture Filtering	8x

General MC settings: Graphics **Fancy**, Render Distance **10 chunks**, Simulation Distance **8 chunks**, Biome Blend **2 blocks**, Mipmap Levels **4x**, Entity Distance **100%**, Fullscreen ON, VSync OFF, Max Framerate **120**. Always Defer Chunk Updates **Enabled**, Use No Error Context **Enabled**.

Distant Horizons handles far rendering — keep vanilla render distance low. DH LODs render through the shader pipeline; start LOD distance at **128** and push to **256** if the FPS budget allows. Monitor VRAM usage (shaders + DH + high-res texture packs can peak past 12GB at 4K). If you experience dips below 70 FPS, first reduce Shadow Resolution to **1024**, then lower Shadow Distance to **8 chunks**.

Texture Packs

Pack	Role
Smooth Font	32x smooth font — clean modern typeface, Faithful 32x-based
Fresh Animations	Living entity animation overhaul — mobs blink, look around, show emotion
Enhanced Boss Bars	Redesigned boss bars with themed designs for vanilla and modded bosses
RAY's 3D Rails	3D block & item textures for all rail types
Better Lanterns	Enhanced 3D look for lanterns and chains
(Bee's) Fancy Crops	Reworked crop models with texture variations — compatible with Farmer's Delight

Pack	Role
Fire Rekindled	Animated fire, lava, water, and portal textures — includes Create mod support
Bushy Pink Petals, Wildflowers & Leaf Litter	3D models for pink petals, wildflowers, and leaf litter on the ground

Smooth Font replaces Minecraft’s pixelated default font with a clean, smooth 32x typeface based on Faithful 32x. Improves readability across menus, chat, and UI while keeping a vanilla-friendly look. Only overrides font textures — safe to use alongside any 16x resource packs. By apt9002. [Source \(CurseForge\)](#)

Fresh Animations adds idle animations, directional looking, sleep/blink cycles, and emotional states to all vanilla mobs. It’s a resource pack overlay (not a mod) — place it high in the pack order. Works on any version with no mod dependencies.

Enhanced Boss Bars replaces boss bar textures with themed designs — covers vanilla bosses and has support for Twilight Forest bosses. For full compatibility with L_Ender’s Cataclysm and The Aether (which use custom boss bar rendering), install the companion [Enhanced Boss Bars mod](#). Pure resource pack for vanilla/Twilight Forest bars.

RAY’s 3D Rails replaces rail block and item textures with 3D models. Supports shaders since v3.0. Compatible with all rail types from Steam ‘n’ Rails and Create.

Better Lanterns gives lanterns and chains an enhanced 3D look with connected chain models. Compatible with Arcane Lanterns and Thin Air mods.

(Bee’s) Fancy Crops reworks vanilla crop models with texture variations and a fluffier look. Built to work with **Farmer’s Delight** (Wave 4) and compatible with any mod that adds crops. 16x, vanilla-friendly. No mods required — pure resource pack. By EverLoste. 6.5M downloads. [Source \(Modrinth\)](#)

Fire Rekindled overhauls almost all animated textures in the game: fire, soul fire, campfires, torches, lanterns, water, lava, magma, nether portal, slime, sculk, and more. Animations are 32-64 frames, loop seamlessly, and stay close to vanilla artstyle. Supports **Create** mod (honey, steam jet particles). All textures include Normal and Specular maps (LabPBR standard) for PBR-compatible shaders. 1.3M downloads, MIT license. By Tectato. [Source \(Modrinth\)](#) — Optional CTM addon pack available.

Bushy Pink Petals, Wildflowers & Leaf Litter replaces flat pink petals, wildflowers, and leaf litter with 3D models. Makes ground cover look thicker and more natural — petals and leaves pile up with visible depth. Apache 2.0 licensed, actively maintained. By godhusk. 178K downloads. [Source \(CurseForge\)](#)

Installation: Install via XMCL — drop [.zip](#) files into the instance's Resource Packs tab, or copy them into the instance's [resourcepacks/](#) folder (access via instance settings). Load order (top = highest priority):

1. Smooth Font
2. Fresh Animations
3. Enhanced Boss Bars
4. RAY's 3D Rails
5. Better Lanterns
6. (Bee's) Fancy Crops
7. Fire Rekindled
8. Bushy Pink Petals, Wildflowers & Leaf Litter

Infrastructure

Mod	Role
Chunk Loaders	Keep chunks loaded across dimensions — essential for Phase 2+ automation
SuperMartijn642's Config Lib	Config library (Chunk Loaders dependency)
SuperMartijn642's Core Lib	Core library (Chunk Loaders dependency)
Chunky	World pregenerator — generates chunks preemptively to eliminate stutter

The essentials that make the game run well. Dependencies, performance, rendering, shaders, chunk loading, and texture packs. Install everything here before launching.

Install

Use **XMCL** — create a 1.21.1 NeoForge instance, install all Wave 0 mods via the Mods tab. See [Wave 0](#) for the full list.

Wave 0.5 (Player Experience) can be installed at the same time — they're independent mods that don't conflict.

First Launch Checklist

1. Video Settings

Setting	Recommended	Notes
Graphics	Fancy	Sodium makes this fast; Fancy (not Fabulous) avoids transparency conflicts with Iris shaders
Render Distance	10 chunks	Distant Horizons handles the rest
Simulation Distance	8 chunks	Fewer distant mobs ticking = better FPS

Setting		Recommended	Notes
VSync		OFF	
Max Framerate		Uncapped	

2. Distant Horizons (LOD Mod)

Open **Video Settings** → [colored tiles icon] next to the FOV slider.

Setting	Value
Enable Rendering	ON
Enable Distant Generation	ON
LOD Render Distance Radius	128–256
Quality Preset	Medium
CPU Load	Aggressive
Enable Cloud Rendering	OFF
Show LOD Gen Progress	ON

Tip: DH generates LOD data as you explore. First visit to an area has temporary pop-in. Terralith + Tectonic terrain takes longer initially. Monitor VRAM — shaders + DH at 4K can push past 12GB.

3. Shader Setup

Drop **Complementary Unbound .zip** into [shaderpacks/](#). In-game: Options → Video Settings → Shaderpacks → select Complementary Unbound. Works with Iris on NeoForge.

Euphoria Patches (optional add-on): Install the patcher mod, it auto-detects Complementary and applies additional visual options. All features disabled by default. Open Shader Options → Popular Settings tab → apply the Popular Settings preset, then browse individual categories to fine-tune.

Settings for RTX 4080 SUPER (16GB VRAM, 4K):

- Visual Style: Unbound, Profile: High, RP Support: Integrated PBR+
- Shadow Resolution: 2048, Shadow Distance: 12 chunks
- Real-Time Shadows: High, Light Shaft Quality: Medium
- Entity Shadows: OFF, Cloud Quality: High, Water Reflection: Medium

General MC: Graphics Fancy, Render Distance 10, Simulation 8, Biome Blend 2, Mipmap 4x, Entity Distance 100%, VSync OFF, Max Framerate 120.

4. Texture Pack Load Order (top = highest priority)

1. Smooth Font

2. Fresh Animations
3. Enhanced Boss Bars
4. RAY's 3D Rails
5. Better Lanterns
6. (Bee's) Fancy Crops
7. Fire Rekindled
8. Bushy Pink Petals, Wildflowers & Leaf Litter

5. Chunky — World Pregeneration

Run before exploring beyond your spawn area to avoid exploration stutter.

```
/chunky radius 5000
/chunky start
```

Let it finish (~15-30 minutes). Check progress with `/chunky status`. The world border expands as chunks generate. Set `"continue-on-restart": true` in `config/chunky/config.json` before you start — this auto-resumes the task every time you load the world, so you can close the game and it picks up where it left off.

Performance Mods — Install and Forget

These mods optimize automatically. No config needed.

Mod	What it does
Sodium	Modern rendering engine — massive FPS boost
Iris	Shader loader (Complementary Unbound)
Lithium	Server-side logic optimization — faster AI, block updates
Distant Horizons	LOD rendering — see 256+ chunks away
ModernFix	All-in-one performance + bug fixes
ImmediatelyFast	Immediate-mode rendering for blocks/items
Entity Culling	Skip rendering off-screen entities
Dynamic FPS	Lower FPS when Minecraft is backgrounded
NoisiumForked	Faster world generation when exploring
FerriteCore	Lower memory usage — fewer OOM crashes

ModernFix critical setting: `mixin.perf.dynamic_resources = true` in `config/modernfix-mixins.properties` — biggest single launch-time speedup. Disable only if you see missing textures.

Infrastructure

Chunk Loaders

Keeps chunks loaded when you're far away or in another dimension.

Type	Coverage	Use
Single Chunk Loader	1 chunk	One machine or farm
Spotloader	Handheld, temporary	Explore nearby
Multi Chunk Loader	3×3 chunks	Factory complex or colony

How: Craft (check EMI — iron + redstone + ender pearl), place, right-click to toggle. Colored border shows loaded area.

Placement priority (late game):

1. Main base (ME drive, Mekanism induction matrix, Create power core) — Multi Chunk
2. MineColonies colony — Single or Multi
3. Mining outposts (Digital Miner) — Single
4. Other dimensions — Single per outpost

No performance concern with a dozen loaders on a modern CPU.

Chunky — World Pregenerator

Chunky pregenerates chunks so your world has no exploration stutter, lag spikes from new terrain, or Distant Horizons LOD gaps. Run once per world before serious play.

First world setup:

Open chat (T) and run:

```
/chunky radius 5000  
/chunky start
```

This generates a 5000-block radius circle around your spawn point. On a modern CPU with NoisiumForked, this takes ~15-30 minutes. The world border expands as it works — you'll see chunk count progress in chat. Let it finish before exploring far from spawn.

Checking status: `/chunky status` shows progress, speed, estimated time remaining.

Pausing/resuming: `/chunky pause` and `/chunky continue`.

Continue on Restart: If you pause or leave a generation task running, Chunky can resume automatically every time you load the world. To enable this, edit `config/chunky/config.json`:

```
"continue-on-restart": false → true
```

Any active task will now resume by itself whenever you open the world. This is the recommended approach for single-player — the task resumes the moment you load the world, no manual `/chunky continue` needed.

Wave 0.5 — Player Experience

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great.

Visual & Client QoL

Mod	Role
LambDynamicLights	Dynamic lighting
BetterF3	Customizable debug HUD
Sound Physics Remastered	Realistic sound
AmbientSounds	Nature ambiance (birds, wind, water)
The Immersive Music Mod	147 new biome/structure-based songs, vanilla silence gaps preserved
Immersive UI	Animated UI, smooth hotbar, item particles
Immersive Hotbar	Animated hotbar slot transitions on pickup/use/swap, XP bar + level-up effects, tooltip animations
Smooth GUI	Slide-in animation when opening GUIs, fade-in background overlay + blur
Dark Mode Everywhere	Shader-based dark mode for all GUIs — 3 darkness levels, toggleable in-game
Status Effect Bars Reforged	Duration bars on status effect icons in HUD and inventory
Traveler's Titles	RPG-style titles entering biomes/dimensions
Better Modlist	Enhanced mod list screen with badges/categories
Better Third Person	Independent 360° camera rotation in third-person view
Countered's Smooth F5	Smooth third-person camera transitions on F5 — cinematic, configurable
Tooltip Overhaul	Modern, sharp tooltip rendering with equipment compare
First Person Model	Full body rendering in first-person view (requires Not Enough Animations)

Mod	Role
3D Skin Layers	Renders player skin layers (jackets, hats) with 3D depth
Presence Footsteps	Block-aware footstep sounds for every surface type
Particular Reforged	Ambient particles: 3D water splashes, fireflies, falling leaves, cave dust, waterfall cascades
Particle Rain	Replaces default weather with particle effects: angled rain, desert sandstorms, fog, wind
Swinging Lanterns	Subtle swinging movement on lanterns and attached chains — makes builds feel alive
Wakes Reforged	Realistic water wakes and splashes when traveling on water
Sounds	170+ new sound effects for UIs, items, blocks, actions
Better Days	Customizable day/night cycle length, enhanced sleep
Better Safe Bed	Prevents mobs that can't reach you from blocking sleep — chill-rule essential
Beautiful Enchanted Books [Mod Edition]	Unique textures per enchantment book type — standalone mod, no resource pack needed
Overflowing Bars	Expanded health, armor, and toughness bars beyond vanilla limits
Leave My Bars Alone	Keeps food/health/armor bars visible while riding mounts — requires Puzzles Lib
Cherished Worlds	Favorite/pin worlds to top of list, prevent accidental deletion
Eating Animation	First-person eating/drinking animation — food moves to mouth and shrinks
Camera Overhaul	Camera tilting, screen shakes, mouselook roll, idle sway — satisfying movement feedback

Mod	Role
Dynamic Crosshair	Context-aware crosshair — hides/shows/changes style based on what you're looking at or holding
[DLS] Dark Loading Screen	Replaces the default bright-red loading screen with a dark version — easier on the eyes
Cosy Critters & Creepy Crawlies	Ambient atmospheric animals — birds, bugs, and critters that bring the world to life
Foxified Dense Flowers	Renders multiple flowers per block in flower fields for denser, more natural-looking flora
Effectual	Atmospheric effects and decorative particles — steam breath, footprints, cave dust, sparks

Inventory & UI

Mod	Role
EMI	Recipe viewer with recipe trees
Jade	Block info HUD
Jade Addons (Neo/Forge)	Extra Jade info panels
JadeColonies	Jade tooltips for MineColonies — worker stats, hut info, DO blocks
JourneyMap	Real-time minimap + full-screen world map, waypoints, entity radar
JourneyMap Integration	Shows Waystones as icons on JourneyMap
AppleSkin	Food hunger/saturation overlay
Immersive Armor HUD	Armor type icons + durability above armor bar — shows what's about to break
Inventory Essentials	Drag-to-transfer, single-item moves, sorting
ClientSort	Sort (4 modes), Fill Stacks, Transfer Matching, customizable button positions — requires Cloth Config API
Advancement Plaques	Glowing plaque advancement popups with copper/silver/gold borders
Shulker Box Tooltip	Preview shulker box contents on hover
Item Borders	Colored borders on items based on rarity
Loot Journal	Animated item pickup notifications

Mod	Role
Day Counter	Day counter HUD — displays at dawn via ActionBar, fades after. Also supports Title/Subtitle/Chat display. Calendar block and reward system available but fully optional.
Hud Texts	HUD text framework — dependency for Day Counter (from V3.4)

Day Counter config (config/daycounter-common.toml — verify keys after first launch):

Setting	Value	Why
New day display	ACTIONBAR	ActionBar messages auto-fade after ~3s
New day sound	false	No sound notification — just the text
Rewards	false	No day-based rewards — too gamey
Week days	false	No day-of-week names — just “Day 42”
Date system	false	No calendar date — just day count
Calendar block/item	N/A	Don’t craft them — they’re items, not mandatory features

This gives you exactly: a brief “Day 42” (or similar) on the ActionBar at dawn that fades. No HUD element, no sounds, no rewards. Days count in the background. The calendar block can be crafted later if you want a physical day tracker for your base.

| [Better Advancements](#) | Overhauled advancements UI with editor/pan/zoom | | [Polymorph+](#) | Choose crafting result when recipes conflict | | [Traveler Tool Belt](#) | Tool radial menu — hold R for quick-swap, 2→9 slots via upgrades | | [Controlling](#) | Search keybinds by name, show only conflicting binds, list available keys — no dependencies |

Storage & Travel

Mod	Role
Sophisticated Storage	Upgradable barrels/chests
Sophisticated Backpacks	Portable inventory
Waystones	Fast travel between waypoints
GraveStone Mod	Keep inventory on death

Utility & QoL

Mod	Role
Easy Anvils	Overhauled anvils — persistent items, reduced prior work penalties, free renaming, repair with blocks

Easy Anvils requires **Puzzles Lib** — already in Wave 5 as a Mutant Monsters dependency. No additional mods needed.

Village & Trading

Mod	Role
Dynamic Villager Trades Neoforge 1.0.0	Dynamically changing villager trades — villagers specialize in what you buy from them, reducing the need for massive trading halls
Improved Village Placement	Forces villages to spawn on flat terrain — prevents cliff-side and underwater villages

Configuration — Village & Trading

Dynamic Villager Trades ([config/dynamicvillagertraders.properties](#)):

- `global_randomness = 0.7` (recommended, default 1.0) — Lower values make villagers settle on specific trades faster; higher values increase time to get desired trades. Multiplied by group randomness in the datapack.
- `refresh_delay = 1` (recommended, default 0) — Number of restocks between trade changes. `1` means trades change every other restock, giving you time to use existing trades before they rotate. Trades always change on level-up regardless.
- `no_book_duplicates = true` — Prevents librarians from selling the same enchant at different levels simultaneously.
- `enchant_repair_compat = false` — Adds Curse of Vanishing to every book sold by librarians (for Enchant & Repair compatibility). Keep disabled unless that mod is added.

Dependencies — Wave 0.5

Mod	Role	Required By
Architectury API	Cross-loader compatibility — lets Fabric mods run on NeoForge	First Person Model, Eating Animation, Effectual, multiple Wave 1-3+ mods

Mod	Role	Required By
Iceberg	Library for config-driven UI mods	Advancement Plaques
Polytone	Custom block colors, lightmaps, biome colors — Optifine format support for resource packs	— (dependency orphaned — Rainbow's Foliage removed)
Lithostitched	Worldgen modifier API — handles village placement hooks	Improved Village Placement
TLib (Take's Lib)	Library for atmospheric effects	Effectual

Lithostitched is a worldgen library that Improved Village Placement uses to hook into the village generation system. Without it, IVP crashes at startup with a `NoClassDefFoundError` for `AddWorldgenModifiersEvent`. This mod was not declared as a dependency in IVP's metadata — install it manually.

Architecture API is a lightweight library with no user-visible features — install once and forget. It provides the abstraction layer that lets Fabric-origin client QoL mods (First Person Model, Eating Animation) run seamlessly on NeoForge. Without it, these mods crash at startup with a missing-dependency error.

Install: In XMCL, search `Architecture API` in the Mods tab and add it. Alternatively, download from CurseForge and drop the `.jar` into `mods/`. No config needed. The game loads it silently — you'll never see it in-game, but EMI's mod list confirms it's present.

Configuration — Wave 0.5

Waystones (`config/waystones-common.toml`):

- `inventoryButton = "any"` — Opens waystone menu from inventory without a keybind.
- `enableCosts = false` — Free teleportation.

Loot Journal (`config/obscuria/loot_journal-client.toml`):

- `trackItemPickups = true` — Logs all item pickups for per-session statistics.
- `trackXpPickups = true` — Logs all XP pickups for per-session statistics.

Better Days (`config/betterdays-common.toml`):

Time speed can be set via three modes (`speedMethod`):

Mode	How it works	Best for
MINUTES (default)	Set <code>daySpeedMinutes</code> and <code>nightSpeedMinutes</code> in real minutes	Simple: 10 min day / 10 min night
RATIO	Multiplier relative to vanilla (1.0 = vanilla 20min cycle)	Fine control: 0.5 = 40 min, 2.0 = 10 min
SEASON	Day length varies by season using <code>seasonDayMinutes</code> and <code>seasonLatitude</code>	Immersive — day shortens in winter, lengthens in summer

Recommended settings (`speedMethod` = "SEASON"):

⚠ CRITICAL: These settings **MUST** be edited directly in `config/betterdays-common.toml`. The in-game config screen (Mod List → Better Days → Config) does not support the `SEASON` mode and **will silently overwrite** your config back to `MINUTES` mode if opened. **Do not open Better Days' in-game config.**

- `speedMethod` = "SEASON" — Day length varies by Serene Seasons sub-season. Summer days are longer, winter days shorter.
- `seasonDayMinutes` = 20.0 — Total day+night cycle in real minutes (vanilla baseline). 20 = same average length as vanilla.
- `seasonLatitude` = 48.0 — Central Europe latitude. Higher values = more extreme day-length swings between seasons.
- `dayStart` = 23500 — Dawn at time 23500 (default, correct range 22300–24000).
- `nightStart` = 13000 — Dusk at time 13000 (default).

Sleep — smooth time acceleration:

- `enableSleepFeature` = true — Smooth acceleration while in bed.

Time effects — all disabled. Leave them off:

- `weatherEffect` = "SLEEPING" — Lets rain/storms finish faster when sleeping.
- `potionEffect` = "NEVER" — Keep off. Speeding potion timers is confusing.
- `hungerEffect` = "NEVER" — Keep off. Speeding hunger is annoying.
- `blockEntityEffect` = "NEVER" — Keep off. Speeding furnaces/hoppers desyncs items and wastes fuel.
- `cropEffect` = "NEVER" — Keep off. Conflicts with Serene Seasons crop cycle timers.
- `randomTickEffect` = "NEVER" — Keep off. Avoids crop desync and random tick inconsistencies.

Dynamic Crosshair (configurable in-game via Mod List → Dynamic Crosshair → Config, or `config/dynamiccrosshair.json`):

Dynamic Crosshair uses **YACL** (already in the pack) for its in-game config screen. Recommended settings:

Setting	Value	Effect
<code>hideWhenNotLookingAtAnything</code>	<code>true</code>	Hides crosshair when pointing at sky/air — clean UI during building/exploration
<code>showWhenLookingAtBlock</code>	<code>true</code>	Reappears when looking at a block — signals interactability
<code>showWhenLookingAtEntity</code>	<code>true</code>	Reappears when looking at mobs/players — signals combat target
<code>showWhenHoldingTool</code>	<code>true</code>	Shows crosshair when mining with pickaxe/axe/shovel — shows exact breaking target
<code>showWhenHoldingWeapon</code>	<code>true</code>	Shows crosshair in combat with any weapon
<code>crosshairStyle</code>	<code>"CONTEXTUAL"</code>	Changes crosshair shape per context (default dot, block outline for mining, reticle for combat)
<code>crosshairColor</code>	<code>"ENTITY_BASED"</code>	White for neutral/block, red for hostile entities, green for interactable entities

No keybind changes needed — the mod works fully through context detection. YACL provides a searchable settings screen.

Dark Mode Everywhere (`config/darkmodeeverywhere-client.toml`):

- `darkModeLevel = 1` — Darkness level: 0 (light), 1 (medium, default), 2 (full dark). Change in-game via the button in the bottom-left corner of any GUI.
- `methodShaderDump = false` — Set to `true` temporarily if a GUI looks broken, to log which methods the shader is targeting.

Known issue — double text: Some mods render text shadows that get doubled by the dark shader. Fix: edit `darkmodeeverywhereshaders.json` (create in `config/` or via a resource pack) and change `darkColorRecplacment` from `16777215` to `4868682` (all 3 occurrences). Only needed if you see doubled text on specific screens.

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great. Install alongside Wave 0.

Visual & Client QoL — What to Expect

These mods change how the game looks and feels. All are client-side and work immediately.

LambDynamicLights

Hold a torch, glowstone, or lantern → it emits light around you. Works in off-hand.

Pro tip: Hold torch in off-hand while mining. No more placing torches every 5 blocks. Toggle with `Ctrl+L`.

BetterF3

Press `F3` for a cleaner debug HUD. Customize in Mod Menu → BetterF3 → Config. Toggle FPS, coords, biome, light level without clutter.

Sound Physics Remastered

Sound behaves realistically — muffled through walls, echo in caves, directional audio. Works automatically.

AmbientSounds

Biome-aware ambient audio — birds in forests, wind in mountains, water near rivers. Automatic, no config.

The Immersive Music Mod

147 new biome/structure-aware songs with natural silence gaps (unlike Constant Music which forces a 2-minute fade). Plays like vanilla but with much more variety.

Immersive UI

Smooth hotbar sliding, item pickup particles, animated tooltips, menu transitions. Pure polish.

Immersive Hotbar

Animated hotbar item slots bounce/shrink on pickup, use, and swap. Animated tooltip on item switch or hover. Animated XP bar and level-up effects make RPG progression feel impactful. Configurable animations via Mod Menu (YACL, already in pack). Complements Immersive UI — IUI animates the hotbar *selector bar*; Immersive Hotbar animates the *slot contents*.

Smooth GUI

Smooth slide-in animation when opening any GUI, with fade-in background overlay and blur. Pure cosmetic — makes inventory/chest/crafting screens feel polished. No dependencies, no config needed.

Status Effect Bars Reforged

Small progress bars on status effect icons showing remaining duration. Config via Cloth Config. Auto-hides for beacon effects.

Traveler's Titles

RPG-style title cards when entering biomes, dimensions, or deep caves/high peaks. No config needed.

Better Modlist

Enhanced mod list screen with badges (library/core/addon), categories, and search.

Better Third Person

Decouples camera from character in third-person. Look around independently while walking any direction. Essential for combat awareness and scenic building.

How: Bind a toggle key in Controls → Better Third Person. Once active, character moves independently of camera.

Tooltip Overhaul

Rounded corners, sharper text, scrollable tooltips, 3D item previews on hover. No config.

First Person Model + 3D Skin Layers

Shows your full body in first-person view, with 3D depth on skin layers (jackets, hats). Requires Not Enough Animations (auto-installed dependency).

Presence Footsteps

Block-aware footstep sounds for every surface — wood creaks, gravel crunches, stone clacks. Works with Sound Physics for realistic acoustics.

Particular Reforged

Ambient visual effects: 3D water splashes, waterfall cascades, fireflies at dusk, falling leaves, cave dust, underwater bubbles, rain ripples. Toggle individual effects in `config/particular.json` if any feel excessive.

Particle Rain

Replaces vanilla rain and snow with angled particle effects (wind-blown during storms), adds sandstorms to deserts and dry biomes (including modded), plus atmospheric fog and wind. Client-side only, no gameplay impact — purely

cosmetic weather enhancement. Configurable via [/particlerain](#) in-game menu or Mod List → Config. By PigCart, same author as Cosy Critters (already in this wave).

Swinging Lanterns

Adds subtle physics-based swaying to hanging lanterns and their chains. Makes builds feel alive and dynamic — lanterns gently rock as if stirred by a breeze. Client-side only, works on vanilla servers. No dependencies, no config needed.

Wakes Reforged

Realistic water wakes and splashes when traveling on water. Works with Small Ships (Wave 2) and vanilla boats.

Sounds

170+ new sound effects for UI clicks, item drops, eating, achievements, block actions. Every vanilla action feels tactile. Customize individually in Mod List → Sounds → Config or [config/sounds/](#). Conflicts with nothing — complements Sound Physics (environment) perfectly.

Better Safe Bed

Mobs that can't reach you no longer block sleep. Chill-rule essential.

Beautiful Enchanted Books [Mod Edition]

Unique textures per enchantment type — Protection books look different from Sharpness books. Standalone mod, no resource pack needed.

Overflowing Bars

Health, armor, and toughness bars expand beyond vanilla limits. Essential once you stack Apotheosis bonuses and skill tree HP.

Leave My Bars Alone

Keeps your food, health, and armor bars visible while riding mounts. Vanilla replaces your bars with the mount's health — this fixes that. Essential for exploration on horses, aircraft, ships. Requires Puzzles Lib (already in pack). No config needed.

Cherished Worlds

Favorite/pin worlds to the top of your list, prevent accidental deletion. Right-click a world in the list to favorite it.

Day Counter

Day counter title overlay at dawn with milestone celebrations (Day 10, 50, 100...). Appears automatically — no config.

Eating Animation

First-person eating/drinking animation — food moves to your mouth and shrinks. Works with Farmer's Delight foods.

Camera Overhaul

Camera tilting on movement, screen shakes on explosions, mouselook roll, idle sway. Satisfying feedback. Configurable in `config/cameraoverhaul-client.toml`.

[DLS] Dark Loading Screen

Replaces the bright-red loading screen with a dark version. No config.

Cosy Critters & Creepy Crawlies

Adds ambient atmospheric creatures: birds flock in the sky and land on elevated blocks, bugs scuttle through grass, and other critters bring the world to life. Birds scatter when you or a vehicle approach. **Client-side only** — no server install needed. All effects individually toggleable via `/cosycritters` command or the mod list config menu. Zero performance impact (97KB).

Foxified Dense Flowers

Renders multiple flowers per block in flower fields — the more flowers adjacent to each other, the denser the display. Makes flower forests and plains feel lush and overgrown without adding new flowers or changing worldgen. **Client-side only**. Works with existing vanilla and modded flowers.

Effectual

Atmospheric particle effects that make the world feel more alive: steam breath in cold biomes, bubbles underwater, drips when surfacing, fire sparks and lamp glows, footprints in snow, cave dust, steam columns where water meets lava, minecart sparks, and more. **Compatible with Particular Reforged** — any overlapping effects (e.g., cave dust) can be disabled individually via `config/effectual-client.toml`. Requires **Architecture API** (already in pack), **Cloth Config API** (already in pack), and **TLib** (auto-installed Wave 0.5 dependency). Client-side only, MIT license. By iMeeTake. 1.6M downloads.

Inventory & UI — Your Daily Tools

EMI

Your primary learning tool. **Hover any item, press R** to see how to craft it. **Press U** to see what it's used in. Every mod in future waves is learned through EMI first. EMI also shows recipe trees — break down complex crafts (Create sequenced assembly, Mekanism ore processing chains) into every step, showing base ingredients and leftovers.

Jade + Jade Addons

Look at any block → HUD shows block name, mod source, machine status, crop growth, chest contents. **Jade Addons** extends this with panels for Create (SU/RPM), Mekanism (energy/gas), AE2 (channel info).

Pro tip: Keep Jade enabled. Fastest way to check if a machine is running or a crop is ready.

Conflict fix: Main Menu → Mods → Jade → Config → Mod Settings → Display Bosses → OFF (avoids conflict with Enhanced Boss Bars).

JourneyMap + JourneyMap Integration

- **Minimap** — top-right corner, auto-maps, shows entity dots
- **Fullscreen map** (**J**) — zoomable world map with biome colors, terrain height, waypoints
- **Waypoints** — **B** at your location, or set on the map. Custom icons, colors, groups
- **Entity radar** — shows nearby mobs/animals on minimap
- **Integration** — your placed Waystones appear as icons on JourneyMap automatically

Pro tip: **J** → right-click to set temporary waypoint. Double-click a waypoint name to edit it.

AppleSkin

Hover over food → see exact hunger and saturation values. Pre-eat preview shows dashed outlines on your hunger bar. Saturation overlay shows the hidden second layer. No config.

Immersive Armor HUD

Miniaturized armor type icons above the armor bar with durability indicators — shows which pieces are about to break at a glance without opening inventory. Vanilla-friendly 7×7 pixel icons. Works with modded armor (Mekanism, Apotheosis gear). No dependencies, no config needed.

Inventory Essentials

Drag-to-transfer, single-item moves, sorting:

- **Ctrl+click** a stack → moves one item
- **Shift+Ctrl+click** → moves all items of that type
- Sort buttons on chest/inventory GUIs

Note: ClientSort (below) handles sorting with 4 sort modes, Fill Stacks, and Transfer Matching. Inventory Essentials remains for drag-to-transfer and single-item moves — its sorting is superseded but not conflicting.

ClientSort

Versatile inventory sorting with 4 sort modes:

- **Sort** (default): creative inventory tab order — middle-click to trigger
- **Shift+Sort**: descending order of quantity
- **Ctrl+Sort**: alphabetical order
- **Alt+Sort**: ascending order of item ID

Additional operations:

- **Fill Stacks**: top up partial stacks in the other inventory using items from this one
- **Transfer Matching**: move all matching items between inventories in one click
- **Button Editor**: right-click any trigger button to reposition or enable/disable

All operations can be triggered via buttons (shown on inventory/crafting GUIs) or keybinds. Requires **Cloth Config API** (already in pack for config). No overlap conflict with Inventory Essentials — IE handles drag-to-transfer and single-item moves; ClientSort handles sorting and bulk operations.

Shulker Box Tooltip

Hover over a shulker box in inventory → preview its contents. No config.

Item Borders

Colored rarity borders on items. Common = gray, Uncommon = green, Rare = blue, Epic = purple, Mythic = gold. Helps spot good drops instantly.

Loot Journal

Animated pickup notifications near your hotbar. Session summary on disconnect shows what you gained/lost. Config in Mod Menu → Loot Journal:

- `trackItemPickups = true`
- `trackXpPickups = true`
- `showSummaryOnDisconnect = true`

Better Advancements

Overhauled advancements UI — larger screen, pan/zoom, tab sorting, search. Open with **L**. Pairs with Advancement Plaques (glowing popup banners).

Advancement Plaques

Glowing advancement plaques replace vanilla popups — copper for Tasks, silver for Goals, gold for Challenges. Requires **Iceberg** (auto-installed). Configurable position, distance, and whitelist filter.

Polymorph+

When two mods have conflicting recipes for the same item, a small button appears next to the crafting output. Click to cycle recipes. Persists per ingredient combination.

Traveler Tool Belt

Quick-swap tool access without opening your inventory. Hold **R** (configurable in Controls) to open a radial menu, then select the tool you need. The belt renders as a subtle layer on your player model.

- Starts with **2 slots**, expandable to **9** by upgrading at a crafting table
- Holds any non-stackable item by default (tools, weapons, weapons)
- Integrates with Curios API belt slot (already in pack) — equip from Curios menu or inventory
- Equip in the belt slot, or keep it in your inventory if Curios is unavailable

Controlling

Search keybinds by name in the Controls menu. “Show Conflicts” button highlights only conflicting binds so you can fix them fast. Lists all available (unbound) keys. Essential for a 150+ mod pack — the vanilla keybind menu without Controlling is a wall of unsortable text. No dependencies, no config needed.

Storage & Travel

Sophisticated Storage + Backpacks

- **Barrel** — craft: 1 chest + 8 planks. Upgrade with iron/gold/diamond/netherite upgrades for more slots, pickup, magnet, filter, composting
- **Backpack** — equip in chest armor slot, press **B** to open. Upgradable like barrels
- **Create Integration** (auto-installed) — Sophisticated barrels and backpacks work fully on Create contraptions: upgrades, automation, inventory settings all function while moving

Waystones

Craft a waystone → right-click to claim. Use a Warp Stone (ender pearl + amethyst) to teleport from anywhere. **No XP costs** (`enableCosts = false` in config). Set keybind to `Ctrl+Shift+W`.

GraveStone Mod

Die → grave spawns with all your items. Right-click the grave to retrieve. No item loss, no despawning, 24-hour protection.

Village & Trading

Dynamic Villager Trades

Villager trades change dynamically — villagers specialize in what you buy from them. No more massive trading halls.

Config (`config/dynamicvillagertraders.properties`):

- `global_randomness = 0.7` — faster specialization
- `refresh_delay = 1` — trades change every other restock
- `no_book_duplicates = true` — no duplicate enchantment books

Improved Village Placement

Forces villages to spawn on flat terrain. No more cliff-side or underwater villages. No config.

Session Plan — Wave 0.5

Session	Goal
Session 1 — Setup	Launch, set keybinds, configure Distant Horizons (LOD 128-256, Medium quality). Install Complementary Unbound shader. Enable Fresh Animations texture pack. Walk around spawn to generate LOD data
Session 2 — Survival basics	Punch trees, build a starter shack, find food, get iron. Place your first Waystone. Craft a Sophisticated Storage barrel. Craft a backpack
Session 3 — Comfort	Explore nearby terrain. Find a village. Set up a bed. Light up your base perimeter. Start noticing Traveler's Titles when entering new biomes

Wave 1 — Tech (Create + Mekanism + AE2)

The classic tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting.

Core Tech

Mod	Role
Create	Mechanical automation — rotational power, water wheels, windmills, gear trains, conveyor belts, deployers, sequenced crafters, trains
Mekanism	Industrial processing — ore multiplication (2x→5x), digital miner, superdense energy storage, MekaSuit (flight, invulnerability)
Mekanism Generators	Power generation — heat, wind, solar, gas-burning, bio generators, fission reactor, fusion reactor, turbine
AE2 19.2.17	Digital storage — ME drives, autocrafting, P2P tunnels, spatial storage. The storage backbone
Applied Mekanistics	AE2-Mekanism integration — autocrafting for chemical/gas/fission/fusion recipes, ME chemical storage, P2P chemical tunnels

Note: Mekanism v10 for 1.21.1 NeoForge is split into separate modules.

Mekanism (core) provides machines, processing, cables, and armor. **Mekanism Generators** provides all power generation. Both must be installed.

Create Addons

Mod	Role
Create Crafts & Additions 1.6.0	Electric motors, alternators, rolling mill, wires/rods, silver/electrum — bridges kinetic SU and Forge Energy (FE)
Create: Power Grid 0.5.5.1	Physics-based electricity simulation — Ohm's law, AC/DC, generators, motors, cables, FE conversion. Separates Create power from SU
Steam 'n' Rails NeoForge 0.2.1	Expanded train system — new tracks (spruce, monorail), semaphores, conductor mob, coupling/decoupling blocks

Mod	Role
Create: The Factory Must Grow (TFMG)	Heavy engineering & oil — crude oil drilling, distillation, diesel/gasoline/LPG engines, steel, aluminum, electricity with voltage
Mekanism TFMG Compatibility	85+ recipe bridges — deduplicates lead/steel/sulfur, integrates aluminum/plastic into Mekanism chains
TFMG Energy Converter 26.03	Bidirectional TFMG electricity ↔ FE/AE2 bridge — lets TFMG diesel engines/generators power non-TFMG machines (Mekanism, AE2). Requires Kotlin for Forge
Create: Gunsmithing	Steampunk firearms integrated with Create — craftable using Create machines. Animated guns, attachments, ammo types. Optional Better Combat compatibility
Create Ornithopter Glider	Mechanical flapping glider with boost-assisted flight — integrates with Create rotational power
Sophisticated Storage Create Integration	Full-featured Sophisticated Storage on Create contraptions — upgrades, automation, inventory settings work while moving
Sophisticated Backpacks Create Integration	Full-featured Sophisticated Backpacks on Create contraptions — upgrades, automation, inventory settings work while moving
Create: Train Utilities 3.0.3	Sliding doors with realistic sounds, platform blocks, station building blocks
Create Railways Navigator	Display boards, route navigation, new schedule entries — search routes and manage train networks
Create: Interiors 0.6.1	Colored chairs with armrest toggles, tray tables, kelp seats — functional furniture for train cars
Create Train Parts 0.4.1	Crossing gates, sliding windows, train steps, slides — decorative and

Mod	Role
	functional blocks for trains and stations
Create: Threaded Trains	Performance — runs train pathfinding on a separate thread, prevents tick lag on complex rail networks
Create Better FPS	Create rendering optimization — up to 50% FPS gain with shaders. Handles addon models, speed controllers, factory gauges, and schematic previews
ColorWheel	Flywheel + Iris compatibility — renders Create contraptions under shaders. Required for Create to work with Iris. Both Complementary Unbound and Euphoria Patches have official support
Create Slice & Dice	Create + Farmer's Delight automation — Slicer (auto Cutting Board), automatic FD Cooking Pot via heated mixing, Sprinkler for fluid-based farming
Create: Dragons Plus 1.11.2b	Fan processing expansion — Bulk Coloring, Bulk Freezing, Bulk Ending, Bulk Sanding. Fluid Hatch for tanks. Adds Aether Bulk Enchanting. Library mod for Create addon devs
Create: Protection Pixel	Create-themed power armor — brass/alloy tiers, per-piece active abilities (AoE, debuff clear, speed→damage), combustion reactor (blaze rods + water). Bridges gear between Mekanism Tools and MekaSuit

Create Addon Details

Create Better FPS optimizes Create's rendering pipeline — handles addon models (e.g., blaze with straw from Create Crafts & Additions), speed controllers, factory gauges, and schematic previews. Benchmarked 7→35 FPS with 512 waterwheels under shaders. Works alongside ColorWheel: ColorWheel handles Flywheel instance compatibility; Create Better FPS handles everything else.

ColorWheel makes Flywheel (Create's rendering engine) compatible with Iris shaders. Without it, Create contraptions either don't render or tank performance under shaders. ColorWheel is the higher-priority install between the two — Create Better FPS is additional optimization on top.

Create Slice & Dice automates Farmer's Delight through Create. The **Slicer** acts as an automatic Cutting Board — place a knife or axe inside (right-click), and it registers all FD cutting recipes as mechanical processing. **Automatic cooking** converts FD Cooking Pot recipes into Create heated mixing recipes, so you can pipe ingredients in and collect meals. The **Sprinkler** distributes fluids below it: water for hydration, lava for damage, potions for effects, and Liquid Fertilizer (new fluid) for automatic bonemeal. This bridges Phase 1's food focus with Phase 2's automation — by the time you have Create infrastructure, meal production becomes fully pipeable.

Create: Dragons Plus is a library that expands Create's fan processing with Bulk Coloring (dye items via Encased Fan + dye fluid), Bulk Freezing (Powder Snow), Bulk Ending (Dragon's Breath / Dragon Head), and Bulk Sanding (quicksand). Also adds a Fluid Hatch for convenient tank I/O and Aether Bulk Enchanting (enchanting recipes, repair, Moa Egg incubation in the Aether). All features can be toggled individually in config. Required dependency for Create: Shimmer.

Create: Protection Pixel

Steampunk power armor built with Create materials. Each armor piece has a unique active ability powered by a waist-slot Combustion Reactor (blaze rods + water as fuel).

Armor pieces (brass → alloy tier progression):

Piece	Ability	Best for
Plague Helmet	Release steam to dispel blindness, darkness, weakness, slowness	Exploration, debuff-heavy mobs
Lancer Helmet	Speed → damage scaling — the faster you move, the harder you hit	Mobility builds
Hunter Helmet	Sensors highlight nearby creatures while sneaking; +luck for finding treasure	Dungeon crawling
Breaker Chestplate	+damage +attack speed while maintaining protection	DPS builds
Magnetic Storm	AoE magnetic burst when taking heavy damage or	Crowd control

Piece	Ability	Best for
	surrounded; pulls ranged attackers	
Float Shield	Energy shell around user when charged — absorbs incoming damage	Tank builds

How it works:

1. Craft armor pieces at Create machines (check EMI — requires brass-tier Create infrastructure)
2. Craft a **Combustion Reactor** (waist slot, check EMI) — this powers all equipped armor
3. Feed the reactor **blaze rods + water** (displayed in minutes in the GUI). More blaze rods = longer runtime
4. If reactor runs out: slowness + mining fatigue + weakness. Keep it fueled!

Armor plates: Stand on an **Armor Load Platform** (requires lava piped in — 100mB per assembly/disassembly). Right-click while sneaking to install/remove armor plates with different stat values. Plates also provide a repair method — no anvil needed.

External devices: Additional Create-themed accessories (back-mounted) that provide utility without consuming armor slots.

Compatibility: Works with Better Combat, Epic Fight, and backpacks (visual space reserved for backpack rendering). Apotheosis affixes and gems should apply — verify after test launch.

Dependencies: [Create](#) (already in pack) + [Kleiders Custom Renderer API](#). For versions below 1.9.0, requires [Curios API](#) instead (already in pack). Currently on v2.1.9+ using Kleiders.

Progression placement: Brass-tier armor is early Phase 2 (requires Create brass). Alloy-tier is mid-Phase 2 (requires Create alloy infrastructure). MekaSuit remains the Phase 3 endgame upgrade. This fills the Phase 2 power-armor gap between basic Mekanism Tools and the MekaSuit.

Mekanism Addons

Mod	Role
Mekanism Tools	Armor, shields, tools, and paxels for Mekanism materials — Osmium, Bronze, Refined Obsidian, Refined Glowstone, Steel. Fills gear gap between netherite and MekaSuit.

Mod	Role
Advanced Finders	Ore detection devices — directional indicators for nearby ore veins, synchronized multiplayer support
Advanced Chimneys	Smoke ventilation systems — redirect factory emissions outdoors, customizable appearance and behavior

Dependencies: Both Advanced Finders and Advanced Chimneys require [ForgeEndertech](#) library.

How They Work Together

Layer	Mod	
Power generation	Create (water wheels, windmills, steam) + Mekanism (advanced generators, fusion reactor)	
Gear progression	Mekanism Tools — Osmium → Bronze → Refined Obsidian → MekaSuit	
Ore processing	Mekanism — up to 5x ore multiplication with chemical processing	
Factory automation	Create — conveyor belts, deployers, sequenced crafters	
Storage	AE2 — ME drives, storage busses, crafting CPUs	
Transport	Create trains + AE2 P2P tunnels	
Endgame	Mekanism fusion reactor + MekaSuit + AE2 autocrafting	### Ores

Mod	Role
Create (built-in)	Zinc, brass, copper
Mekanism (built-in)	Osmium, tin, lead, uranium, fluorite

Configuration — Wave 1

Mekanism ([config/Mekanism/general.toml](#)):

- `allowChunkloading = false` — Disable Mekanism’s built-in chunk loading. Prevents conflict with the dedicated Chunk Loaders mod and keeps chunk management unified.

Ore generation: All default rates. Osmium at iron-like frequency, tin/lead moderate, uranium/fluorite rarer. Default distribution aligns with the pack’s progression curve.

Interdependency notes for Wave 1:

- Mekanism's universal cables can power AE2 controllers via the Energy Acceptor. No adapter mod needed.
- **Create Crafts & Additions** electric motors and alternators create a direct kinetic↔FE bridge, replacing the Create Alternator workaround. Silver and electrum add new material tiers that integrate with Mekanism's ore processing line.
- **Steam 'n' Rails** expands Create's train system with new tracks, semaphores, conductor mobs, and coupling blocks. Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.
- **Create train ecosystem:** SnR provides track types (monorail, wooden, wide gauge), semaphores, and the conductor mob. **Train Utilities** adds station architecture (sliding doors, platform blocks). **Railways Navigator** adds display boards and route management. **Interiors** makes train cars usable with seats and tray tables. **Train Parts** adds crossing gates, windows, and steps for detailing. **Threaded Trains** keeps the whole network performant by running pathfinding on a separate thread. Together they cover tracks → stations → navigation → interiors → detailing → performance without overlap.
- Mekanism's 5x ore processing (Chemical Dissolution Chamber → Washer → Crystallizer) outclasses Create's Crushing Wheels (1.5x) and Millstone (1x) for ore multiplication. This is intentional — Create handles mechanical automation and component crafting, Mekanism handles industrial processing. Neither obsoletes the other.
- AE2's Storage Bus on a Sophisticated Storage barrel makes early-mid game storage migration seamless. Place a Storage Bus on your barrel wall, configure it to show contents in the ME terminal, and transition gradually to full digital storage.
- **Sophisticated Storage Create Integration** and **Sophisticated Backpacks Create Integration** (both by P3pp3rF1y) enable full backpack and barrel functionality on Create contraptions — upgrades, automation features (pickup, magnet, feeding), and inventory settings all work while the contraption is in motion. Without these addons, Sophisticated inventories on contraptions are limited to basic item transport.
- **Applied Mekanistics** bridges AE2 autocrafting with Mekanism's chemical processing — ME Chemical Output Buses, ME Chemical Import Buses, and P2P tunnels for chemicals/gases enable autocrafting of Mekanism recipes (including fission/fusion products) directly through the AE2 crafting CPU. Without this mod, Mekanism chemical steps are manual bottlenecks.
- **Create: The Factory Must Grow** adds oil drilling, distillation towers, blast furnaces, and diesel engines — expanding Create's mechanical toolkit into heavy industry. Its steel replaces Mekanism's steel (TFMG's blast furnace is the intended steel source; Mekanism's steel recipe is disabled by the Mekanism TFMG Compat mod). TFMG aluminum and plastic feed into Mekanism's advanced processing chains.

- **Mekanism TFMG Compatibility** resolves 85+ recipe overlaps between TFMG and Mekanism — lead, sulfur, and steel are unified under TFMG's production lines, while Mekanism retains its chemical processing advantage. Both EMI item blacklist config and per-recipe toggles are available for pack tuning.
- **Create: Protection Pixel** adds steampunk power armor that bridges Create's brass-tier infrastructure with the power fantasy of Mekanism's MekaSuit. Brass-tier armor requires Create brass (early Phase 2); alloy-tier requires additional Create alloy processing (mid-Phase 2). The Combustion Reactor's blaze rod fuel can be farmed via Mekanism spawner systems, Apotheosis spawner upgrades, or vanilla blaze grinders. Blaze automation is a natural Phase 2 milestone that unlocks full power armor capability. MekaSuit remains the Phase 3 ultimate upgrade — Protection Pixel fills the gap, it doesn't compete.

The core tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting. Install all three core mods plus addons at once.

Core Tech

Create — Mechanical Automation

Create is about **rotational power (SU)**. Machines consume SU to run. If consumption exceeds generation, machines stop.

Learning with Ponder: Hold **W** while hovering over any Create block/item in EMI. An animated tutorial plays showing exactly how it works. **Use Ponder constantly** — it's better than any written guide.

First Create setup:

1. Mine **andesite** — the most important early resource
2. Craft **andesite casing** (andesite + wood nugget in EMI)
3. Craft **cogwheel** and **large cogwheel** — transfer rotational power
4. Place a **hand crank** → attach cogwheel → attach **millstone** → right-click crank to grind wheat/ore manually
5. Build a **water wheel** in flowing water (2–3 block drop) → attach shaft → power 2–3 basic machines (256 SU)

Brass progression (mid-Create): Find zinc ore → mix zinc + copper in a **heated mixer** → brass ingots → brass casing → unlock:

- **Deployer** — places blocks/uses items from inventory. Automates crafting
- **Sequenced Crafter** — multi-step autocrafting sequence
- **Mechanical Arm** — moves items between inventories
- **Chute & Funnel** — vertical item transport with filtering
- **Train Station** — build tracks, assemble trains

Blueprint system: Craft **Blueprint and Quill**. Select an area, save as a blueprint. Place a **Deployer** with the blueprint to auto-build it. Or use the **Schematicannon** — load a blueprint, feed it materials from chests, and it builds the structure for you automatically. This is your “here’s a blueprint → build it” tool.

Keybinds:

- Ponder: **W** (hold over block)
- Wrench: right-click
- Goggles: **G** (hold over machine) — shows SU, RPM, stress info
- Toolbox: **T** (hold)

Goggles are essential — craft early, check EMI.

Mekanism — Industrial Processing

Mekanism v10 for 1.21.1 is split into two required modules:

- **Mekanism** (core) — machines, processing, cables, armor
- **Mekanism Generators** — all power generation (heat, wind, solar, gas-burning, bio, fission, fusion)

Finding Osmium: Generates at all underground levels, similar rarity to iron. Mine with stone pickaxe or better.

Tier system: Every machine has 4 tiers. Upgrade by shift-right-clicking with the tier upgrade item.

Tier	Color	Speed	Upgrade
Basic	Orange	1x	Starting
Advanced	Red	2x	Basic + osmium + redstone
Elite	Dark Red	4x	Advanced + diamond + obsidian
Ultimate	Purple	8x	Elite + refined obsidian + alloy

First Mekanism setup:

1. **Heat Generator** — place lava source below it for early power
2. **Metallurgic Infuser** — combine osmium + carbon (coal) → **steel ingots**
3. **Enrichment Chamber** — 1 ore → 2 dust → smelt → 2 ingots (2x processing)

Processing multipliers:

- **2x:** Enrichment Chamber only
- **3x:** Enrichment Chamber → Crusher → Enrichment Chamber again
- **5x:** Full chemical line — Chemical Dissolution Chamber (sulfuric acid) → Washer → Crystallizer → 5 dust

Gas infrastructure (for 5x processing):

- Hydrogen/Oxygen: Electrolytic Separator (split water)

- Sulfuric Acid: Chemical Infuser (sulfur + oxygen)
- Deuterium/Tritium: Fusion reactor fuel (late game)

Gases move through **pressurized tubes**. Use a **Rotary Condensentrator** to convert gas ↔ liquid.

Digital Miner (mid-game): Automated mining machine. Set radius (up to 64 blocks) and filters (whitelist ores). Needs significant power. Tip: set to mine only ores to avoid filling with stone.

Fusion Reactor (endgame): Millions of Joules per tick. Fuel chain: water → Electrolytic Separator → hydrogen → deuterium → tritium → feed both into reactor. Requires significant infrastructure — a Phase 3 project.

Config: `config/Mekanism/general.toml` → `allowChunkloading = false` (delegated to Chunk Loaders mod).

AE2 — Applied Energistics 2

Digital item storage and autocrafting. Everything lives in an electronic network accessed from a single terminal.

Finding Certus Quartz:

- **Meteorites** — circular surface depressions. Dig down to find certus quartz blocks + sky stone chest with free AE2 components
- **Ore veins** — certus quartz ore underground

Processing: certus quartz block → mine → item → craft into dust → smelt → crystal → charge in **AE2 Charger** → charged certus quartz

Channel system: AE2 limits devices per cable. This is the hardest concept for new players.

Cable type	Channels	Notes
ME Glass Cable	8	No indicator
ME Smart Cable	8	Colored lines show usage
ME Dense Smart Cable	32	Thicker, for backbone

- Each ME Controller face provides 32 channels
- Each device (drive, terminal, bus) consumes 1 channel
- Exceed channels → devices show “no power”

Use ME Smart Cable everywhere — colored lines instantly show usage.

First network:

1. **Power:** Build a **vibration chamber** or **energy acceptor**, connect any power source (Mekanism generator works)

2. **Controller:** Craft **ME Controller**, place it, connect power. One controller = 32 channels
3. **Storage:** Craft **ME Drive + 1k storage cells** (charged certus + redstone + glass), place next to controller
4. **Terminal:** Craft **ME Crafting Terminal** — essential, lets you craft from stored items

Connecting to the world:

- **ME Import Bus** — pulls items from adjacent inventory into network
- **ME Export Bus** — pushes items from network into adjacent inventory
- **Storage Bus** — treats adjacent inventory as part of network (no import/export needed)
- **ME Interface** — exports AND accepts items for autocrafting

Autocrafting:

1. Craft **Patterns** (blank pattern + items in grid)
2. Put patterns in **ME Pattern Provider**
3. Place provider next to **ME Molecular Assembler**
4. Request item from terminal — AE2 pulls ingredients, crafts, returns result

For Mekanism/Create machines: use **ME Interface** instead of pattern provider. Put pattern in interface, it exports ingredients to adjacent machine and accepts the result.

Subnetworks (advanced): When you exceed 32 channels, add a second controller connected via **ME P2P Tunnel** or **ME Quantum Link** for wireless.

Applied Mekanistics

Bridges AE2 autocrafting with Mekanism's chemical processing. Adds:

- ME Chemical Output Buses / Import Buses
- P2P tunnels for chemicals/gases
- Autocraft Mekanism recipes (including fission/fusion products) through AE2 crafting CPU

Without this mod, Mekanism chemical steps are manual bottlenecks.

Create Addons

Create Crafts & Additions

Electric motors and alternators create a direct **kinetic SU ↔ Forge Energy (FE)** bridge. Also adds:

- **Rolling mill** — processes metal into wires/rods

- **Silver and electrum** — new material tiers that integrate with Mekanism ore processing

Use: Connect an electric motor to your Create network to power it with FE from Mekanism. Or connect an alternator to generate FE from Create rotational power.

Create: Power Grid

Physics-based electricity simulation — Ohm's law, AC/DC, cable loss, transformers. Bridges to FE so Power Grid can power Mekanism/AE2 and vice versa.

Key concepts:

- Generators produce voltage, cables have resistance, machines draw current
- AC travels farther but needs rectification for DC machines
- Use transformers to step voltage up for long runs, then step down locally
- Built-in FE conversion via FE Converter block

When to build: Mid-Phase 2, after you have basic Create power and have outgrown SU. Diesel engines from TFMG feed into Power Grid naturally.

Steam 'n' Rails

Expanded train system:

- New track types: spruce, monorail, wide gauge
- Semaphores for signal control
- Conductor mob (rides trains, interacts with stations)
- Coupling/decoupling blocks

Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.

Create: The Factory Must Grow (TFMG)

Heavy engineering and oil processing. Drill for crude oil, refine it, burn fuels in diesel engines.

Oil processing chain:

1. Find oil — surface seepage in desert/plains biomes, or underground pools
2. **Pumpjack** — extracts oil automatically (crafted from brass and steel)
3. **Distillation Tower** (multi-block, 4+ blocks tall) — refines crude oil into diesel, gasoline, LPG, lubricant, asphalt
4. **Cracker** — breaks heavy fractions into lighter fuels
5. **Desulfurizer** — cleans fuel for more efficiency

Diesel engines:

Engine	Fuel	SU Output
Small Diesel	Diesel/gasoline	~2,000 SU
Large Diesel	Diesel/gasoline	~8,000 SU
Gas Turbine	LPG	~16,000 SU

Progression: water wheel → steam engine → diesel engine.

Steel Blast Furnace: Multi-block structure for industrial-scale steel. Faster than Mekanism's Metallurgic Infuser once you have oil power.

Aluminum: Bauxite ore (mid-to-high Y-levels) → multi-block smelter → aluminum ingots. Used in advanced TFMG machines and feeds into Mekanism alloy processing.

TFMG Energy Converter: Bidirectional bridge between TFMG electricity ↔ FE/AE2. Wire diesel generators into your ME system or Mekanism induction matrix. Requires Kotlin for Forge (auto-installed Wave 0 dep).

Mekanism TFMG Compatibility: Resolves 85+ recipe overlaps between TFMG and Mekanism. Steel, lead, sulfur unified under TFMG. Mekanism's steel recipe disabled — use TFMG Blast Furnace instead. Config in [config/mekanismtfmgcompat-common.toml](#).

Create: Gunsmithing

Steampunk firearms crafted using Create machines. Animated guns with attachments and ammo types.

Getting started:

1. Build a **Gunsmithing Table** (check EMI — uses Create brass + wood)
2. Craft **gun frames** — pistol, rifle, shotgun, each with different stats
3. Craft **barrels, grips, magazines** at the mechanical press
4. Assemble at the Gunsmithing Table
5. Craft **ammo** — each gun type uses different ammo (pistol rounds, rifle cartridges, shotgun shells)

Attachments: Scopes, extended magazines, suppressors, bayonets. Attach at the Gunsmithing Table.

Compatibility: Built-in Better Combat support — guns use appropriate animations.

Dependencies: NTGL (gun framework), GeckoLib (3D animation), Framework (utility library) — all auto-installed.

Create Ornithopter Glider

Mechanical flapping glider powered by Create rotational power.

How: Craft the glider (check EMI — brass + leather + feathers). Equip in chest slot. Hold jump to flap (consumes SU from a connected power source or internal buffer). Boost-assisted flight for short bursts.

Use: Early flight before MekaSuit. Pairs with Create windmills or water wheels for power.

Create Train Ecosystem

Six mods work together for a complete train network:

Mod	Role
Steam 'n' Rails	Track types, semaphores, conductor, coupling
Train Utilities	Station architecture — sliding doors, platform blocks
Railways Navigator	Display boards, route management, schedule entries
Interiors	Train car furniture — chairs, tray tables, kelp seats
Train Parts	Crossing gates, sliding windows, train steps, slides
Threaded Trains	Performance — train pathfinding on separate thread

Together: tracks → stations → navigation → interiors → detailing → performance, with no overlap.

Mekanism Addons

Advanced Finders

Ore detection devices with directional indicators for nearby ore veins.

How: Craft a finder (check EMI — tiers for different ores). Hold it — arrows point toward nearest vein, distance shown on HUD. Synchronized in multiplayer.

Advanced Chimneys

Smoke ventilation systems for factories. Redirect emissions outdoors.

How: Craft chimney blocks, place above furnaces/generators. Smoke renders realistically and vents upward. Customizable appearance.

Dependencies: Both require ForgeEndertech library (auto-installed).

How Wave 1 Mods Work Together

Layer	Mod	What's Happening
Power generation	Create + Mekanism Generators	Water wheels → windmills → steam → heat/gas/solar → fission → fusion
Ore processing	Mekanism	2x→3x→5x ore multiplication
Factory automation	Create	Conveyors, deployers, sequenced crafters, mechanical arms
Storage	AE2	ME drives, storage busses, crafting CPUs
Transport	Create trains + AE2 P2P tunnels	Bulk item movement across bases
Endgame	Mekanism fusion + MekaSuit + AE2 autocrafting	God-tier power and automation
Oil/steel	TFMG	Distillation → diesel engines → blast furnace steel
Power bridge	Crafts & Additions + TFMG Energy Converter	Kinetic ↔ FE ↔ TFMG voltage
Rail expansion	Steam 'n' Rails + train ecosystem	Full train network with stations and interiors
Colony defense	Create Big Cannons (Wave 5)	Artillery for base perimeter

Integration highlights:

- Mekanism universal cables power AE2 controllers via Energy Acceptor
- AE2 Storage Bus on Sophisticated Storage barrel = seamless early-mid storage migration
- Sophisticated Storage/Backpacks Create Integration = full backpack/barrel functionality on moving contraptions
- Applied Mekanistics bridges AE2 autocrafting with Mekanism chemical processing
- TFMG steel replaces Mekanism steel (blast furnace is the intended source)

Session Plan — Wave 1

Session		Focus		What to Build	
Session 1 — Create basics		Mine andesite. Hand crank + millstone. Water wheel. Grind wheat and ore		256 SU power, basic ore doubling	
Session 2 — Mekanism start		Find osmium. Heat Generator + Enrichment Chamber + Metallurgic Infuser. 2x ore processing		Basic Mekanism line, first steel	
Session 3 — AE2 discovery		Find certus quartz meteorite. Mine it. Craft dust → crystal → charge. Build AE2 Charger		First certus, charge infrastructure	
Session 4 — AE2 network		ME Controller + ME Drive + 1k cells + ME Crafting Terminal. Storage Bus on Sophisticated barrel		First digital storage network	
Session 5 — Mid Create		Find zinc. Make brass. Deployer + Sequenced Crafter + Mechanical Arm. Automate a simple recipe		Component automation, brass-tier	
Session 6 — Mekanism expand		Add Crusher for 3x ore. Electrolytic Separator for gas. Upgrade to Advanced tier		3x multiplication, gas infrastructure	
Session 7 — Integration		Import Bus on Enrichment Chamber → AE2. Export Bus to Create machines. Connect alternator to Mek power		Cross-mod automation bridge	
Session 8 — TFMG start		Find oil (desert/plains pools). Pumpjack + Distillation Tower. Small Diesel Engine		Oil refining, massive SU upgrade	

Session		Focus		What to Build	
Session 9 — TFMG expand		Build Steel Blast Furnace. Mass-produce steel. Find bauxite for aluminum		Industrial steel, aluminum	
Session 10+ — Scaling		Digital Miner. Expand AE2 drives. Add Create train between base and outpost. Work toward 5x processing		Automated mining, bulk transport	

Pacing tip: You don't need to finish Wave 1 before starting Wave 1.1 (megabuilding), Wave 1.5 (colony), or Wave 2 (exploration). Dip into exploration between factory sessions. The colony takes real-world days to grow — start it early. TFMG is mid-game — start after you have brass Create and basic Mekanism.

Wave 1.1 — Megabuilding

Schematics and blueprint tools for building at scale in survival. Import community-made structures, place them layer-by-layer with Forgematica's holographic overlay, and construct anything from starter homes to mega-factories without creative mode or guesswork. This wave unlocks the Phase 3 megabuild fantasy early — you can plan and start placing mega-structures from day one, even if the resources take weeks to gather.

Megabuilding Mod

Mod	Role
Forgematica	Client-side schematic mod — import <code>.litematic</code> blueprints, display holographic overlay, place layer-by-layer in survival

Forgematica is the unofficial NeoForge port of Litematica. It lets you load schematics (`.litematic` files) and shows a translucent hologram of the build in-world, complete with material lists per layer. You place blocks manually while the overlay guides every position — no auto-build, fully survival-legal. Essential for mega-scale projects that would be impossible to freehand.

- **Version:** 0.4.3+mc1.21.11 (NeoForge, 1.8 MB)
- **Environment:** Client-side only
- **Dependencies:** None (standalone — no MaLiLib required)
- **License:** LGPLv3
- **Downloads:** 1.6M+ total

Shader compatibility: Forgematica rendering may conflict with shader packs. If the schematic overlay looks broken, toggle shaders off temporarily (`K` by default in Iris). On 1.21.1 (this pack's version), Iris compatibility is better than newer MC versions. Switch back when done placing.

Keybinds (set in Options → Controls → Key Binds after installing):

Action	Suggested Key	Purpose
Open Schematic Manager	<code>M</code>	Load, manage, and switch between schematics
Execute Schematic Placement	<code>Ctrl+Enter</code>	Locks in the current layer after verifying alignment
Toggle Schematic Rendering	<code>Ctrl+M</code>	Show/hide hologram (useful for screenshots)
Toggle Material List	<code>Ctrl+Shift+M</code>	Show/hide required materials HUD

How it works:

1. Download [.litematic](#) schematics and place them in your instance's [schematics/](#) folder.
2. Open Schematic Manager ([M](#)), select a schematic, choose placement position and rotation.
3. The holographic overlay appears at the target location. Use the material list to gather blocks per layer.
4. Place blocks manually following the overlay. Cycle through layers as you build up.
5. When a layer is complete, press [Ctrl+Enter](#) to lock it in and show the next layer.

No blocks are placed automatically — Forgematica is a guide, not a world editor. Every block must be placed in survival. This respects the pack's survival-only constraint while enabling builds at a scale that would normally require creative mode.

Schematic Recommendations

These schematics complement the Driftwood progression. All are free Litematica-format downloads from [Abfielder.com](#) — one of the largest schematic libraries (700+ schematics, all free). Each schematic is picked to serve a specific purpose in your journey from Cozy Roots through Industrial Ascent to Perpetual Ascent.

#	Schematic	ID	Phase	Purpose	Style
1	Starter Home	2818	1	Cozy first shelter — compact, complete, no wasted space	Oak & cobblestone cottage
2	Medieval Starter House	6457	1	Early game base with living space and storage	Timber-framed medieval
3	Cozy Windmill	5959	1	Decorative windmill — pairs with Create water wheels	Sails & stone tower
4	Simple Starter House	13174	1	Minimal starter — place in first 30 minutes	Spruce & stone
5	Windmill Farmhouse (FURNISHED)	10955	1–2	Farmer's Delight hub with enclosed mill, kitchen, and crop plots	Furnished medieval farm

#	Schematic	ID	Phase	Purpose	Style
6	Medieval Windmill	4216	1-2	Standing windmill — landmark and Create rotational power aesthetic	Tall stone & wood tower
7	Warehouse, Auto-sorting	11682	2	Multi-chest sorting hall — pre-AE2 bulk storage	Industrial timber frame
8	Medieval Stone Factory	13065	2	Factory shell for Create sequenced lines and Mekanism machines	Stone & deepslate industrial
9	Steampunk Factory	10175	2	Themed factory — fits Create brass-tier and TFMG oil refining	Brass, copper & brick
10	Industrial Factory <i>(Blackstone/ Nether Brick)</i>	2899	2-3	Late-game factory enclosure — Nether-themed Mekanism wing	Blackstone & nether brick
11	Large Underground Base <i>(Full Interior + Netherhub)</i>	12756	2-3	Massive underground complex with portal hub — ultimate tech bunker	Subterranean sprawl
12	Dark Fantasy Castle	7162	3	Endgame colony fortress — 62K+ DLs, fully detailed exterior	Dark stone & spires
13	Skyreach Castle	12188	3	Skyborne castle — Phase 3 “I’ve earned this” megabuild	Floating fantasy fortress
14	Wizard Watchtower	12509	1-3	RPG-themed tower — fits	Mage spire

#	Schematic	ID	Phase	Purpose	Style
				Apotheosis enchanting room	
15	Stone Bridge	10587	1–3	Decorated bridge — connect base sections or span colony rivers	Arched stone bridge
16	Wall + Tower	8559	2–3	Colony perimeter wall segment with integrated guard tower	Defensive medieval wall
17	Grand Dwarven Gate	11546	3	Epic mountain-face entrance — mega-scale statement piece	Dwarven stonework
18	Minecraft Wooden Mansion	12525	2	Large furnished mansion — Phase 2 upgrade from starter home	Spruce & dark oak estate

Download: Visit <https://abfielder.com/Products/ProductDetails.php?id=<ID>> for each schematic. Click “Download” for the `.litematic` file. Place all `.litematic` files in your instance’s `schematics/` folder.

Finding more: Browse <https://abfielder.com/Products/BrowseProducts.php?sort=trending&game=minecraft&type=Schematic> — filter by tag (House, Castle, Factory, Bridge, Mega Build, Farm, Medieval, Industrial) for additional options. Planet Minecraft (planetminecraft.com/projects/tag/litematica/) and MineMev (minemev.com/explore/) have thousands more schematics if you want alternatives. Search for tags: `medieval`, `survival`, `industrial`, `fantasy`, `steampunk`.

Style guide: When browsing additional schematics, stick to materials already present in the modpack — stone, deepslate, oak/spruce/dark oak, brick, terracotta, copper, iron, and Create decorative blocks. Avoid schematics that depend on mods not in this pack (looking for Quark, Chipped, or BYG blocks in the material list is a red flag). Medieval, steampunk, and dark fantasy styles all fit the Driftwood aesthetic. Modern, sci-fi, and glass-heavy builds don’t.

Interdependency Notes — Wave 1.1

- **Forgemtica × Create:** Factory schematics are designed as shells — you install Create machinery and Mekanism lines inside them. They provide the structure; you wire the automation.

- **Forgemata × MineColonies:** Castle and wall schematics make excellent colony defense perimeters. Wall + Tower (ID:8559) segments can be tiled to enclose an entire colony. Rebuild colony buildings inside them for a unified architectural style.
- **Forgemata × AE2:** The Warehouse Auto-sorting schematic (ID:11682) is designed for pre-AE2 storage. Once you transition to ME drives, repurpose it as bulk item overflow or decorative archive hall.
- **Forgemata × Apotheosis:** The Wizard Watchtower (ID:12509) includes space for an enchanting setup — drop your Apotheosis enchanting table and bookshelves in the top room.
- **Shader caveat:** Forgemata hologram rendering may flicker with complementary shaders. Toggle shaders off (**K**) while placing schematics, then re-enable. The 1.21.1 compatibility is better than newer MC versions — distortion is usually minor.
- **Survival constraint:** Forgemata is purely a visual guide. No blocks are auto-placed. Every resource must be gathered and placed by the player in survival mode. This respects the pack's survival-only constraint.

Install: In XMCL, search [Forgemata](#) in the Mods tab, or download the [.jar](#) from CurseForge and drop into [mods/](#). Create a [schematics/](#) folder in your instance directory, then drop downloaded [.litematic](#) files there. Forgemata will pick them up automatically on next launch.

Session Plan — Wave 1.1

Session	Focus	What to Build
Session 1 — Pick a starter	Download 3 Phase 1 schematics. Place Starter Home (ID:2818).	First schematic build, learn Forgemata flow
Session 2 — Farm hub	Place Windmill Farmhouse (ID:10955). Plant Farmer's Delight crops around it.	Farm-to-table setup next to base
Session 3 — Bridge it	Place Stone Bridge (ID:10587) between base areas or over a river.	Connect your growing compound
Session 4+ — Factory prep	Download Phase 2 schematics. Scout a flat area for the factory district.	Site planning for industrial ascent

Session		Focus		What to Build	
Ongoing		As your tech tree advances, place larger schematics. Wall+Tower around colony. Underground base for AE2. Castle for endgame.		Megabuilding scales with progression	

Tip: Schematics eat materials fast. The Windmill Farmhouse alone needs several stacks of spruce and stone. Set up a Create tree farm and cobblestone generator early — you'll need them. Treat schematic building as a multi-session project, not a speedrun.

Wave 1.5 — Colony Management

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base.

Colony Automation

Mod	Role
MineColonies	Autonomous NPC colony — builders, miners, farmers, crafters, guards, research tree
Structurize	Structure building/placement library (MineColonies dependency)
MultiPiston	Multi-block piston library (MineColonies dependency)
BlockUI	UI library (MineColonies dependency)
Domum Ornamentum	Decorative block library (MineColonies dependency)

Colony Addons

Mod	Role
Byzantine Styles Pack	3 build styles — Byzantine, Shogun, Nile. Visual variety for colony buildings
Stylecolonies	Official MineColonies blueprint addon — 10+ styles (Steampunk, Frontier, High Magic, FairyTale, etc.)
Create: Colony Logistics	Bridges MineColonies warehouses to Create logistics — belts, chutes, trains for colony supply
ColonyLink	Connects MineColonies builders to AE2 — monitor resources, trigger autocrafting from ME terminal

Style packs coexist: Byzantine and Stylecolonies cover completely different themes (Byzantine/Shogun/Nile vs Steampunk/Frontier/FairyTale/etc.) — no overlap. Both can be installed simultaneously. FairyTale style requires Twilight Forest + Create (both in pack) and Quark (not in pack — that style simply won't appear).

Create: Colony Logistics and **ColonyLink** serve complementary roles — Colony Logistics handles Create-based colony restocking (belts, trains), while ColonyLink handles AE2-based resource monitoring and autocrafting. Both integrate MineColonies into the tech loop.

Interdependency notes for Wave 1.5

- MineColonies guards (deployed via the Town Hall's "Guard Tower" building) patrol your base perimeter, reducing the need to secure every wall gap manually. Their effectiveness scales with the Guard Tower level and the colony's

skill research. This creates a natural progression — automated base defense grows as your colony does.

- MineColonies Colony Research milestones signal Phase 2 progression readiness. For example, the “Industrial Revolution” research (Colony level 3, 5000 research points) serves as a soft gate — by the time you reach it, you have the colony infrastructure to support Mechanism and AE2. Hard recipe gating (e.g., Atomic Alloy locked behind research) requires a data pack or scripting mod and is not implemented in this pack — the progression pillars in AGENTS.md define the intent, and playtesting determines if soft gating is sufficient.
- Chunk Loaders (Wave 0) are essential for MineColonies — they keep workers active while you explore other dimensions. Place at least one at your colony after setting up your base.
- Create: Colony Logistics lets Create’s logistic network (belts, chutes, depots, trains) interact directly with MineColonies warehouses. Automate material delivery to builder huts without manual restocking.
- ColonyLink lets you monitor colony resource needs from your AE2 terminal. When a builder runs out of materials, ColonyLink triggers AE2 autocrafting — the loop closes: AE2 crafts → Create delivers → MineColonies builds. This is the Phase 2+ integration sweet spot.
- Byzantine Styles and Stylecolonies are purely visual — pick a style pack, apply to a hut, and the build theme changes. No gameplay impact. Mix and match across different huts in the same colony.

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base. Grows alongside your factory.

Getting Started

1. Craft the **Town Hall** and place it on flat ground
2. Right-click → colony GUI opens
3. Hire a **Builder** from the Citizens tab
4. Place a **Builder’s Hut** near the Town Hall
5. Put wood, cobblestone, and tools in the builder’s chest
6. The builder starts constructing from blueprints

Core Worker Huts (build in this order)

Hut	Resource	What it does
Mine	Coal, iron, stone	Digs shaft mine. Passive ore supply
Lumberjack	Wood	Chops trees, replants. Infinite wood
Farmer	Wheat, carrots, potatoes	Plants, grows, harvests. Feeds colony
Guard Tower	Defense	Patrols base perimeter. Scales with colony size

Hut	Resource	What it does
Crafter	Recipes	Crafts items from storage materials

Research Tree

Research	Colony Level	Points	Unlocks
Basic Industry	2	2000	Better mines, furnace automation
Advanced Smithing	3	4000	Better worker tools
Industrial Revolution	3	5000	Atomic Alloy recipe (Mekanism gate)
Logistics	4	8000	Warehouse, courier routes

Tech gating: Atomic Alloy (Mekanism ultimate-tier) requires “Industrial Revolution” research. This prevents skipping from Phase 1 straight to Phase 3.

Guard Scaling

- 1 Guard Tower = 20-block radius coverage
- Each additional tower adds overlapping coverage
- Guard equipment upgrades with colony level (leather → iron → diamond)

Integration With Your Tech Base

- **AE2 Storage Bus** on colony warehouse → all colony resources in your ME terminal
- **ColonyLink** on AE2 network → monitor builder resource needs, trigger autocrafting when supplies run low
- **Create belts/trains via Colony Logistics** → automated material delivery to builder huts and worker chests
- **Create: Colony Logistics** on colony warehouse → colony resources flow through Create’s logistic network (belts, depots, trains, stockpile switches)
- **Farmer’s Delight meals** supply colony kitchen for worker happiness bonuses
- **Chunk Loaders** keep colony running when you’re in other dimensions

Session Plan — Wave 1.5

Session	Focus	What to Do
Session 1 — Foundation	Town Hall, Builder’s Hut, supply materials	Colony established, first build started
Session 2 — First workers	Mine hut + Lumberjack hut	Passive coal/iron/wood income

Session	Focus	What to Do
Session 3 — Farm + Guard	Farmer hut + Guard Tower	Colony self-sufficient, base defended
Session 4 — Research	Basic Industry research, Crafter hut	First tech unlocked
Ongoing	Check in every few sessions. Upgrade buildings, expand guards, unlock research	Atomic Alloy ready by Phase 2

Tip: The colony is a long-term background investment. Start it early, check periodically, let it grow.

Wave 2 — Exploration (World & Dimensions)

Overworld Worldgen

Mod	Role
Terralith	Overhauled Overworld biomes
Tectonic	Better terrain shape — deeper valleys, taller mountains, cliffs

Environment

Mod	Role
Serene Seasons	Seasonal foliage colors, temperature shifts, crop growth cycles

Serene Seasons adds four seasons (Spring, Summer, Autumn, Winter) with visual foliage changes, temperature shifts, and seasonal crop growth modifiers. Crop impact is enabled — planting in the wrong season reduces yield, encouraging greenhouse planning. Temperature damage is **disabled** — no freezing or heat damage, preserving the chill rule.

Underground & Ocean

Mod	Role
TerraBlender (NeoForge) — REQUIRED	Worldgen biome library — required by Darker Depths 2.1+ for biome placement
Darker Depths	Underground expansion — 3 new cave biomes, mobs, Forsaken Bronze, Living Crystals
Upgrade Aquatic	Ocean expansion — Thrasher mob, Ocean Ravines, Prismarine Coral, renewable sand
Hybrid Aquatic	Marine creatures — sharks, crustaceans, otters, nautilus, diving armor, corals

Interdependency note — Hybrid Aquatic + Upgrade Aquatic: Both expand oceans but cover different niches. Upgrade Aquatic adds mob mechanics (Thrasher) and ocean-floor features. Hybrid Aquatic adds creature variety (sharks, otters, crustaceans) and diving gear. They layer without conflict — more ocean life, more reasons to dive.

Interdependency note — Hybrid Aquatic + Starcatcher (Wave 4): Starcatcher has built-in compatibility with Hybrid Aquatic. Hybrid Aquatic fish are automatically added as catchable entries with the corresponding entity and bucketed fish versions. No config needed.

Interdependency note — Hybrid Aquatic + Better Fishtanks (Wave 4): Better Fishtanks can house Hybrid Aquatic creatures in its aquariums. Use Fish Bags to catch Hybrid Aquatic mobs and place them in tanks.

Interdependency note — Darker Depths + TerraBlender: Darker Depths 2.1+ requires a worldgen API to register its cave biomes. Either **TerraBlender** (NeoForge) or **Biolith** must be installed. TerraBlender NeoForge is the recommended choice (available for 1.21.1 on both CurseForge and Modrinth). Without it, Darker Depths crashes on startup with [Missing Worldgen API!](#) [Install TerraBlender](#) or [Biolith](#).

Interdependency note — Darker Depths + YUNG's Better Caves: Darker Depths adds its own cave biomes (Molten Cavern, Sandy Catacombs, Crystal Cavern) that generate alongside YUNG's Better Caves' enlarged cave systems. Darker Depths biomes replace vanilla cave sections with themed variants, while YUNG's Better Caves changes the shape and connectivity of caves — they layer without direct conflict.

Interdependency note — Upgrade Aquatic + YUNG's Better Ocean Monuments: Upgrade Aquatic adds ocean mobs, coral variants, and ocean-floor features. It does not modify Ocean Monuments — YUNG's Better Ocean Monuments replaces monument structure generation without conflict. Upgrade Aquatic items (thrasher teeth, prismarine coral) integrate naturally with Ocean's Delight (Wave 4) for seafood recipes.

Structures

Mod	Role
YUNG's Better Dungeons	Overhauled dungeons
YUNG's Better Mineshafts	Improved mineshafts
YUNG's Better Strongholds	Larger, more complex strongholds
YUNG's Better Desert Temples	Multi-room desert temples
YUNG's Better Jungle Temples	Jungle temples with traps
YUNG's Better Ocean Monuments	Redesigned ocean monuments
YUNG's Better Witch Huts	Expanded witch huts
YUNG's Better Nether Fortresses	Improved nether fortresses
YUNG's Better Caves	Larger, more varied caves
Structory	New hand-crafted structures
Towns & Towers	Better village generation

Dimensions

Mod	Role
The Aether	Sky dimension — safe islands, gear progression, dungeons with bosses
Deep Aether	Aether addon — 5 new biomes, Skyjade/Stratus gear, new mobs, poison swamps. Gives the Aether the “Nether Update” treatment
Umbral Skies	Aether × Twilight Forest fusion — Aether-themed structures in TF, Aether boss trophies, TF-material gloves
Deeper and Darker	The “Otherside” dimension — deep dark themed, warden gear
Twilight Forest	Classic dimension — 6+ bosses with progression gates, unique loot, enchanted forest biomes

End Access & Structures

Mod	Role
End Remastered v6.3.0	Portal progression overhaul — 16 custom Ender Eyes hidden across vanilla structures and bosses. Find 12+ to open the End portal
Moog's End Structures v2.0.3	New structures to fill the End dimension
Nullscape (data pack)	End terrain overhaul — surreal alien landscapes, new sky colors

End Remastered replaces vanilla Ender Eyes with 16 custom eyes scattered across vanilla structures and bosses. The Cryptic Eye comes from enchanting, the Evil Eye from master clerics, the Wither Eye from the Wither — each eye has a unique source. Once you have 12+, follow them to a stronghold and insert all 12 to open the portal. This turns reaching the End from a grind (kill Endermen for pearls) into an exploration-driven scavenger hunt.

Nullscape: A data pack that transforms the End’s terrain into alien shapes (floating islands, spires, arches, voids) with unique sky colors per biome. Nullscape handles terrain shape and sky only — it does not add biomes. Install as a data pack in the world’s [datapacks/](#) folder.

Navigation

Mod	Role
Explorer's Compass	Craftable structure/biome finder

Mobility

Mod	Role
Immersive Aircraft 1.1.5	Flyable aircraft — biplane, airship, cargo airship, warship, gyrocopter. Upgradable, colorable
Small Ships 2.0.0-b2.1	Sailable ships — brigantine, cog, drakkar, galley. Build, crew, and navigate; moveable bases on water. Beta (NeoForge)
Hang Glider v21.1.0	Craftable hang glider — glide from any height. No fuel, no power. Phase 1 accessible exploration aid
Man of Many Planes 0.2.1	Immersive Aircraft addon — two Create-themed aircraft. Requires Immersive Aircraft

Field Guide

Opens a new way to experience the world. Craft a spyglass, then look at any plant, animal, or monster to catalog it in your Field Guide. Each entry you discover adds to your collection with lore about the creature or plant. Works with modded mobs and biomes — automatically discovers content from every mod in the pack.

Recommended companion mods:

Mod	Feature Added
Immersive Overlays	Biome displays — shows spawn details for entities
Item Descriptions	Entry descriptions — auto-fills entry lore
Mod Descriptions (resource pack)	Modded support — adds descriptions for modded entries

Configuration — Wave 2

Aether (`aether-common.toml`):

- `"Use default Curios' menu" = true` — Uses Curios API menu instead of Aether's separate accessories screen.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Grants Book of Lore and Golden Parachutes on first entry.
- `"Show Patreon message" = false` — Removes Patreon nag screen.
- `"Add Temporary Freezing automatically" = false` — Disables Aether's temporary freezing datapack.
- `"Add Ruined Portals automatically" = false` — Disables ruined portal datapack.

Interdependency note — Deep Aether + Aether: Deep Aether is an addon that expands the Aether dimension with 5 new biomes, Skyjade/Stratus gear, and new mobs. It requires The Aether and Aeroblender (worldgen biome API) — Aeroblender comes **pre-bundled** with Deep Aether, no separate install needed. Aeroblender itself depends on TerraBlender (already in Wave 0.5 dependencies). Stratus armor grants slow-fall and dash abilities, extending the Aether’s power fantasy progression. All content is within the Aether dimension — no new dimension added.

Interdependency note — Umbral Skies + Aether + Twilight Forest: Umbral Skies connects Aether and Twilight Forest. Aether-styled Druid Huts and Wells generate in Twilight Forest. Aether boss trophies appear in TF’s trophy display style. Gloves can be crafted from TF materials (ironwood, fiery, knightmetal, phantom). No config needed — install and both dimensions gain crossover content.

Interdependency note — Immersive Aircraft + Aether: Aircraft can fly in the Aether dimension. An airship is a natural way to navigate Aether islands — it pairs well with the dimension’s sky-island geography.

Interdependency note — Man of Many Planes + Immersive Aircraft: Man of Many Planes is a direct addon for Immersive Aircraft that adds two Create-themed aircraft. It requires Immersive Aircraft to function. The addon planes share the same fuel, parking, and dimension traversal behavior as base Immersive Aircraft vehicles.

Interdependency note — Hang Glider + Create Ornithopter Glider: The Hang Glider provides simpler, earlier flight than the Create Ornithopter Glider (Wave 1). The Hang Glider requires only wood and leather (Phase 1 accessible, no power), while the Ornithopter Glider needs Create rotational power (Phase 1.5+). Both coexist as a natural progression: Hang Glider for early-Phase-1 exploration, Ornithopter Glider once Create infrastructure is online, Immersive Aircraft for powered mid-game flight.

Interdependency note — Twilight Forest + Apotheosis: Apotheosis boss affixes (Wave 3) apply to Twilight Forest bosses. A Naga with the “Tough” affix is significantly harder. This is intended — it keeps dimension bosses relevant even with late-game gear. If a boss feels too strong, the player can return after better equipment or use Apotheosis gems/runes for the edge.

Interdependency note — Towns & Towers + Cristel Lib: Towns & Towers requires **Cristel Lib** (included in Wave 2 Dependencies above). Without it the game will crash at startup. Cristel Lib provides structure config and runtime datapack support that Towns & Towers relies on for its biome-specific village variants.

The world is dramatically transformed. Terralith overhauls biomes, Tectonic reshapes terrain, YUNG's expands every structure, and three new dimensions open up.

Overworld

Terrain — Terralith + Tectonic

Aspect	What changed	What to look for
Mountains	Taller, steeper. Painted mountains (orange/white layers), volcanic peaks	Build in mountain valleys
Oceans	Deeper, trench-like drops offshore	Use Small Ships (below) to cross
Biomes	80+ overhauled — redwood forests (200+ block trees), cherry groves, floating islands	Cherry groves for vibes, redwoods for wood
Caves	YUNG's Better Caves — wider tunnels, deeper chasms	Easier to navigate, more impressive
Rivers	Deeper valleys from Tectonic, shattered savannahs	Follow rivers for scenic travel

Pro tip: These worldgen mods work best in newly generated chunks. Every direction has something different.

Environment — Serene Seasons

Four seasons with visual foliage changes and crop growth modifiers:

- **Spring/Summer/Autumn/Winter** — foliage colors shift realistically
- **Crop impact ON** — planting in wrong season reduces yield. Plan greenhouses for off-season crops
- **Temperature damage OFF** — no freezing or heat damage (chill rule preserved)

Integration: Farmer's Delight crops are affected by seasons. Build glass greenhouses or use Season Sensor blocks to track optimal planting times.

YUNG's Overhauled Structures

Every vanilla structure is expanded:

Structure	What's new	Loot
Better Dungeons	Multi-room with spawners, traps, treasure	Early iron, gold, enchanted books

Structure	What's new	Loot
Better Mineshafts	Multi-level with support beams, loot caches	Rails, minecarts, ore samples
Better Strongholds	Deeper portal room, more libraries/prisons	Enchanted books, mid-game exploration
Better Desert Temples	Multi-room underground, traps, vault	High-value vault loot
Better Jungle Temples	Vertical temple, redstone puzzles	Puzzle-solving loot
Better Ocean Monuments	New rooms, more elder guardians	Better sponge room loot
Better Witch Huts	Expanded hut with potion lab	Brewing supplies, potions
Better Nether Fortresses	Districts (blaze, wither skeleton, strider)	Nether wart, blaze rods, skulls

Structory + Towns & Towers

- **Structory** — dozens of hand-crafted structures: camps, shrines, castles, towers, walled towns
- **Towns & Towers** — villages generate with unique architecture per biome style. Each biome gets its own building designs

What to do: Explore! Mark interesting structures on your JourneyMap (fullscreen map → right-click to set waypoint). Return with an Explorer's Compass.

Explorer's Compass — Navigation Tool

1. Craft: 1 compass + 4 iron ingots in a + pattern
2. Right-click → GUI opens
3. Search tab: type biome or structure name (partial matching — “cherry” finds all cherry biomes)
4. Select an entry → compass points to nearest instance, distance shown
5. Follow the arrow in real-time
6. When found, add a JourneyMap waypoint

Configuration (<config/explorerscompass-common.toml> → [General] section):

Setting	Default	Recommended	Why
<code>maxRadius</code>	10000	5000	Halve search radius — you must explore ~5K blocks before the compass finds a structure. Prevents instant discovery from spawn

Setting		Default	Recommended	Why
<code>maxSamples</code>		100000	50000	Fewer search samples = lower search accuracy at range. Combined with reduced radius, structures won't be found until you're meaningfully close
<code>maxNextSearches</code>		100	10	Limit "find next" to 10 — once you've cleared the 10 nearest instances, explore further and search again
<code>displayCoordinates</code>		true	true	Show X/Z coords and distance on HUD — essential QoL
<code>defaultXpLevels</code>		0	0	No XP cost for searching — chill pack, navigation shouldn't gate

Effect: With `maxRadius` = 5000, the compass scans a ~10K×10K area centered on you instead of 20K×20K. You'll need to explore new regions to find new structures — no more "find the nearest End City from spawn."

Underground & Ocean

Darker Depths — Expanded Underground

3 unique cave biomes deep underground:

Biome	Depth	What to expect	Key resources
Molten Cavern	Deep (Y < 0)	Lava pools, geysers, amber clusters	Living Crystals (feed diamond to grow), Crystal Melons
Sandy Catacombs	Mid (Y 0-20)	Petrified roots, aridrock, Void Soul mobs	Void Souls (bottle them), Forsaken Bronze
Crystal Cavern	Any depth	Growing crystals, glowing ambience	Crystal materials for decorative/utility

Living Crystals: Found dead in Molten Cavern. Feed a diamond → reawakens, grows, spreads to nearby diamond ore. Crystal Melons (off-hand) = unbreakable tools with +20% speed/damage for 5 minutes.

Void Souls: Kill a Void Soul mob → becomes invulnerable orb (60 seconds). Capture in glass bottle. Use amber to prevent expiration. Forsaken Bronze from void-touched materials has death-related abilities.

Compatibility: Darker Depths biomes generate inside YUNG's Better Caves systems — they layer without conflict.

Required dependency: TerraBlender NeoForge (auto-installed). Darker Depths 2.1+ crashes without it.

Upgrade Aquatic — Richer Oceans

Feature	What it does
Thrasher	Fantasy shark-like mob. Uses sonar, thrashes prey. Drops thrasher teeth for trident crafting
Ocean Ravines	Underwater ravines with Prismarine Coral — harvest for decorative blocks
Nautilus	Passive mob. Drops shells for conduit crafting
Renewable sand	Bubble columns below sandstone convert stone to sand

Integration: Thrasher teeth and prismarine coral integrate with Ocean's Delight (Wave 4) for seafood recipes. Compatible with YUNG's Better Ocean Monuments.

Hybrid Aquatic — Marine Creatures

Expands underwater life with realistic marine creatures and diving equipment.

Creatures added:

Category	Examples
Fish	Reef fish, clownfish, groupers, puffers
Sharks	Hammerhead, great white — neutral until provoked
Crustaceans	Crabs, lobsters, shrimp — drop food ingredients
Critters	Otters (tameable!), nautilus, seahorses
Mini-boss	Deep sea guardian — found in ocean caves

Blocks & items:

Item	Use
Diving Armor	Full set — extended underwater breathing, faster swim speed
Sea Sponges	Absorb water, decorative blocks
Fishing Hooks	Specialty hooks for better catches
Crates	Storage blocks with ocean aesthetic
Anemones	Decorative coral-like blocks
New Corals	Additional coral variants for ocean decoration

Getting started:

1. Explore oceans — creatures spawn naturally in rivers, swamps, and oceans
2. Kill crustaceans → food ingredients for Farmer's Delight recipes
3. Find deep sea guardian in ocean caves → challenging mini-boss fight
4. Collect coral and anemones → decorate your underwater base
5. Craft diving armor → extended ocean exploration becomes viable

Integration with Starcatcher (Wave 4): Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

Integration with Better Fishtanks (Wave 4): Use Fish Bags to catch Hybrid Aquatic creatures and place them in aquariums. Otters, nautiluses, and reef fish make excellent tank inhabitants.

Integration with Ocean's Delight (Wave 4): Hybrid Aquatic crustacean drops (crab meat, lobster tail) integrate as seafood ingredients.

Mobility

Immersive Aircraft

Vanilla-style flyable aircraft. Mid-game (requires iron + basic Create/Mek materials).

Vehicle	Fuel	Seats	Cargo	Best for
Biplane	Coal/charcoal	1	None	Fast personal flight, combat
Airship	Coal/charcoal	3	9 slots	Exploration with friends
Cargo Airship	Coal/charcoal	1	27 slots	Material transport
Warship	Coal/charcoal	3	9 slots	Armed combat, defense
Gyrocopter	Coal/charcoal	1	None	Agile short-range flight

How: Craft at table (check EMI — iron + wool/leather), place on ground, right-click to enter. [W](#) to ascend, [S](#) to descend, [A/D](#) to rotate. Any furnace fuel works.

Integration: Aircraft work in the **Aether** — fly between sky islands without bridges. Park indoors or on high platforms (mobs damage parked aircraft).

Small Ships

Buildable, sailable ships. **Phase 1 accessible** — requires wood, wool, iron.

Ship	Crew	Cargo	Best for
Brigantine	1	18 slots	All-purpose sailing

Ship		Crew	Cargo	Best for	
Cog		2	27 slots	Cargo transport	
Drakkar		2	9 slots	Fast travel, combat	
Galley		1	9 slots	Early personal vessel	

How: Craft hull (wood) + mast (wood + wool) + helm. Right-click water to place. Right-click helm to board. [W/S](#) to move, [A/D](#) to steer. Speed depends on wind direction. Dismount to park — ships float where left. Right-click with lead to tow.

Note: Beta for NeoForge (2.0.0-b2.1). Core features work; minor bugs expected.

Hang Glider

Simple fabric-and-wood glider. **Phase 1 accessible** — requires only wood and leather, no fuel or power.

- Craft the glider (check EMI — wood + leather + string)
- Equip in your hand and jump off any height to glide
- Hold forward to maintain speed, steer with mouse
- Dismount by pressing sneak — you'll drop safely to the ground
- Requires **Puzzles Lib** (already in pack from Wave 5 / Easy Anvils)

Progression: The Hang Glider is the earliest flight option. It provides gentle gliding for early exploration before the player has Create power for the Ornithopter Glider (Wave 1) or fuel infrastructure for Immersive Aircraft. It cannot gain altitude — it's pure glide, not powered flight. Ideal for scouting Terralith biomes from mountain peaks.

Config: No changes needed — default behavior is ideal. Gliding is inherently chill (no fuel, no maintenance, no risk of crashing unless you fly into terrain).

Man of Many Planes — Immersive Aircraft Addon

Adds two Create-themed aircraft to Immersive Aircraft. **Phase 2 accessible** — requires Immersive Aircraft (iron + wool/leather) plus Create materials (andesite alloy, brass).

- **Two new aircraft** — brass-and-mechanical themed designs
- Uses the same fuel, parking, and controls as base Immersive Aircraft
- Works in the Aether dimension like all Immersive Aircraft
- Craft recipes available in EMI

Integration: These aircraft offer a visual and thematic bridge between Create (Wave 1) and Immersive Aircraft (Wave 2). A player deep in Create will appreciate aircraft that match their factory's aesthetic.

Dimensions

The Aether — Sky Dimension

Safe, beautiful sky islands. Gentle adventure curve.

Portal: 4×5 glowstone frame (like Nether portal) → fill with water bucket → step through.

Progression:

- **Phase 1:** Spawn on sky island. Collect **ambrosium** (Aether coal) and **zanite** (Aether iron — mining level increases as durability drops)
- **Phase 2:** Bronze dungeon → defeat **Slider** (hit when blue, avoid when red) → Golden Amber
- **Phase 3:** Silver dungeon → defeat **Valkyrie Queen** → Valkyrie tools (flight-enabled weapons), **Gravitite** (floats upward when right-clicked)
- **Phase 4:** Highest crystal islands → **Sun Spirit** — bring fire resistance

Tip: Not required for any tech progression. Visit when you want a change of pace. MekaSuit trivializes all Aether combat.

Deep Aether expansion: Installed alongside — it adds 5 new biomes (Aerglow Forest, Aerlavender Fields, Yagroot Swamp, Golden Heights) that generate organically in the Aether. **Skyjade** tools/armor (stronger at full durability) appear in holystone. **Stratus** armor (slow-fall + dash — press R with full set) is crafted from Sterling Aerclouds found above Y=200. The Yagroot Swamp biome introduces poison liquid, Virulent Quicksand, and the Spooky Ring (night damage bonus). All content is purely additive — the base Aether progression is unchanged.

Config (`aether-common.toml`):

- `"Use default Curios' menu" = true` — uses Curios API menu
- `"Gives player Aether Portal Frame item" = false` — build portal manually
- `"Gives starting loot on entry" = true` — Book of Lore + Golden Parachutes on first entry
- `"Show Patreon message" = false`
- `"Add Temporary Freezing automatically" = false`
- `"Add Ruined Portals automatically" = false`

Twilight Forest — Progression Dimension

Dark, enchanted dimension with strict boss progression. Each boss kill unlocks the next area.

Portal: Find a 2×2 water pool → surround with 12 flowers (any type) → throw a **diamond** into the pool → lightning strikes → step into the water.

Note: One-way down. To return, craft a **Twilight Portal** block or use a Waystone (place one immediately on arrival).

Boss progression:

Order	Boss	Location	Strategy	Unlock
1	Naga	Naga Courtyard	Strafe charges, hit tail segment	Lich Tower
2	Lich	Lich Tower	Dodge projectiles, hit during cooldown. Shield reflects projectiles	Twilight Swamp
3	Minoshroom	Labyrinth	Find minotaur room, kill it, grab Meef Stroganoff	Firefly Forest
4	Hydra	Hydra Lair	Dodge fireballs, melee heads. Drops Fiery Tears	Ur-Ghast fight
5	Ur-Ghast	Dark Tower	Shoot tentacles, then main body. Dark Forest has blindness — bring milk	Snowy Forest + Highlands
6	Snow Queen	Aurora Palace	Push into pillars to break shield. Avoid ice arrows	Final Castle
7	Final Castle	Highlands	Multi-boss fight. Needs weapons from all prior bosses	Giant's Pick (5×3 mining area)

Key items:

- **Ironwood** gear — smelt ironwood ingots from iron + magic beans. Good starter armor
- **Fiery** gear — iron + blaze powder. Mid-tier weapons
- **Knightmetal** — from Lich Tower
- **Phantom armor** — creative flight for short bursts. From Ur-Ghast tentacles
- **Giant's Pick** — 5×3×5 mining area. From Final Castle

Tips:

- Always twilight — no day/night cycle. Mobs spawn any time

- Place **Waystones** frequently
- **Twilight Forest Maze Map** (paper + torch) reveals maze/castle layouts
- Bring **milk buckets** for Dark Forest blindness
- Your AE2 wireless terminal, Mekanism gear, and Create tools all work here

Integration: Apotheosis boss affixes apply to Twilight Forest bosses. A Naga with “Tough” affix is significantly harder — intended to keep dimension bosses relevant even with late-game gear.

Deeper and Darker — The Otherside

Deep-dark-themed dimension accessed through the Deep Dark.

How to enter:

1. Find a **Deep Dark** biome (Y=-40 or below)
2. Locate an **Ancient City** (Explorer’s Compass → search “ancient city”)
3. Kill the **Warden** — drops a **Warden Heart**
4. Craft: Warden Heart + 8 sculk → **Otherside Portal Catalyst**
5. Build a 4×5 frame of **reinforced deepslate** → right-click catalyst on frame → step through

Warden strategy: Detects by sound. Use wool to silence footsteps. Sneak near sculk shriekers. Throw snowballs to distract. 500 HP, slow attacks — hit and run.

What to expect:

- 4 new biomes — sculk forests, frozen deep, expanded ancient cities
- 8 new mobs — sculk variants, stalker (ranged), shriek worm (ambush), **Sculk Leech** (heals on damage)
- **Miniboss:** Shattered Warden — drops **Sculk Transmitter** components
- **Key loot:** **Sculk Transmitter** — Elytra upgrade for mid-flight teleportation. Combine at smithing table

Gear: Warden-themed armor/tools ≈ diamond tier with unique abilities (sound-based attacks, silence fields).

Difficulty: Harder than Aether, easier than Twilight Forest’s Dark Forest. Come with iron-tier armor + decent weapon.

End Access — End Remastered

The End portal now requires a scavenger hunt instead of a grind.

Old method: Kill Endermen → craft Ender Eyes → find stronghold → insert 12 eyes.

New method: Find 16 unique custom Ender Eyes scattered across the world:

Eye	Source	Eye	Source
Old Eye	Desert Pyramids	Cursed Eye	Bastion treasure
Nether Eye	Nether Fortresses	Exotic Eye	Craft (aquatic items)
Cold Eye	Igloos	Evil Eye	Master Cleric trade
Rogue Eye	Jungle Pyramids	Undead Eye	Craft (Undead Soul from skeleton horse)
Black Eye	Buried Chests	Cryptic Eye	Random chance when enchanting
Magical Eye	Evoker drop	Wither Eye	Wither drop
Lost Eye	Mineshafts	Guardian Eye	Elder Guardian drop
Corrupted Eye	Pillager Outposts	Witch Eye	Craft (Witch Pupil)

How it works: Find 12+ unique eyes. Hold one to locate the stronghold (like vanilla Ender Eyes). Insert all 12 into the portal frame to open it. If a vanilla eye is already in a frame, right-click with a custom eye to replace it.

Tip: This naturally paces your journey to the End — you'll need to explore deserts, swamps, nether fortresses, ocean monuments, woodland mansions, and bastions before you can fight the dragon. By the time you open the portal, you'll have gear, experience, and world knowledge that makes the End a natural next step rather than a rushed target.

End Dimension

Moog's End Structures adds towers, ruins, and dungeons with loot better than vanilla End cities. Nullscape (data pack) overhauls terrain into surreal alien shapes — floating islands, spires, arches, void pits, unique sky colors per biome. Install in [datapacks/](#) folder.

Ender's Delight (Wave 4) adds food cooked from End ingredients — useful status effects for End exploration.

Progression:

- Post-Dragon: Explore outer islands for new structures and loot
- Mid-End: Tackle Moog's structures for unique weapons
- Endgame: Build an End base — Nullscape terrain makes for dramatic architecture

Tip: Place a Waystone immediately. Build a small AE2 subnet with wireless terminal to access main base storage.

Session Plan — Wave 2

Session	Focus	What to Do
Session 1 — Scout	Craft Explorer's Compass. Search for cherry grove or redwood forest. Mark waypoints on JourneyMap. Loot a YUNG's dungeon. Craft a Hang Glider for early aerial scouting	First taste of transformed world
Session 2 — Aether	Gather glowstone. Build Aether portal. Explore first sky island, collect ambrosium and zanite	Safe new dimension
Session 3 — Twilight Forest	Find 2×2 pool, surround with flowers, throw diamond. Enter, place Waystone. Defeat the Naga	First Twilight boss
Session 4+ — Deep exploration	Work through Twilight Forest bosses. Visit Deeper and Darker (find Ancient City). Explore End post-Dragon	Boss loot, dimension resources
Session 5 — Aircraft	Craft a biplane or airship (iron + wool). Explore Terralith biomes from the air. Craft Man of Many Planes aircraft for Create-themed flight	Fast travel, scenic views

Pacing tip: Explore between factory builds. Grab your compass and walk in a random direction when waiting for Mekanism to process or Create to assemble.

Wave 2.5 — Building Your Home

You’ve explored the world, found stunning biomes, and gathered materials. Now it’s time to stop surviving and start **living**. This section covers how to build a home you’re proud of — no creative mode required, no artistic talent assumed. Just a few principles that separate “dirt hut” from “place I actually want to come back to.”

Building is a core pillar of this modpack. At any point you can stop optimizing factories and just build something beautiful. This guide works at any stage — but it’s especially satisfying now, when you have enough materials to be selective but haven’t yet automated everything.

Finding Your Build Site

Before placing a single block, find a location that inspires you. You have tools for this:

Tool	How	Why
Explorer’s Compass	Craft: 1 compass + 4 iron (+pattern). Right-click → search “cherry”, “redwood”, “meadow”	Find specific biomes by name
JourneyMap	Press J → explore the fullscreen map	See terrain, biomes, and your marked waypoints
Small Ship	Sail along coastlines	Discover waterfronts you’d miss on foot
Biplane	Fly low over terrain	Spot valley builds, mountain perches, river mouths

What to look for:

- **Waterfront** — river banks, ocean cliffs, lake shores. Water reflects builds beautifully with shaders
- **Mountain valleys** — enclosed, dramatic, defensible. Terralith makes these spectacular
- **Forest clearings** — surrounded by trees feels cozy and sheltered
- **Cliff faces** — cut into the hillside for a build that looks permanent, not placed
- **Cherry groves** — pink petals + mountain backdrop = instant atmosphere

Pro tip: Place a **Waystone** at your chosen site before you start building. You’ll want to teleport back with materials.

The Block Palette Rule

The single biggest upgrade from “meh” to “wow” is using **3–5 complementary blocks** instead of one. Every great build you’ve ever seen uses a palette. Here’s the formula:

Role	Percentage	Purpose
Primary	~50%	Dominant wall material — defines the build
Secondary	~30%	Complementary texture — similar tone, different surface
Accent	~20%	Detail material — corners, trim, highlights, pops of color

Finding Palettes: [BlockPalettes.com](https://blockpalettes.com)

BlockPalettes.com — the single best resource for block combination inspiration. Browse hundreds of community-created palettes, each showing exactly which blocks work together. Filter by style, color, or mood.

How to use it:

1. Open blockpalettes.com/palettes
2. Browse featured palettes or filter by color/theme
3. Click a palette to see all blocks in the combination
4. Test it: place a 5×5 section in a flat area, step back, see if it works
5. If any block fights the others, swap it out

Starter Palettes for This Modpack

These palettes use blocks you can gather in Waves 0–2. Each fits a different biome and vibe:

Medieval Stone — stone brick + cracked stone brick + mossy stone brick + cobblestone (corners/base)

Cozy Cottage — oak planks + spruce planks + cobblestone (chimney/base) + oak leaves (roofline trim)

Desert Sandstone — sandstone + smooth sandstone + cut sandstone + terracotta accents

Nordic Cabin — spruce planks + stripped spruce log + stone + dark oak stairs (roof)

Cherry Zen — cherry planks + cherry log + stone brick + pink petals (garden)

Mountain Fortress — stone brick + deepslate tiles + tuff bricks + copper (accents/patina)

Tropical — jungle planks + bamboo mosaic + mangrove propagules + sand

Endstone Palace — purpur block + purpur pillar + end stone bricks + chorus flower (End builds)

Tip: Open BlockPalettes.com in your browser while you build. When you're stuck on what goes together, browse palettes until something clicks.

Depth and Layering

Flat walls are the second most common beginner mistake (after single-block builds). A flat wall looks like a placeholder. A layered wall looks **built**.

Five Techniques That Transform Flat Walls

Technique	How to do it	Why it works
Inset windows	Push glass back one block into the wall instead of flush with the surface	Creates shadow lines that frame windows
Outcropped corners	Extend corner blocks one block forward from the main wall plane	Breaks up 90-degree edges, adds structural feel
Alternating depth	Every 3–4 blocks along a wall, push one column back or pull one forward	Wall reads as textured rather than uniform
Stair/slab detailing	Upside-down stairs under windowsills, stairs as roofline trim, slabs as half-height ledges	Catches light and shadow in ways full blocks can't
Overhanging roofline	Always extend the roof 1–2 blocks past the wall on all sides	Creates shadow line that defines where building ends and sky begins

Fastest single improvement: Add vertical **log pillars** every 3–5 blocks along exterior walls. Logs imply structural support and instantly break up flat surfaces. Works with every style.

Roof Design

A bad roof ruins a good build. A good roof makes even a simple box look intentional.

Three Roof Styles (Pick One)

Style	Shape	Best for	How
A-frame (pitched)	Triangle cross-section	Cottages, cabins, medieval	Stairs ascending from each wall meeting at a ridge.

Style		Shape		Best for		How	
						Overhang 1 block on all sides	
Flat with parapet		Flat top, raised edges		Modern, desert, fortress		Slabs or full blocks, 1-block wall around perimeter	
Stepped pyramid		Layered squares getting smaller		Asian, temples, fantasy		Each layer 2 blocks smaller than the one below	

Material rule: Roof material should **contrast** with wall material. Dark roof on light walls, or vice versa. Spruce stairs on oak walls. Deepslate on stone brick. Blackstone on sandstone.

Common mistake: Roof too small. The roof should be wider and taller than feels “necessary.” An oversized roof looks dramatic; an undersized one looks like a hat that doesn’t fit.

Interior Design

An empty interior is as bad as a bad exterior. Minecraft has no furniture mod yet (Wave 6 adds Handcrafted), but vanilla blocks create convincing rooms.

Lighting (Critical)

Never place torches on interior walls. They look improvised. Better options:

Light source	Look	How
Lanterns	Warm, architectural	Hang from ceiling via fence post
Glow berries	Organic, trailing	Place on ceiling, let vines trail down
Sea lanterns	Clean, neutral	Modern or sci-fi builds
Jack o’lanterns	Hidden under carpets	Floor lighting without visible sources
Froglights	Subtle, colorful	Pearlescent/verdant/ochre variants from Aether or Overworld

Functional Rooms

Room	Key blocks	Details
Kitchen	Smoker, campfire, barrels, chests	Barrel = cabinets. Chest with food item frame = pantry. Cooking pot over campfire

Room	Key blocks	Details
Bedroom	Bed, crafting table, bookshelf, lantern	Banner on wall above bed as headboard. Carpet at foot of bed
Library	Bookshelves, lectern, quill	Cover walls with bookshelves for instant warmth
Storage	Sophisticated Storage barrels, chests	Organized rows. Sign labels on each barrel
Workshop	Crafting table, furnace, anvil	Industrial feel — stone walls, iron accents

Pro tip: Place **item frames** on walls with maps, banners, or shields inside. They function as paintings with more variety.

Blending Into the Landscape

A build that looks dropped from the sky is unfinished, no matter how detailed the structure itself is.

Key principles:

- **Use biome-appropriate materials** — spruce in cold biomes, sandstone in desert, dark oak near jungle edges. The build should look like it was built from local materials, not imported
- **Never leave flat grass at the base** — add a slight slope toward the entrance, gravel/cobblestone paths leading away, small dirt mounds or rock outcroppings at corners
- **Plant trees near the build** — a cluster of 2–3 trees on one side breaks the silhouette and gives context. Use the biome’s native trees
- **Cut into hillsides** — a build emerging from a hillside looks permanent; one sitting on top looks temporary
- **Paths and surroundings** — gravel paths, stone brick walkways, and lantern posts leading away from the entrance sell the idea that people live here
- **Never run more than 7 blocks in a straight line** when shaping terrain — curves look natural, straight lines look artificial

Modpack-Specific Building Tools

You have tools that vanilla players don’t. Use them:

Tool	What it does for building
Waystones	Place one at your build site. Teleport home for materials, teleport back. No more carrying 3 inventories of blocks across 500 blocks

Tool	What it does for building
Sophisticated Storage	Barrels near your build site hold materials organized by type. Label with signs
Backpack	Carry your palette blocks in your backpack while building. Press B to swap materials on the go
Create Blueprint	Craft Blueprint and Quill. Select your finished build → save as blueprint → use Schematicannon to auto-rebuild it elsewhere with materials from chests
Building Wands (Wave 6)	Place walls, fill areas, build circles/cylinders in bulk. Wait for Wave 6 for this one
JourneyMap waypoints	Mark your build site, mark material locations, mark inspiration you find while exploring
Dynamic Lights (Ctrl+L)	Hold a lantern while building at night. No more stopping to place torches

Quick-Start: Your First Real House

If you've never built anything beyond a dirt hut, follow these steps for your first proper home:

1. **Pick a palette** — Open [BlockPalettes.com](https://blockpalettes.com), find 4 blocks you like. Test in a 5×5 section
2. **Lay the foundation** — Cobblestone or stone brick ring, 7×9 interior minimum (small but not cramped)
3. **Raise the walls** — Use your primary block. Leave gaps for windows (2-block wide minimum)
4. **Add depth** — Push windows back 1 block. Add log pillars at corners and every 4 blocks
5. **Build the roof** — A-frame with stairs. Overhang 1 block. Use a contrasting material
6. **Add the door** — Center of one wall. Stairs as a small porch step in front
7. **Light it up** — Lanterns from ceiling, not torches on walls
8. **Furnish basics** — Bed, crafting table, furnace, 2–4 barrels for storage
9. **Blend the exterior** — Paths leading away, trees nearby, flowers at the entrance, leaves around the roofline
10. **Step back and admire** — You just built something worth coming home to

Community Resources

Resource	URL	What it's for
BlockPalettes	blockpalettes.com	Block color combinations and palette inspiration

Resource	URL	What it's for
Building Guide App	buildingguide.app	Step-by-step tutorials with materials lists (128+ builds)
r/MinecraftBuilds	reddit.com/r/MinecraftBuilds	Community showcase, inspiration, feedback
r/MinecraftHouses	reddit.com/r/MinecraftHouses	House-specific builds and ideas
r/MinecraftTutorials	reddit.com/r/MinecraftTutorials	How-to guides for specific techniques
Minecraft Wiki — Building	minecraft.fandom.com/wiki/Building	Official guides on terraforming, furniture, layout

Session Plan — Wave 2.5

Session	Focus	What to Build
Session 1 — Site selection	Explore with Explorer's Compass. Find a biome you love. Place Waystone. Mark on JourneyMap	Your future home location
Session 2 — Palette + foundation	Browse BlockPalettes.com. Pick 4 blocks. Lay foundation ring. Raise walls to 4 blocks high	Shell of your first real house
Session 3 — Roof + depth	Build A-frame roof. Add log pillars. Inset windows. Add door	Recognizable house shape
Session 4 — Interior	Lanterns from ceiling. Bed, crafting table, furnace. Barrels for storage. Item frames on walls	Livable interior
Session 5 — Landscaping	Paths, trees, flowers, leaf trim around roofline. Step back and see the difference	Home in a world

Tip: Don't aim for perfection on your first build. Aim for "better than a dirt hut." You can always expand, renovate, or rebuild later. The skills transfer — your second build will be twice as good.

Wave 3 — Equipment Magic & RPG

RPG Progression

Mod	Role
Skill Tree (RPG Series)	XP leveling — spend points on attributes and passives
Pufferfish's Skills	Underlying skill system framework

Equipment

Mod	Role
Simply Swords	60+ unique melee weapons with passive/activated abilities
Too Many Bows	30+ unique bows with special abilities and custom attributes
Spartan Weaponry Unofficial	Spears, halberds, pikes, lances, throwing knives — fills the polearm gap
Relics RPG	Legendary accessories with unique abilities
Runes	Socketable runes for weapons/tools
Curios API	Accessory slots (rings, amulets, belts, gloves)

Loot & Affixes

Mod	Role
Apotheosis	Rarity tiers, random affixes, gem socketing, enchanting overhaul. Replaces Savaru's Affixology with more depth

Configuration — Wave 3

Apotheosis — the most config-heavy major mod. Apotheosis is split into modules, each with its own config:

Adventure module — affixes, gems, boss modifiers, rogue spawners:

- Keep all affix types enabled: Ruthless (crit), Arcane (spell), Sturdy (armor), Nimble (speed), Plentiful (mining), Titanic (HP). Each supports a different playstyle.
- Gem tiers enabled: Rough, Cut, Flawless. The gem progression (Rough → Cut → Flawless) provides a satisfying vertical upgrade path. Each tier unlockable through Apotheosis's reforging mechanic.
- Gem types cover all primary stats: melee damage, projectile damage, magic damage, armor, max HP, mining speed, movement speed, luck. This ensures every build archetype benefits from gem-socketed gear.

- Socket count: Gear drops with 1-2 sockets by default. Reforging at the Smithing Table can add a 3rd socket at high rarity (Mythic+). This prevents socket overload while keeping the system meaningful.
- Boss modifiers: all active. Apothic Invaders and Elites spawn naturally with random affixes. They always drop affix items — this is the primary source of mythic gear.
- Rogue Spawners: enabled. Underground spawner+chest combos. Risk vs reward.
- Affix item probability: default rates (~5% base for rare+, scaling with luck and difficulty). Common-enough to feel rewarding, rare-enough to keep each upgrade exciting.

Enchanting module — [config/apotheosis/enchanting.cfg](#):

- Keep all enchantment overhauls enabled: higher level cap (up to level 150+), tiered bookshelves (Hell, Seashelf, Endshelf), Library of Alexandria, new enchantments (Chromatic, Mounting, Stable Footing, etc.).
- The Library of Alexandria stores all enchanted books in one block — essential for enchanting organization at scale.
- Enchantment max levels increased for many vanilla enchantments: Protection can reach level 8, Sharpness level 6, Fortune level 5, Looting level 5, etc. These higher caps match the power fantasy curve — gear enchanted at max level should feel mythic.

Potion module:

- Potion Charms enabled: craft a charm from blaze powder + potions, equip in offhand for passive effect. Reduces inventory clutter from potion bottles.
- Potion augmentation allowed: glowing, brewing, changing base type into splash/lingering at standard ratios. Augmented potions can have higher level caps (e.g., Potion of Strength III).

Spawner module:

- Spawner capture and modification enabled. Silk-touch a spawner with a pickaxe, then modify it (entity type, spawn delay, player range, min/max entities). Key for XP farms and mob-based automation.
- Spawner stats are fully configurable after placement through the spawner's GUI (right-click with an empty hand). Players can fine-tune delay, count, range, and conditions per spawner — no external tool needed.

Village module:

- Wandering trader improvements enabled. Traders can sell Apotheosis items (gems, affix gear, enchanted books) at elevated prices. Adds a secondary acquisition path for players who prefer trading over combat.

Deadly module: Enabled. Apotheosis’s Deadly module adds randomly generated death-traps to the world (sudden spikes, poison darts, instakill rooms, lava traps). This adds risk-reward tension to structure exploration — the deadliest rooms hold the best loot. The Deadly module’s traps are telegraphed and avoidable with caution, and GraveStone Mod keeps your items safe if you do trigger one, so the chill rule is preserved.

Reforging costs — Apotheosis’s reforging mechanic (reroll affixes with gem dust) uses default material costs. As the player progresses, reforging costs scale:

- Common→Uncommon: cheap (a few gem dust)
- Rare→Epic: moderate (dust + some ingots)
- Epic→Mythic: expensive (dust + rare materials like netherite scrap/echo shards)

This cost curve encourages the player to settle for “good enough” gear until they’re ready to invest in perfect rolls.

Interdependency notes for Wave 3

- **Spell Engine** is required by both Relics RPG and Skill Tree (RPG Series) — it was previously considered for removal but must be retained as a mandatory dependency of these core RPG mods.
- **Ranged Weapon API** is shared between Relics RPG and Skill Tree — install once, used by both.

Every weapon rolls with random affixes, every accessory provides a unique ability, and a skill tree gives every action meaning. This wave makes you **feel** your character grow alongside your tech.

Skill Tree (RPG Series) — Character Growth

Every action earns XP. Spend points on permanent stat bonuses.

Getting started:

1. Controls → Skill Tree → set a key ([P](#) recommended)
2. Press key → skill tree GUI opens
3. Spend first points on **Strength** (+melee damage, +mining speed) or **Vitality** (+max HP, +regen)

Attribute	Effect	Best for
Strength	+melee damage, +mining speed	Everyone
Vitality	+max HP, +natural regen	Everyone
Agility	+movement speed, +attack speed	Combat, kiting
Magic	+spell damage, +mana regen	If using spell mods
Defense	+armor toughness, +damage reduction	Tank builds

How XP works: Every ore mined, mob killed, item crafted, crop harvested generates XP. Scales with action complexity.

Pufferfish's Skills: The underlying framework powering Skill Tree. Handles XP tracking and attribute modification behind the scenes. Auto-installed dependency — no separate config needed.

Tip: Set the keybind and start earning XP from minute 1. Lowest-investment, highest-return mod in this wave.

Simply Swords — Unique Weapons

60+ unique weapons with passive and activated abilities. Drop from mobs (~2% chance), chests, and bosses.

Category	Examples	Playstyle
Heavy	Greathammer, Greatsword, Glaive	Slow, high damage, knockback. Two-handed
Medium	Longsword, Katana, Claymore	Balanced speed/damage. Versatile
Light	Rapier, Dagger, Sai, Cutlass	Fast, low damage per hit. Armor penetration on rapiers
Polearms	Spear, Halberd, Pike	Extended reach (2-3 blocks). Throwable spears
Special	Scythe (AoE), Soul Harvester (lifesteal)	Unique mechanics

Weapon abilities (hold [Ctrl](#) while hovering to see):

Ability	Effect	Found on
Whirlwind	Spin attack, hits all nearby	Greathammer, Greatsword
Impale	Charge + release for piercing thrust	Spear, Pike
Backstab	+300% damage from behind	Dagger, Sai
Lifesteal	Heal 10% of damage dealt	Soul Harvester, Scythe
Execute	+50% damage to targets below 30% HP	Axe variants

Finding weapons:

- Mob drops: ~2% chance. Zombies = common, skeletons = ranged-adjacent
- Chests: Dungeon/mineshaft/YUNG's structure chests have higher rates
- Trading: Some from weaponsmith villagers
- Bosses: Higher guaranteed drops from Twilight Forest and Cataclysm bosses

Too Many Bows — Unique Ranged Weapons

30+ unique bows with special abilities and custom attributes.

How they work: Each bow has a unique passive or activated ability. Check EMI for recipes and stats. Some shoot multiple arrows, some inflict status effects, some have special draw mechanics.

Finding them: Dungeon chests, mob drops, and boss loot. Combine with Apotheosis gem socketing for even more damage.

Integration: Simply Swords + Too Many Bows covers melee and ranged. Better Combat (Wave 5) provides appropriate animations for both.

Spartan Weaponry Unofficial — Polearms & Throwing

Spears, halberds, pikes, lances, throwing knives — fills the polearm gap that Simply Swords doesn't fully cover.

Key feature: Extended reach on all polearms (hit from 2-4 blocks away). Throwing weapons return after being thrown.

Finding them: Crafted at table (check EMI — wood/stone/iron/gold/diamond/netherite tiers). Lower tiers craftable early; higher tiers gate behind material progression.

Relics RPG — Legendary Accessories

Unique accessories for **Curios slots**. You have 6 slots: Ring (2), Amulet (1), Belt (1), Gloves (1), Back (1), Charm (1). Press Curios keybind ([Y](#) recommended) to open inventory.

Relic	Slot	Ability	Where to find
Monkey Talisman	Charm	Double jump	Vanilla dungeon chests
Holy Water	Charm	Right-click cleanses negative effects	Desert temples, churches
Lucky Coin	Charm	+luck for better loot	Mineshafts, outposts
Captain's Hook	Gloves	Pull entities toward you	Shipwrecks, ocean ruins
Sacred Wardstone	Amulet	Absorbs X damage then breaks	Stronghold chests
Magic Mirror	Back	Teleport to spawn/home	End cities, libraries
Heart of the Void	Ring	+max HP, slow regen	Nether fortresses, bastions
Blazing Feather	Back	+15% movement speed	Jungle temples, pagodas

Relic		Slot		Ability		Where to find	
Drowned Belt		Belt		Underwater speed boost		Ocean monuments, shipwrecks	

Stacking: Two rings = two ring relics. Wear multiple different types.

Finding strategy: Use Explorer's Compass → find YUNG's structures → loot them. Relics have higher drop rates in YUNG's structures than vanilla ones.

Runes — Socketable Weapon Modifiers

Swappable modifications for weapons/tools. Change them between fights.

How to get:

- Loot chests in any structure
- Craft at **Rune Altar** (4 cobblestone 2×2 + lapis block on top, right-click with rune fragment)
- Low drop chance from hostile mobs

Rune	Effect	Best on
Fire	On-hit fire damage (3 sec)	Fast weapons (daggers, rapiers)
Ice	On-hit slow (2 sec)	Kiting weapons (spears, scythes)
Lightning	Chance to strike lightning	Heavy weapons (single big hits)
Poison	On-hit poison (5 sec)	DoT builds, boss kiting
Lifesteal	Heal 5% of damage dealt	Tank builds, slow weapons
Critical	+10% crit chance	Medium weapons

Using runes:

1. Craft **Rune Pouch** (leather + string + lapis) — stores all runes
2. Place weapon + rune at **Rune Altar** (4 stone + 1 lapis block)
3. Rune applied. To swap: place weapon back with different rune

Tip: Keep 2-3 loadouts. Fire for general combat, Lifesteal for bosses, Ice for kiting.

Apotheosis — Loot & Enchanting Overhaul

The single largest change to how gear feels.

Rarity System

Every item drop rolls with rarity:

Rarity	Affix slots	Color	Source
Common	0	White	Basic mobs, mining
Uncommon	1	Yellow	Hostile mobs, basic chests

Rarity		Affix slots	Color	Source	
Rare		2	Cyan	Dungeon chests, mini-bosses	
Epic		3	Purple	Bosses, late-game chests	
Mythic		4	Gold	Endgame bosses, Cataclysm	

Higher rarities = more gem sockets + better affixes.

Affixes — Random Stats

Affix		Effect		Gear type	
Ruthless		+crit damage, +crit chance		Melee weapons	
Arcane		+spell damage, +mana		All weapons	
Sturdy		+armor, +knockback resist		Armor	
Nimble		+movement speed, +attack speed		Boots, leggings	
Plentiful		+mining speed, +fortune		Tools	
Titanic		+max HP, +natural regen		Chestplate	

Tip: Check every drop. A Common helmet with a good affix can beat an Epic with useless stats. Salvage unwanted gear at a salvaging table for gem dust.

Gem System

Socketable permanent stat boosts:

Tier		How		Effect	
Rough		Mob/chest drops		Basic stat boost	
Cut		Rough + Gem Cutting Station + dust		+50% stat value	
Flawless		2 Cut gems of same type + station		+100% stat value	

Gem Cutting Station: Craft (check EMI), place rough gem + dust → cut gem. Dust from breaking unwanted cut gems.

Gem Socketing Table: Craft (check EMI), place item + gem in socket slot. Limited sockets per gear piece — expand with socketing tools.

Enchanting Overhaul

- **Higher level cap:** Sharpness VIII, Protection VIII, Fortune V, Looting V possible
- **Tiered bookshelves:** Hell, Seashelf, Endshelf enable higher enchantment levels
- **Library of Alexandria:** Stores all enchanted books in one block
- **30+ new enchantments:** Check EMI for full list

Boss Affixes

Boss mobs spawn with modifiers under their health bar:

Modifier	Effect	Strategy
Tough	+50% HP, damage reduction	High-damage weapons + Lifesteal
Blinding	Blinds on hit	Blindness-resistant gear or milk
Clumsy	Disarms on hit	Speed — don't get hit
Plagued	Poison/wither on hit	Antidotes, regeneration
Huge	2× size, more knockback	Stay mobile
Chaotic	Random effects per hit	Unpredictable — bring all potions

Tip: Boss affixes are the main source of Mythic gear. Hunt affixed bosses for endgame upgrades.

Spawner Module

Silk-touch a spawner with a pickaxe → modify it (entity type, spawn delay, player range, min/max entities). Right-click placed spawner with empty hand to fine-tune. Essential for XP farms and mob automation.

Deadly Module

Enabled. Adds telegraphed death-traps to the world (spikes, poison darts, instakill rooms, lava traps). Deadliest rooms hold the best loot. Traps are avoidable with caution. GraveStone Mod preserves your items if you trigger one.

Session Plan — Wave 3

Session	Focus	What to Do
Session 1 — Skill tree	Set keybind (P). Spend first points on Strength or Vitality	Permanent progression started
Session 2 — First affix loot	Check every gear drop. Look for Rare+ with useful affixes	Learn the loot system
Session 3 — Weapon hunt	Explorer's Compass → YUNG's dungeon → loot for Simply Swords weapon. Try Rapier (fast) or Greathammer (AoE)	First unique weapon
Session 4 — Gem cutting	Collect rough gems. Build Gem Cutting Station. Cut Rough → Cut. Socket into gear	Permanent gear upgrade

Session	Focus	What to Do
Session 5 — Relic hunting	Monkey Talisman (dungeons) + Lucky Coin (mineshafts)	Accessory slots filled
Session 6 — Boss affixes	Search for affixed bosses (colored name above health bar). Best source of Mythic gear	Endgame gear progression
Session 7 — Rune loadout	Craft Rune Pouch + Rune Altar. Apply Fire rune to fast weapon, Lifesteal to slow weapon	Swappable combat modifiers

Tip: Wave 3 gear changes how every other wave feels. A Rare+ weapon with the right affix makes Twilight Forest and Cataclysm bosses dramatically easier. Invest in gear before tackling tough content.

Wave 4 — Food, Farming & Comfort

Mod	Role
Farmer's Delight 1.3.2	Cooking hub — cutting board, cooking pot, stove, new crops
Extra Mod Integrations	Adds compat recipes for 50+ mods — Farmer's Delight, Create, Mekanism, AE2
Brewin' And Chewin' 4.5.0	Drinks and fermentation — beer, wine, mead, whiskey
Storage Delight 1.3.0	Kitchen furniture — drawers, cabinets, shelves for ingredient storage
More Delight 26.04.18a+1.21-neoforge	Extra meals, sandwiches, toasts, and knife variants
My Nether's Delight 1.10.2	Nether-themed food ingredients and recipes
Ocean's Delight 1.0.4	Ocean-themed seafood dishes
Ender's Delight v1.3.0	End-themed food for Farmer's Delight (moved from Wave 2)
Twilight's Flavors & Delight 3.2.2	Twilight Forest × Farmer's Delight crossover — 35+ FD-style TF foods, 4 TF knives, maze stove, 5 potion effects, trophy dismantling
Neo Bee Fix	Fixes bee AI bugs — bees leave hives, don't fall from sky, no NW wander bias
Comfortable Campfires	Campfires grant configurable status effects — sit by the fire, get a buff
Starcatcher	Fishing overhaul — 100+ fish, skill minigame, trophies, tournaments, catalogue
Better Fishtanks	Functional aquariums — fish bags, decorations, feeding system, Create/AE2 compat

Interdependency note — Starcatcher + Farmer's Delight: Starcatcher has a dedicated Farmer's Delight addon ([Starcatcher's Delight](#)) that adds fish-related cooking recipes. Install alongside Farmer's Delight for seafood dishes.

Interdependency note — Starcatcher + Hybrid Aquatic (Wave 2): Starcatcher automatically adds Hybrid Aquatic fish as catchable entries with biome/weather/time restrictions. Hybrid Aquatic baits attract Hybrid Aquatic fish. No config needed.

Interdependency note — Better Fishtanks + Create (Wave 1): Better Fishtanks is designed for tech mod automation. The Fish Tank and Aquarist Table inventories are fully accessible via Create pipes. Use fluid pipes to fill tanks with water. Automate fish feeding with Create’s mechanical systems.

Interdependency note — Better Fishtanks + Hybrid Aquatic (Wave 2): Better Fishtanks can house any fish mod’s creatures. Use Fish Bags to catch Hybrid Aquatic mobs and place them in tanks as living displays.

Interdependency note — Twilight’s Flavors & Delight + Twilight Forest (Wave 2) + Farmer’s Delight: This mod bridges Twilight Forest and Farmer’s Delight. TF mob drops and foraged items become cookable into 35+ FD-style meals. The Maze Stove (crafted from maze ooze) accelerates cooking in TF biomes. 4 TF-material knives (ironwood, fiery, knightmetal, phantom) have unique abilities. Boss trophies can be dismantled for precious loot — giving each TF boss run lasting value beyond the first kill. Optional Neapolitan compat adds 19 more food types (Neapolitan not in pack — 35+ FD meals still available without it).

Farmer’s Delight brings a full cooking ecosystem. Optional — you can survive on baked potatoes — but cooking gives powerful status effects, better healing, and a cozy kitchen to build around.

Farmer’s Delight — Cooking Hub

New Crops

Crop	Grows on	Use
Tomato	Farmland. Bush grows, pick when red	Soups, sandwiches, salads
Cabbage	Farmland. Full head when ready	Coleslaw, cabbage wraps
Onion	Farmland. Pick when shoots show	Savory dishes, soups
Rice	Shallow water (1 block deep on dirt/mud)	Hearty meals, sushi

Wild crops: Break wild variants in the world for seeds:

- Wild tomatoes — plains/forests
- Wild cabbage — beaches
- Wild onions — taiga
- Wild rice — swamps

Cutting Board

Craft: 1 knife + 2 planks (horizontal row). Place on ground, right-click to open GUI.

Place ingredient in left slot → processed output in right slot.

Input		Output		Use	
Cooked meat		Meat strips		Sandwiches, salads	
Cabbage		Cabbage leaves		Wraps, coleslaw	
Bread		Bread slices		Toast, sandwiches	
Carrot		Carrot slices		Salads, soups	
Apple		Apple slices		Desserts, snacks	

Tip: Knife in off-hand while chopping lasts longer (uses knife durability). Iron knife = dozens of chops.

Cooking Pot

Craft: 2 iron + 2 planks + 1 bowl. Place over a heat source:

- **Stove** (fastest) — 4 bricks + 4 iron nuggets
- **Campfire** (medium) — 3 logs + 3 sticks + 1 coal
- **Lava** (slowest) — pot must sit directly above lava source

Cooking process:

1. Right-click pot → open GUI
2. Fill water slot (bucket or bottle)
3. Add up to 6 ingredients in any order
4. Wait for progress bar
5. Collect meal — most give **6 servings** with bowl stacking

Meal categories:

- **Soups** — Tomato soup, beef stew. Good saturation, easy ingredients
- **Sandwiches** — Chicken sandwich, BLT. On-the-go eating
- **Salads** — Mixed salad, fruit salad. Good for hot biomes
- **Pasta** — Pasta with meat sauce. Filling, complex ingredients
- **Rice meals** — Fried rice. Requires rice farming

Kitchen Storage

Craft **oak/pantry cabinets** (check EMI — varies by wood). Function as chests with kitchen aesthetics.

Tip: Set up kitchen near your AE2 network. Cook manually or stock ingredients in ME drives for autocrafting.

Brewin' And Chewin' — Drinks & Fermentation

Alcohol brewing with aging mechanics. Grow hops, barley, grapes on farmland.

Brewing Station

Craft (check EMI — barrel-like recipe). Right-click to open GUI.

1. Add **water** + main ingredient:
 - Barley → beer base (wort)
 - Grapes → wine base (must)
 - Honey → mead base
 - Potatoes → spirits base (mash)
2. Wait for fermentation
3. Collect **fermented mixture**

Aging Barrel

Craft (check EMI — wooden barrel). Place fermented mixture inside, right-click to seal.

Ages one stage per in-game day: Young → Matured → Aged → Vintage (4 stages). Each stage increases status effect potency and duration.

Optimal: Let everything reach **Aged** (stage 3). Push to Vintage (stage 4) only for special occasions.

Drink	Matured effect	Vintage effect
Beer	Resistance I (2 min)	Resistance II (5 min)
Wine	Haste I (2 min)	Haste II (4 min)
Mead	Speed I (2 min)	Speed II (4 min)
Whiskey	Strength I (30 sec)	Strength II (1 min)
Vodka	Night Vision (3 min)	Night Vision + Fire Resist (5 min)

Practical use: Brew a batch of each, age to Matured/Aged, store in kitchen. Pop a drink before boss fights for free buffs. Mead stacks with Swiftiness potions.

Bottle return: Empty bottles are returned after drinking. Reuse at brewing station.

Storage Delight — Kitchen Furniture

Decorative kitchen storage that works as functional inventory:

Block	Slots	Use
Cabinet	27	Counter-height storage
Drawer	18	Fits under counters
Shelf	9	Wall-mounted spice display

Tip: Use for early-game ingredient storage before AE2. Later, place an ME Storage Bus on a cabinet to bridge into your digital network.

More Delight — Extra Meals & Knives

- **Extra meals:** Sandwiches, toasts, stuffed mushrooms

- **Knife variants:** Diamond and netherite knives (much longer lasting than iron)
- **New cooking pot recipes:** Stews, soups, pasta dishes

All recipes visible in EMI. Uses existing cutting board and cooking pot.

My Nether's Delight — Nether Food

Nether-themed ingredients and recipes:

- **Nether crops:** Grow in soul sand or warped/crimson nylium
- **Ingredients:** Glowstone berries, soul berries, nether-root vegetables
- **Recipes:** Spicy stews (Fire Resistance), soul-force snacks (Night Vision)

Tip: Set up a Nether farm for combat-relevant status effects. Glow berry pie gives Haste II — useful before mining expeditions.

Ocean's Delight — Seafood

Seafood dishes using ocean mob drops:

- **Ingredients:** Squid tentacles, fish fillets, seaweed, turtle eggs
- **Recipes:** Seafood stew, grilled fish, kelp salads, sushi rolls
- **Effects:** Water Breathing + Dolphin's Grace — perfect for ocean monument raids

Tip: Build a fish farm or farm kelp for steady supply. Combined with My Nether's Delight, prepare meals for any environment.

Ender's Delight — End Food

End-themed food for post-Dragon exploration:

- **Ingredients:** Chorus fruit variants, end-themed crops
- **Recipes:** End stews and snacks
- **Effects:** Status boosts useful for End dimension exploration

Integration: End ingredients cook into Ender's Delight meals. Plan an End farming outpost with Farmer's Delight crops.

Neo Bee Fix

Fixes bee AI — bees properly leave hives, don't fall from sky, no northwest wander bias. No config. Install and forget.

Integration: Bees produce honey for Brewin' And Chewin' mead. Farm bees near your kitchen for renewable brewing ingredients.

Starcatcher — Fishing Overhaul

A complete fishing overhaul with 100+ fish, skill-based minigame, progression system, and tournaments.

Core mechanics:

Feature	What it does
Fishing Minigame	Skill-based reel-in — timing matters, not just patience
Catalogue	Records every catch — biome, weather, time conditions documented
Trophies	Mount your best catches as wall decorations
Tackle Box	Store hooks, bobbers, baits, and fish — portable fishing station
Aquariums	Multi-block fish tanks — place caught fish inside, scaled by size/weight
Tournaments	Compete against friends for biggest catch
Selling Bin	Sell fish for emeralds — price based on size, weight, rarity
Cosmetics	Fisherman hats, rod skins — found in shipwrecks and treasure

Getting started:

1. Craft a **fishing rod** (3 sticks + 2 string) — Starcatcher replaces vanilla fishing
2. Cast near water → minigame triggers on bite
3. Complete the minigame → fish caught, added to catalogue
4. Craft a **Tackle Box** (check EMI) → store fish and accessories
5. Find **baits** in loot chests or craft them → attract specific fish types

Fish conditions: Each fish has specific requirements:

Condition	Examples
Biome	Ocean fish, river fish, swamp fish
Weather	Rain-only, clear-only, storm-only
Time of day	Day fish, night fish, dawn/dusk
Elevation	Surface vs. deep water

Golden Fish: Rare variants of every fish — higher value, trophy-worthy. Hunt these for the ultimate collection.

Integration with Farmer's Delight: Install [Starcatcher's Delight](#) for fish-related cooking recipes. Fish fillets, seafood stews, sushi — all visible in EMI.

Integration with Hybrid Aquatic (Wave 2): Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

Better Fishtanks — Aquariums

Functional aquariums for displaying your catches. Build the tank of your dreams.

Core blocks:

Block	Function
Fishtank	Glass tank — house fish, configurable water levels
Aquarist Table	Crafting station for tanks, decorations, and supplies
Fish Bag	Catch and transport fish — right-click fish in world, right-click tank to place
Food Container	Store fish food in bulk — flakes, kelp, shrimp, pellets
Decorations	Coral, shipwrecks, stones, branches, towers, kelp — 30+ variants

Getting started:

1. Craft an **Aquarist Table** (check EMI — iron + glass + planks)
2. Craft a **Fishtank** (glass + iron ingots) — place it down
3. Craft a **Fish Bag** (leather + string) → right-click any fish in the world to catch it
4. Right-click the Fishtank with a filled Fish Bag → fish placed inside
5. Add **decorations** from the Aquarist Table → customize the tank

Fish Eggs: Tanks passively produce fish eggs in 6 rarity tiers (Common → Legendary). Crack them in the Aquarist Table for loot.

Aquarist Villager: A new villager profession. Trades fish supplies and rare eggs. Assign by placing an Aquarist Table near an unemployed villager.

Tech integration (Create, AE2):

- Fishtank and Aquarist Table inventories are **fully accessible** via Create pipes
- Use fluid pipes to fill tanks with water automatically
- Automate fish feeding with Create mechanical systems
- Connect to AE2 network for remote inventory access

Config options:

Setting	Default	What it does
Require Food	ON	Fish need feeding to survive — disable for pure aesthetics
Fish Bag Captures	Normal fish only	Enable to capture Guardians, Drowned, Whales
Rendering Mode	Vanilla Waterlogging	Toggle experimental entity rendering for water

Integration with Starcatcher: Better Fishtanks displays what Starcatcher catches. Starcatcher provides the fishing gameplay; Better Fishtanks provides the aquarium showcase. Different systems, complementary purposes.

Wave 4.5 — Quests

Authored quest book with chapters covering every major mod in the pack. Guides progression without forcing it — quests are optional breadcrumbs, not mandatory checklists.

Questing Foundation

Mod	Role
FTB Library	Core library for all FTB mods — configuration, GUI framework, team management
FTB Quests	Quest book framework — chapters, task types, rewards, in-game visual editor
FTB Teams	Quest progress tracker — in single-player, you're automatically a team of 1
KubeJS	JavaScript scripting engine — recipe modification, worldgen tweaks, event hooks
FTB XMod Compat	Bridges FTB Quests with KubeJS, JEI/REI compat layer (works with EMI)

Quest Addons

Mod	Role
MineColonies Questline	Pre-written 40+ quest chain covering MineColonies from Town Hall to endgame colony
ExtraQuests	Additional task/reward types — key-value tracking, progressive tasks, daily/weekly quests, console command rewards

MineColonies Questline is the only pre-written content — it covers colony progression from first Town Hall through Industrial Revolution research. All other quests (Create, Mekanism, AE2, Apotheosis, combat, dimensions, food) must be authored using FTB Quests' in-game editor ([/ftbquests editing_mode](#)).

How It Works

1. Press the quest key (configurable, default `-`) to open the quest book
2. Chapters organized by mod: **Create, Mekanism, AE2, MineColonies, Apotheosis, Combat, Exploration, Food & Farming**
3. Quests auto-detect progress — craft an item, enter a dimension, kill a boss → quest completes
4. Rewards include items, XP, and knowledge — some powerful items are quest-gated to soft-enforce progression

- The quest book is **always available** but **never mandatory** — ignore it when you want to chill

Pillar Alignment

- **Power fantasy:** Quests celebrate milestones. “You built a fusion reactor!” with a meaningful reward feels earned.
- **Easy curve:** New players follow quest breadcrumbs. “Craft enrichment chamber → craft crusher → 2x ore processing” teaches Mekanism step by step.
- **200h+ content:** Quests surface hidden content. “Have you visited the Aether?” “Try sequenced crafting with Create.” Players discover systems they might miss.
- **Chill living:** The quest book key is configurable. There’s no on-screen HUD element. No time-limited quests. No penalties for ignoring it. The book waits for you — not the other way around.

Authoring Plan

Phase	Quests to Write	Effort
MineColonies	Pre-written (MineColonies Questline)	0 hours — included
Create	Water wheel → brass → sequenced crafting → trains	~15 quests
Mekanism	Ore doubling → tripling → quintupling → digital miner → fission → fusion	~20 quests
AE2	Certus quartz → ME drive → autocrafting → quantum network	~15 quests
Apotheosis	Affix gear → gems → enchanting → mythic tier	~10 quests
Combat & Exploration	Bosses (Cataclysm, Aether, Twilight Forest), dungeons, dimensions	~15 quests
Food & Farming	Farmer’s Delight meals, brewing, fishing, animal husbandry	~10 quests

Total: ~85 authored quests + ~40 pre-written colony quests = ~125 quests. Author gradually — one chapter at a time. The pack is fully playable without quests; they’re enhancement, not foundation.

Session Plan — Wave 4 & 4.5

Session	Focus	What to Build
Session 1 — Kitchen tools	Cutting board + cooking pot + stove. Find wild crop seeds	Functional kitchen
Session 2 — Farm	Plant tomato, cabbage, onion, rice. Cook first meal (tomato soup is easiest)	Sustainable ingredients
Session 3 — Brewing	Find barley/grapes/honey. Build brewing station. Start first batch. Place aging barrel	First drink fermenting
Session 4 — Kitchen expansion	Storage Delight cabinets/drawers. Try More Delight recipes. Collect Nether ingredients	Full kitchen
Session 5 — Questing	Open quest book (- key). Browse MineColonies questline. Complete “Town Hall” quest	Questing unlocked
Ongoing	Age drinks for boss buffs. Explore oceans for seafood. Stock AE2 with ingredients for autocrafting	Buff food always ready

Wave 5 — Combat & Mobs

Mod	Role
Better Combat	Melee combat animations — weapon reach, sweeping, attack speed
Mutant Monsters	Mutant zombie, skeleton, creeper, enderman — unique drops
L_Ender's Cataclysm	Epic boss fights — Leviathan, Netherite Monstrosity, Ender Guardian (CurseForge, NeoForge 1.21.1)
When Dungeons Arise	Roguelike combat structures with loot
Enchantment Descriptions	Shows enchantment effects in tooltips
Create Big Cannons	Artillery engineering — siege cannons for colony defense and spectacle
Cut Through	Combat QoL — swing through transparent blocks (tall grass, cobwebs) without breaking them
Torchmaster	Territory pacification — Mega Torch suppresses hostile spawns in a radius; each torch gated behind a Cataclysm boss conquest (see below)

Apotheosis also adds boss modifiers and elite mob affixes — partially covers what Champions would have done.

Your powerful gear finally has worthy opponents. Better combat mechanics, tougher mobs, epic boss fights, roguelike dungeons.

Better Combat — Weapon Feel

Replaces Minecraft's single swing with weapon-specific movesets:

Weapon	Speed	Reach	Animation	Best for
Daggers	Very fast	2 blocks	Quick stab	Debuff stacking, fast crit
Swords	Fast	3 blocks	Horizontal arc	General combat
Rapiers	Fast	3.5 blocks	Thrust (ignore shields)	Shield-breaking
Greatswords	Slow	4 blocks	Wide sweep	Group clearing
Axes	Medium	3 blocks	Overhead chop	Armor penetration
Spears	Medium	4.5 blocks	Forward thrust	Kiting
Hammers	Slow	3.5 blocks	Overhead slam	Knockback, stun
Greathammers	Very slow	4 blocks	Ground pound	AoE knockback

Config: Mod Menu → Better Combat. Tweak reach, sweep radius, attack speed, visual effects per weapon type.

Integration: Simply Swords weapons have built-in Better Combat data files — each category uses correct animation, reach, and speed automatically. Too Many Bows and Spartan Weaponry also integrate.

Enchantment Descriptions

Hover over any enchantment in tooltip → detailed description of what it does. Works in enchanting table GUI, on items, and in books. No config.

Cut Through — Clean Combat

Swing through transparent blocks (tall grass, vines, cobwebs) without breaking them. Hit the mob behind. No more breaking your grass farm or getting stuck on cobwebs while skeletons shoot you.

No config. Client-side only. Works with Better Combat and all other combat mods.

Mutant Monsters — Boss Variants

Tougher versions of vanilla mobs. Drop unique gear.

Mutant	Spawns	What to expect	Drops
Mutant Zombie	Overworld, night	Tankier, ground-pound AoE, summons zombies	Hulk Hammer (AoE knockback)
Mutant Skeleton	Overworld, night	Rapid-fire weakness arrows, teleports	Mutant Skeleton Armor (projectile resist)
Mutant Creeper	Overworld, night	2× explosion radius, drops mini-creepers on death	Creeper Minion (follows, explodes on command)
Mutant Enderman	The End	Aggressive teleporting, throws blocks, projectile resist	Enderman gear (teleportation items)

Spawn rate: Adjust in Mod Menu → Mutant Monsters → spawn weight if too frequent/rare.

Strategy: Same as vanilla but more extreme. Zombie → kite. Skeleton → close fast. Creeper → ranged. Enderman → water + fire resistance.

L_Ender's Cataclysm — Epic Boss Fights

Hand-animated bosses with unique mechanics. **Opt-in** — you must build a summoning structure or find a specific location.

Boss	Location	How to Summon	HP	Recommended Gear
Leviathan	Deep ocean (Y<40)	Prismarine + sea lantern altar + Prismarine Shard	~200	Diamond, Depth Strider, door trick
Netherite Monstrosity	Nether wastes	Enter central arena in nether brick lair	~300	Full diamond/netherite, fire resistance
Ender Guardian	End (outer islands)	Eye of Ender on pedestal after Dragon	~400	Endgame gear, Elytra + Slow Falling
Ancient Remnant	Desert	Enter summoning room in large pyramid, insert Nether Star	~350	Blast protection, blocks to pillar
Ignite	Nether (basalt deltas)	5×5 hollow obsidian sphere + Blaze Rod center, wait 10 sec	~250	Fire resistance, water buckets

General strategy:

1. Scout the arena first
2. Bring food buffs (Farmer's Delight + Brewin' And Chewin' stack)
3. Apply runes (Lifesteal or Ice)
4. Set a Waystone nearby
5. AE2 wireless terminal for mid-fight resupply

Leviathan: Phase 1 — dodge charges, attack body, jump shockwaves. Phase 2 (50%) — avoid whirlpools, attack head between spawns. Drops Leviathan Blade (trident-like water weapon).

Netherite Monstrosity: Phase 1 — attack legs to slow. Phase 2 (60%) — enrages, faster attacks + fire breath. Sprint behind pillars. Drops Monstrosity tools (efficient netherite-level mining).

Ender Guardian: Phase 1 — dodge tracking energy bolts, hit wall crystals to stun. Phase 2 (50%) — teleports, summons endermite, break all crystals to lower

shield. Phase 3 (20%) — energy nova, pillar up to avoid ground AOE's. Drops high-damage weapons + teleportation gear.

Ancient Remnant: Phase 1 — sandstorm blinds, attack during cooldown. Phase 2 (40%) — sinks into sand, pops up elsewhere, shoot exposed weak point. Drops tools with unique enchantability.

Ignite: Phase 1 — fireball barrage, block with shield or dodge behind pillars. Phase 2 (40%) — absorbs lava from arena, stay on high ground, attack when emerging. Drops fire weapons (permanent fire aspect).

When Dungeons Arise — Roguelike Structures

Large multi-floor combat structures across the Overworld.

Structure	Floors	Difficulty	Biome	Loot
Abandoned Temple	3	Easy	Plains, forests	Iron gear, early rare items
Small Castle	4	Medium	Plains, hills	Rare weapons, gold
Large Castle	6	Hard	Mountains	Epic gear, diamonds
Barracks	2	Easy	Any	Armor, shields
Scorched Fort	4	Medium	Desert, badlands	Fire-related items
Warped Fortress	5	Hard	Any (rare)	Endgame gear, unique loot
Foundry	3	Medium	Underground	Ores, mining gear
Bandit Camp	1	Easy	Forest, taiga	Early weapons, food

Finding: Explorer's Compass → "When Dungeons Arise" filter → select structure.

Strategy: Clear floors systematically. Loot scales with depth. Bring blocks to pillar, torches to light hallways, decent gear for deep floors.

Note: Structures are finite — once looted, they don't respawn unless chunks reset. Mark looted ones on JourneyMap.

Create Big Cannons — Artillery Engineering

Functional cannons built from Create materials, using rotational power for assembly and gunpowder for ammo.

Getting started:

1. Craft cannon components: barrel (bronze/steel), mount (Create materials), charger

2. Build: place mount, attach barrel, connect charger
3. Arm: insert powder charges + cannonballs into GUI
4. Fire: right-click or redstone signal

Ammo	Effect	Use
Solid cannonball	High damage, block-breaking explosive	Siege, defense
Canister shot	Spread of small projectiles	Anti-personnel
Grape shot	Medium spread	Crowd control
Incendiary	Fire on impact + lingering fire	Area denial
Smoke shell	Smoke screen	Concealment

Integration: Gated behind bronze/steel (Create progression) + gunpowder. Mount on colony walls for defense. Redstone-controlled pulse extender for automatic salvo fire.

Dependency: Ritchie's Projectile Library (auto-installed) — provides custom trajectories and hit detection. No separate config.

Torchmaster — Conquering the Land

Torchmaster makes **peace a reward, not a default**. Its Mega Torch suppresses hostile mob spawning in a 48-block radius — but the torch itself requires a **Cataclysm boss conquest** to craft (KubeJS gate, [kubejs/server_scripts/wave5_torchmaster.js](#)):

Item	Gated Recipe	Gate
Mega Torch	Ignitium Ingot (Cataclysm Ignite drop) + Create brass ingots + blaze rods	Phase 2 — first Cataclysm boss + brass age

The conquest loop — each torch is one earned region:

1. **Scout** — fly your plane (Immersive Aircraft / Man of Many Planes) or roll out (Create train, Small Ships) to a hostile region
2. **Clear** — fight through the structures with your guns (Create: Gunsmithing), artillery (Create Big Cannons), and gear
3. **Harvest** — defeat the Cataclysm boss, collect the Ignitium Ingot
4. **Craft** — return home, craft the Mega Torch in your factory (Create brass)
5. **Plant** — place it at the region's center. The area is now *yours* — permanently peaceful, safe to build in

Why it's gated: Cataclysm bosses are re-fightable (arena respawner blocks accept an Eye item to re-summon), so conquest is **renewable** — no torch farming loophole, no world exhaustion. Every peaceful biome is an earned trophy, and the

200h+ content runway survives because you can always find another arena to conquer.

Config (<config/torchmaster-common.toml>): Mega Torch radius 48 blocks (village-sized zone). Feral Flare Lantern (radius 24) suppresses only dark-spawned mobs — useful for cave/ruin clearing without full pacification. Mega Torch blocks natural spawning only, not spawners.

Chill Rule check: Conquered land is permanent — you can log off, ignore progression, or decorate for a session without a single mob interrupting. This is the *payoff* of the pack's chill pillar, earned through the power fantasy.

Session Plan — Wave 5

Session	Focus	What to Do
Session 1 — Combat feel	Better Combat works automatically. Craft/find a Simply Swords weapon. Test daggers (fast stab) vs greatswords (wide sweep). Equip a rune	Learn your weapon
Session 2 — First mutant	Find Mutant Zombie at night (listen for roar). Kite it, learn patterns. Collect unique drop	First mini-boss
Session 3 — First dungeon	Explorer's Compass → When Dungeons Arise small castle/abandoned temple. Clear floors	Roguelike dungeon complete
Session 4 — First Cataclysm boss	Easiest = Leviathan (deep ocean) or Netherite Monstrosity (Nether). Prepare: good weapons, fire resistance, healing food	Epic boss beaten
Session 5 — Cannons	Craft bronze (zinc + copper in heated mixer). Build Small Cannon. Test fire. Mount near base entrance	Artillery online

Session		Focus		What to Do	
Session 6+ — Boss gauntlet		Remaining Cataclysm bosses. Each drops unique endgame materials. Revisit Twilight Forest bosses with Wave 5 gear		Full endgame gear	
Session 7 — First conquest		Scout a region by plane, clear its structures with guns/artillery, defeat the Cataclysm boss, craft the Mega Torch from its Ignitium Ingot, plant it. The region is yours		First pacified territory	

Tip: Better Combat changes melee fundamentally. Don't skip Session 1 even as a ranged player.

Wave 6 — Building & Decoration

Core Decoration

Mod	Role
Rechiseled	600+ decorative block variants with connected textures
Supplementaries	Vanilla+ decoration — jars, signposts, faucets, weather vanes
Macaw's Bridges	Bridges in all wood types
Macaw's Doors	Vanilla-style doors for every wood
Macaw's Roofs	Roofing blocks
Macaw's Windows	Window variants
Building Wands	Faster large-scale construction
[Reforged] Accurate Block Placement	Precision block placement — place on the exact face you're looking at
Handcrafted	Furniture — chairs, tables, shelves, curtains
The Magic Mirror	Reflective mirror block that shows the nearest player's reflection
Fetzi's Displays	Item displays — glass cloches, katana stands, wall holders, racks
Laser Bridges & Doors	Laser platforms, bridges, doors, and fences powered by redstone — length scales with signal strength, dye-customizable colors

Interdependency note — Laser Bridges & Doors + AE2 (Wave 1): Laser sources require an End Crystal (post-End-dragon). Redstone control pairs naturally with AE2 level emitters and Create redstone circuits — automate laser gates that open when your ME system detects specific conditions.

Interdependency note — Fetzi's Displays + Apotheosis (Wave 3): Fetzi's Displays is ideal for showcasing Apotheosis gear. Glass cloches and katana stands let you display mythic drops and perfect gem rolls as trophies. Wall holders with plaques let you name your legendary weapons.

Interdependency note — Lili's Pottery + Handcrafted: Both add decorative blocks but cover different aesthetics. Handcrafted is rustic furniture (chairs, tables, shelves). Lili's Pottery is earthenware (pots, vases, urns). Together they provide a full interior decoration toolkit.

Rechiseled Addons

Mod	Role
Rechiseled: Create	100+ decorative variants of Create blocks + mechanical chisel
Rechiseled: Applied Energistics 2	AE2 autocrafting for Rechiseled blocks via Chiseling Pattern Encoder

Tools and blocks to build the base you deserve.

Rechiseled — Decorative Block Variants

600+ decorative block variants with connected textures.

How:

1. Craft **chisel** (iron + 2 sticks, check EMI)
2. Right-click with chisel → chiseling screen
3. Place base block in input slot → browse all variants
4. Click variant to preview (cycles through single block, row, 3×3 grid)
5. Take result from output slot
6. **Left-click** placed blocks with chisel to cycle through random variants in-world

Key features:

- **Connected textures** — toggle with bottom-right button
- **Stairs & slabs** — every variant has stair/slab forms
- **No durability** — chisel never breaks

Block types (30+ variants each):

- Stone (bricks, tiles, carved, pillars, panels, runed, mosaic)
- Glass (all colors + connected clear, borders, panels)
- Wood (grain variants, carved panels, ornate borders)
- Wool & Carpet (woven, knitted, cross-stitched)
- Concrete (smooth, paneled, tiles, bordered)
- Nether materials (glowstone, nether bricks, blackstone)

Addons:

- **Rechiseled: Create** — 100+ decorative Create block variants + mechanical chisel
- **Rechiseled: AE2** — autocraft Rechiseled blocks via AE2's Chiseling Pattern Encoder

Supplementaries — Vanilla+ Functional Decoration

Decoration blocks that serve a purpose:

Block	Function	How
Jar	Stores 1 item visually	Right-click with item to put in, empty hand to take out
Signpost	Directional sign pointing to named location	Right-click with tag item to set destination name. Arm points toward Waystone
Faucet	Fluid flow decoration	Place above fluid tank. Right-click to toggle flow
Weather Vane	Shows wind direction/speed	Place outside. Rotates in real-time
Item Shelf	Wall-mounted display (3 items)	Right-click with item to place, sneak-right-click to remove
Rope	Climbable, place in mid-air	Right-click to place knot, add more rope below. Breaks when top knot broken
Flute	Craftable instrument	Right-click to play note. Villagers react

Practical builds:

- Kitchen: Item shelves for ingredients, jars for flour/sugar, faucet over sink
- Entrance: Signposts to “Mine”, “Nether Portal”, “Village”
- Garden: Weather vane, rope swing from tree
- Study: Jars with glowing items as lamps

Macaw’s — Complete Building Set

Five modules covering structural elements in every wood type.

Bridges: 6 slabs + 1 stick in bridge pattern. Right-click to extend, shift-right-click to shorten. Straight or rope variants. All 10+ wood types.

Doors: Special doors per wood — glass, slatted, stable (top half opens independently), cottage, shop, paper. Craft with matching wood + relevant material.

Roofs: Roofing blocks — slopes, ridges, gutters. 3 slabs in triangle pattern for slope piece. Auto-connects corners.

Windows: Diamond pane, framed, barred, tropical, stained glass. Craft with glass + wood.

Fences: Picket, stockade, horse, wired, highley gate. Each distinct look.

Building tip: Macaw’s roofs + Rechiseled stone + Macaw’s windows = cohesive aesthetic.

Building Wands — Large-Scale Construction

Place many blocks at once. Closest thing to WorldEdit without a server plugin.

Craft: Stick + 2 blocks of same type in + pattern. Block type determines tier (stone = basic, iron = better, diamond = best).

Modes (right-click wand to open GUI):

Mode	Does	When to use
Build Wall	Wall between 2 clicked points	Enclosing rooms, perimeters
Build Line	1-block-wide line	Railings, borders, paths
Fill Area	3D cuboid between 2 corners	Foundations, clearing rooms
Build Circle	Ring of blocks	Towers, wells
Build Cylinder	Filled cylinder	Silos, water tanks
Build Sphere	Hollow or filled sphere	Domes, decorative globes

How: Left-click to set first corner (green sparkle). Right-click to set second corner (structure builds). Uses blocks from your inventory.

Tips:

- Shift+Right-click to undo last placement
- Use **Undo wand** (stick + slimeball) to reverse recent actions
- Combine with AE2 wireless terminal for inventory access while building

Handcrafted — Furniture

Furniture	Function	Craft
Chairs	Right-click to sit	Slabs + planks
Tables	Right-click to place items on top	Varies by type
Shelves	Wall-mounted item display	Wood + relevant material
Curtains	Right-click to open/close	Wool + stick
Desks	Table + drawer storage	Slabs + planks
Cupboards	Wall-mounted aesthetic storage	Wood + glass
Sofa	Multi-block seating	Wool + wood
Dresser	Tall drawer storage	Wood + relevant material

Tip: Place in your most-used rooms — kitchen, dining, bedroom. Use shelves to display favorite loot (Twilight Forest trophies, Apotheosis mythic gear, rare Simply Swords weapons).

The Magic Mirror

Reflective mirror block showing the nearest player's reflection.

How: Craft (check EMI — glass + frame materials). Place on wall. Shows your character model in real-time, including armor and held items.

Use: Decorative vanity mirrors in bedrooms and dressing rooms. Purely aesthetic — no functional gameplay effect.

Fetzi's Displays — Item Showcases

Display your hard-earned items in style. Craft a **Workbench** (check EMI) to make all display types.

Display Type	What it does	Best for
Glass Cloches	Glass dome on pedestal — 16 colors	Trophy items, rare drops, gems
Katana Stands	Vertical weapon rack	Simply Swords weapons, boss drops
Wall Holders + Plaques	Wall-mounted display with name tag	Named weapons, “GOD SWORD” vibes
Racks (4 types)	Horizontal/vertical item holders	Tool organization, collection display

Scale, rotation, position: Items in displays can be scaled, rotated, and repositioned. Check the [Wiki](#) for exact controls.

Integration with Apotheosis (Wave 3): Glass cloches and katana stands are ideal for showcasing mythic drops. Mount your perfect gem rolls and affix-rolled weapons as trophies. Wall holders with plaques let you name your legendary weapons — “Flamebane, Bane of the Ender Dragon.”

Lili's Pottery — Decorative Earthenware

Part of the [Let's Do](#) cozy mod series. Adds pottery blocks for interior decoration.

What it adds:

Category	Variants
Clay	6 new clay types found in various biomes
Pots & Vases	7 variants in old and new clay styles
Urn & Amphore	Functional storage — save items from death

How:

1. Find **new clay types** in world (beaches, rivers, various biomes)
2. **Knead** clay at crafting table → raw pot forms
3. **Bake** in furnace → hardened pottery
4. **Paint** patterns → finished decorated pottery

Integration with Handcrafted: Both add decorative blocks but cover different aesthetics. Handcrafted is rustic furniture (chairs, tables, shelves). Lili's Pottery is earthenware (pots, vases, urns). Together they provide a full interior decoration toolkit.

Laser Bridges & Doors — Energy Barriers

Futuristic laser barriers created by powering Laser Source blocks with redstone.

What it adds:

Block	Function
Laser Source Block	Projects a horizontal laser bridge when powered by redstone
Laser Fence Source Block	Projects a vertical laser barrier

How it works:

1. Craft a **Laser Source Block** — End Crystal + 7 iron ingots + glass (check EMI)
2. Power with **redstone** — laser length scales with signal strength (max 15 blocks by default, configurable)
3. **Dye** with right-click to customize color — any dye works, dispenser-compatible too
4. Convert between laser bridge ↔ laser fence in a crafting grid

Key features:

- **Dynamic length** — changes in real-time as redstone signal strength varies (use comparators, levers at different levels)
- **Blocks cut through** — configurable list of blocks lasers can replace. Default: air only. Tune in [config/laserbridges.json](#)
- **Dispenser dyeing** — automate color changes with redstone and dispensers

Recipe gate: Requires an End Crystal — this is post-End-dragon content. Appropriate for Phase 2+ base building.

Integration with Redstone & Tech:

- **AE2 level emitters** — emit redstone when item quantities hit thresholds → auto-close laser gates when storage is low or auto-open when conditions are met
- **Create redstone links** — wireless redstone across your base to control laser arrays from a central panel
- **Comparators** — use container fullness to control bridge length dynamically

Chill living fit: Purely decorative/functional — no combat interaction, no progression pressure. Dye-customizable to match base aesthetics (warm amber lasers for cozy builds, cool blue for tech areas).

Session Plan — Wave 6

Session	Focus	What to Build
Session 1 — Rechiseled basics	Craft chisel. Convert stone, wood, glass into decorative variants. Try connected textures	Decorative block library
Session 2 — Structural elements	Craft Macaw's roofs, windows, doors in your base wood type. Build a cohesive room	Cohesive room aesthetic
Session 3 — Furniture	Handcrafted chairs, tables, shelves in kitchen and bedroom. Item shelves for loot display	Functional interior
Session 4 — Large-scale	Building Wands — wall mode for perimeter, fill mode for foundation. Build a tower with circle mode	Rapid construction
Session 5 — Details	Supplementaries jars, signposts, weather vanes. Laser Bridges & Doors for energy barriers and decorative lighting. Rechiseled: Create for factory aesthetics	Decorative details

Wave 7 — Space Exploration

Core Space Mod

Mod	Role
Stellaris 1.4.23	Space exploration — rockets, Moon/Mars/Venus/Mercury/Glacio, oxygen system, machines, energy

Stellaris (by ExodusTeam) is the successor to Beyond Earth and the premier space mod for NeoForge 1.21.1. It features 4 rocket tiers, a rover, oxygen management, custom gravity per planet, oil/water/fuel processing, radioactives, and alien mobs. Machines include the Oxygen Distributor, Water Separator, Solar Panel, Fuel Refinery, and Pump Jack. 250+ building blocks across planet-specific sets.

Integration with Tech Mods

Layer	Stellaris ↔ Tech Mod Interaction
Power	Mekanism cables can power Stellaris machines. Create rotational → Mekanism → Stellaris
Energy	Mekanism universal cable connects to Stellaris machines (energy API compat)
Storage	AE2 ME drives store Stellaris materials; autocrafting builds rocket components
Transport	Create trains move space resources from landing pad to base. Sophisticated Backpacks carry oxygen tanks
Ore Processing	Mekanism 5x ore multiplication works on Stellaris ores (desh, ostranium, etc.)
Food	Farmer's Delight meals eatable in space when oxygen is present (v1.2.1+)
Gear	Apotheosis affixes may apply to Stellaris space suit (if item tags align)

Progression Placement

Phase 2 — Industrial Ascent (hours 40–200). Rockets require advanced materials that gate them behind Mekanism mid-game processing and AE2 infrastructure. Oxygen systems and fuel refining add a parallel supply-chain puzzle that rewards industrial automation.

Mod Count Summary

Wave		Mods	Deps	Total	Notes
Wave 0 — Foundation		14	10	24	Dependencies (12), Performance & Rendering (10), Shaderpack (1 mod: Euphoria Patches), Infrastructure (3) — all the nuts & bolts that make the game run (+1 shaderpack, +11 resource packs)
Wave 0.5 — Player Experience		57	—	57	Visual & Client QoL (33), Inventory & UI (20), Storage & Travel (6), Utility & QoL (1: Easy Anvils), Village & Trading (2) — everything that makes the game feel good
Wave 1 — Tech		19	5	24	Create + 13 addons (incl. Power Grid, Gunsmithing, Protection Pixel, Ornithopter Glider, Train Utilities, Railways Navigator, Interiors, Train Parts, Threaded Trains), Mekanism + Generators, AE2, Advanced Finders, Advanced Chimneys + NTGL, GeckoLib, Framework, ForgeEndertech, Kleiders deps
Wave 1.5 — Colony		9	—	9	MineColonies (4 deps counted as mods — they're library mods; CurseForge-only) + 4 addons: Byzantine Styles, Stylecolonies, Create: Colony Logistics, ColonyLink

Wave		Mods	Deps	Total	Notes
Wave 2 — Exploration		26	6	32	YUNG's (9), Terralith/ Tectonic, Serene Seasons, Darker Depths, Upgrade Aquatic, dimensions (3), End overhaul (2 + Nullscape dp + 6 deps), navigation, aircraft, ships, hang glider, IaA addon
Wave 3 — Equipment Magic		9	10	19	Skill Tree, Simply Swords, Too Many Bows, Spartan Weaponry, Relics, Runes, Curios, Apotheosis + 10 deps (Placebo, Apothic modules, Patchouli, Simply Tooltips, Fzzy Config, Ranged Weapon API, Spell Engine, Bundle API)
Wave 4 — Food & Farming		9	—	9	Farmer's Delight + 6 addon mods + Neo Bee Fix + Comfortable Campfires
Wave 4.5 — Quests		6	1	7	FTB Questing ecosystem: FTB Library + FTB Quests + FTB Teams + KubeJS + FTB XMod Compat + MineColonies Questline + ExtraQuests (Modrinth). Bountiful removed.
Wave 5 — Combat		8	6	14	Better Combat, Mutant Monsters, Cataclysm, Dungeons Arise, Big Cannons, Cut Through, Torchmaster, Enchantment Descriptions + RPL lib, playerAnimator, Puzzles Lib, Citadel, Lionfish-API, Bookshelf

Wave		Mods	Deps	Total	Notes
Wave 6 — Building		12	3	15	Rechiseled, Supplementaries, Macaw's (4), Building Wands, Reforged: Accurate Block Placement, Handcrafted, The Magic Mirror, Fetzi's Displays, Lili's Pottery, Laser Bridges & Doors + Rechiseled: Create, Rechiseled: AE2 + Moonlight, Resourceful, Fusion
Wave 7 — Space Exploration		1	1	2	Stellaris, Potentials API
Wave 8 — Programmable Computers		1	0	1	CC:Tweaked
Total		165	42	207	All confirmed NeoForge 1.21.1

Space travel becomes real. Build rockets, manage oxygen, explore alien worlds, bring back exotic resources.

Stellaris — Interplanetary Travel

Complete space exploration mod. Build rockets piece by piece, manage oxygen and fuel, travel to 5 planets.

Video tutorial: [Stellaris Getting Started Guide](#)

Core concepts:

Concept	What it means
Rocket Components	Build engines, fuel tanks, capsules, fins separately → assemble at Rocket Station
Oxygen	Most planets have no atmosphere. Space suit + oxygen required
Fuel	Rockets need fuel (oil → refinery). Higher-tier rockets need more fuel
Planet Tiers	Basic rocket = Moon only. Upgraded rockets reach Mercury, Venus, Mars, Glacio

Concept	What it means
Gravity	Each planet has different gravity. Low = higher jumps. High = weighted movement

Phase 1: Getting to Space

Step 1 — Gather resources:

- Find **oil** in Overworld (pools in deserts/plains)
- Mine **steel** (from Mechanism Metallurgic Infuser or TFMG Blast Furnace)
- Collect iron, copper, redstone, glass

Step 2 — Build basic machines:

Machine	Function	Priority
Rocket Station	Assembly block — build components here	First
Fuel Refinery	Oil → rocket fuel	Second
Oxygen Distributor	Fills space suits with oxygen	Third
Water Separator	Water → oxygen (for Distributor)	Fourth
Solar Panel	Power generation on planets	Fifth

Step 3 — Build first rocket:

1. Open Rocket Station GUI
2. Craft **Tier 1 Engine, Tier 1 Fuel Tank, Capsule, Fins**
3. Rocket Station assembles them into **Tier 1 Rocket**
4. Right-click assembled rocket to create launch pad

Step 4 — Prepare for launch:

1. Fill fuel tank from Refinery (fluid pipes or buckets)
2. Equip **Space Suit** (check EMI — steel + glass + rubber/leather)
3. Fill suit with oxygen at Oxygen Distributor
4. Stock supplies: food (Farmer's Delight meals work with oxygen), blocks, torches, weapons

Step 5 — Launch:

1. Enter rocket (right-click door)
2. Press launch button in GUI
3. Watch ascent animation

The Moon — First Destination

Safest first destination. Low gravity (~17% Overworld), no atmosphere, basic resources.

What to do:

- Mine **moon stone** and **cheese ore** (smelts into cheese ingots)
- Collect **desh** — Moon's primary rare resource. Used for Tier 2 rocket components
- Explore structures — crashed spaceships, moon bases, landing pads
- Place a **Waystone** immediately
- Set up base: solar panel + oxygen distributor + chest
- Look for rich ore veins — higher concentrations of certain Overworld ores

Returning: Enter rocket on Moon → launch → returns to original launch pad.

Other Planets

Planet	Tier	Gravity	Hazards	Key Resource	Notes
Moon	1	Low	None (oxygen required)	Desh	First destination. Safe practice
Mercury	2	Low	Extreme heat, fire damage	Ostranium	Bring fire resistance
Venus	2	High	Acid rain	Tharsite	Dense, heavy movement
Mars	3	Medium	Cold, dust storms	Desh (rich)	Best for permanent off-world base
Glacio	4	Low	Extreme cold	Glacio crystals	Ice planet. Endgame destination

Off-World Machines

Machine	Function	Why you need it
Solar Panel	Power from sunlight	Most planets have more sun than Earth
Oxygen Distributor	Breachable air	Required on every planet
Water Separator	Water → oxygen	Feed oxygen into Distributor
Fuel Refinery	Oil → rocket fuel	Refine on destination planet if oil found
Pump Jack	Extracts oil	Find oil pools on other planets

Integration With Tech Base

System	How it connects
Power	Mekanism generators at Moon base. Universal cables power Stellaris machines

System	How it connects
Ore processing	Bring planet ores (desh, ostranium, tharsite) back to Mekanism line. 5x works on Stellaris ores
Storage	AE2 ME drives store Stellaris materials. Autocraft rocket components. Quantum link or P2P tunnel to off-world base
Transport	Create trains move rocket materials from launch pad to base. Conveyors feed Rocket Station
Fuel	TFMG diesel/gasoline for industrial use. Stellaris uses its own fuel crafting (oil → refinery)
Backpacks	Netherite-upgraded backpacks carry oxygen tanks and supplies
Food	Farmer's Delight meals eatable in space with oxygen present
Gear	Apotheosis affixes may apply to space suit (if item tags align)

Progression Placement

Space belongs in **Phase 2** (hours 40–200). Prerequisites:

- Mekanism basic processing (Enrichment Chamber)
- Small AE2 network
- Create power infrastructure
- Good weapons/armor (Wave 3/5 gear helps with planet mobs)

Start with Moon missions, establish Tier 1 base, then Mercury/Venus. Mars and Glacio are late Phase 2.

Session Plan — Wave 7

Session	Focus	What to Build
Session 1 — Ground prep	Find oil. Build Fuel Refinery + Rocket Station. Craft Tier 1 components. Assemble rocket	Launch infrastructure
Session 2 — Moon landing	Craft Space Suit. Fill with oxygen. Stock supplies. Launch	First off-world base, desh
Session 3 — Moon base	Solar panel + oxygen distributor. Waystone. Mine desh for Tier 2. Chunk Loader	Sustainable Moon operation

Session		Focus		What to Build	
Session 4 — Next planets		Tier 2 rocket → Mercury or Venus. Unique hazards (heat/acid). Temporary base		New resources	
Session 5+ — Deep space		Tier 3 → Mars (permanent base). Tier 4 → Glacio (endgame crystals). Digital Miner for auto-extraction		Interplanetary logistics	

Tip: Space is the most resource-intensive activity. Don't rush it. Each planet should feel like a meaningful expansion. MekaSuit trivializes spacewalks — jetpack flight + environmental protection = no oxygen worries.

Wave 8 — Programmable Computers

Programmable computers, autonomous turtles, and Lua scripting. Write programs to automate your base in ways Create and Mekanism can't — custom logic, conditional automation, self-optimizing systems.

Mod

Mod	Role
CC: Tweaked	Programmable computers, turtles, and Lua scripting. Wireless networking, peripheral API for block interaction, redstone integration

Source note: CC:Tweaked's devs recommend Modrinth for the latest NeoForge builds. If CurseForge has an equivalent version, either source works.

No additional dependencies — CC:Tweaked is self-contained.

Recipe Gates

CC:Tweaked's default recipes use vanilla materials and would be craftable in Phase 1. To enforce the Phase 2 progression gate, a KubeJS script ([kubejs/server_scripts/wave8_cc_tweaked.js](#)) overrides the following:

Item	Gated Recipe	Gate
Advanced Computer	Basic Computer + Enriched Alloy (Mekanism) + Redstone Block	Phase 2 — Metallurgic Infuser
Turtle (all variants)	TFMG Steel ingots + Computer + Chest	Phase 2 — Blast Furnace
Wireless Modem	Create Brass casings + Computer + Ender Pearl	Phase 1.5-2 — Brass age
Basic Computer	Stone + Redstone + Glass Pane (vanilla, unchanged)	Phase 1 — harmless without peripherals
Wired Modem	Stone + Redstone (vanilla, unchanged)	Phase 1 — local only

Configuration

CC:Tweaked's config is world-specific (`<world>/serverconfig/computercraft-server.toml`). Tracked override at `config/computercraft-server.toml`:

Setting		Value	Why
<code>[http].enabled</code>		<code>false</code>	Single-player only — no web API needed
<code>command_computers</code>		<code>false</code>	Survival-only constraint
<code>[turtles].need_fuel</code>		<code>true</code>	Keeps turtles balanced — coal/lava are abundant
<code>[turtles].fuel_limit</code>		<code>200000</code>	Generous but finite. A lava bucket (1000 moves) fits 200 times

Copy `config/computercraft-server.toml` to `<world>/serverconfig/` after first world creation.

Crafting Your First Computer

Step 1: Materials

Item	Recipe	Phase
Basic Computer	4 stone + 1 redstone + 1 glass pane (crafting table, 3×3)	Phase 1
Advanced Computer	1 Basic Computer + 4 Enriched Alloy + 1 redstone block (see Recipe Gates above)	Phase 2
Monitor	3 stone + 1 glass pane (top row)	Phase 1

Step 2: Place it

Place the computer block on the ground. Right-click it to open the interface. You'll see a black screen with a prompt:

```
CraftOS 1.9
>
```

This is the **shell** — your command-line interface to the computer. Type `help` and press Enter to see available commands.

Step 3: Attach a monitor (optional)

Place a **Monitor** adjacent to the computer. Wrap it in a Lua program with `peripheral.wrap("right")` (or whichever side it's on). More on peripherals later.

The Shell

The shell is where you type commands. Here are the essential ones:

Command	What it does
<code>help</code>	List all commands
<code>help <command></code>	Show help for a specific command

Command	What it does
<code>ls</code>	List files in current directory
<code>cd <dir></code>	Change directory. <code>cd ..</code> = up one level
<code>mkdir <name></code>	Create a new directory
<code>rm <file></code>	Delete a file
<code>cp <src> <dst></code>	Copy a file
<code>mv <src> <dst></code>	Move/rename a file
<code>edit <file></code>	Open the built-in text editor
<code>pastebin get <code> <file></code>	Download a program from pastebin
<code>reboot</code>	Restart the computer
<code>shutdown</code>	Turn off the computer
<code>id</code>	Show this computer's ID number
<code>programs</code>	List available built-in programs
<code>clear</code>	Clear the screen

Tab completion: Start typing a filename or command, press Tab to autocomplete.

Ctrl+T: Terminates any running program immediately. If you're stuck in a loop, Ctrl+T is your escape hatch.

The Editor

The built-in editor opens when you run `edit <filename>`.

Key	Action
<code>Ctrl+S</code>	Save file
<code>Ctrl+E</code>	Exit editor (back to shell)
<code>Ctrl+P</code>	Print file to screen
Arrow keys	Move cursor
<code>Ctrl+A</code>	Select all
<code>Ctrl+C</code> / <code>Ctrl+V</code>	Copy / Paste
<code>Page Up</code> / <code>Page Down</code>	Scroll

Write your code, Ctrl+S to save, Ctrl+E to exit, then type the filename (without `.lua`) to run it.

Tip: Create a `programs/` directory for your scripts with `mkdir programs`. Keep the root clean — put test scripts and experiments in folders.

Your First Program

Let's write "Hello, world!":

```
> edit hello
```

This opens a blank editor. Type:

```
print("Hello, world!")
```

Press **Ctrl+S** to save, then **Ctrl+E** to exit. Now run it:

```
> hello  
Hello, world!
```

Congratulations — you just wrote and ran your first Lua program!

How programs run

- Type the filename (without `.lua`) and press Enter
- The computer creates a `.lua` file when you save from the editor
- To stop a running program: **Ctrl+T**
- Programs run in the `/` (root) directory. Use `cd programs` to switch to your programs folder
- The startup program is `/startup.lua` — create it to run code automatically when the computer boots

A slightly bigger program

```
-- greeting.lua  
local name = "Adventurer"  
print("Welcome, " .. name .. "!")  
print("It is day " .. os.day() .. " of your journey.")  
  
if os.day() < 10 then  
    print("Early days – build your shelter!")  
else  
    print("You've been here a while. Time for industry!")  
end
```

Save as `greeting`, run with `greeting`. The `..` operator concatenates strings. `os.day()` returns the Minecraft day number.

Lua Syntax Primer

Lua is a small, simple language. Here's everything you need to write useful programs:

Variables

```
local x = 10      -- local variable (preferred – stays in scope)
y = 20            -- global variable (accessible everywhere – use sparingly)
```

Types

```
local a = nil      -- nothing (unset variable = nil)
local b = true     -- boolean
local c = 42       -- number (no integer/float distinction)
local d = "hello"  -- string
local e = {1, 2, 3} -- table (arrays and dictionaries)
local f = function() end -- function
```

Check a type with `type(value)`: `type(42) → "number"`.

Conditionals

```
if fuel < 100 then
    turtle.refuel()
elseif fuel > 10000 then
    print("Fuel is full!")
else
    print("Fuel level: " .. fuel)
end
```

Comparisons: `==` (equal), `~=` (not equal), `<`, `>`, `<=`, `>=`. Combine with `and`, `or`, `not`.

Loops

```
-- Numeric for (counted)
for i = 1, 10 do
    print("Iteration " .. i)
end

-- While (condition)
local n = 10
while n > 0 do
    print(n)
    n = n - 1
end

-- Repeat-until (always runs at least once)
repeat
    turtle.dig()
until turtle.detect() == false

-- Generic for (iterate over a table)
local items = {"stone", "dirt", "iron"}
```

```

for _, name in ipairs(items) do
    print(name)
end

-- Break out early
for i = 1, 100 do
    if turtle.getFuelLevel() < 10 then break end
    turtle.forward()
end

```

Functions

```

-- Define a function
local function greet(name)
    return "Hello, " .. name .. "!"
end

-- Call it
print(greet("Steve"))    -- "Hello, Steve!"

-- Functions are values – store in tables
local commands = {
    dig = function() turtle.dig() end,
    place = function() turtle.place() end,
}
commands.dig()

```

Strings

```

local s = "Minecraft"
print(string.len(s))      -- 9
print(string.sub(s, 1, 4)) -- "Mine"
print(s:len())            -- 9 (method syntax)
print(s:sub(1, 4))        -- "Mine"

-- Concatenation
local msg = "Day " .. os.day() -- "Day 42"

-- Convert
print(tonumber("42"))      -- 42
print(tostring(42))        -- "42"

```

Tables

Tables are Lua's only data structure. They work as arrays, dictionaries, or both:

```

-- Array (indexed from 1, not 0)
local ores = {"iron", "gold", "diamond"}
print(ores[1])    -- "iron"

```

```

print(#ores)                -- 3 (length)
table.insert(ores, "emerald") -- append

-- Dictionary
local config = {
    depth = 64,
    width = 3,
    fuelThreshold = 100,
}
print(config.depth)          -- 64 (dot syntax)
print(config["depth"])       -- 64 (bracket syntax)
config.height = 5            -- add new key

-- Iterate
for k, v in pairs(config) do
    print(k .. " = " .. v)
end

```

Math

```

print(math.random(1, 100)) -- random 1-100
print(math.random())       -- random 0-1
print(math.floor(3.7))     -- 3 (round down)
print(math.ceil(3.2))      -- 4 (round up)
print(math.max(2, 8, 5))   -- 8
print(math.min(2, 8, 5))   -- 2
print(math.abs(-5))        -- 5

```

The Event Loop

Computers in CC:Tweaked are event-driven. Programs don't just run from top to bottom and exit — they wait for things to happen.

Pulling events

```

-- Block until ANY event happens
local event, param1, param2 = os.pullEvent()
print("Event: " .. event)

-- Wait for a specific event type
local event, key = os.pullEvent("key")
print("You pressed: " .. key)

-- Wait for a timer
os.startTimer(5) -- fire timer in 5 seconds
local event, timerID = os.pullEvent("timer")
print("5 seconds passed!")

```

Event types

Event	Parameters	Fires when...
key	key code (number)	A key is pressed
char	character (string)	A text character is typed
timer	timer ID (number)	A timer expires
redstone	—	Any redstone input changes
peripheral	side (string)	A peripheral is attached
peripheral_detach	side (string)	A peripheral is removed
turtle_inventory	—	Turtle inventory changes

Running multiple things at once

```
-- Run two functions, stop when EITHER finishes
parallel.waitForAny(
    function() while true do print("A") sleep(1) end end,
    function() sleep(5) print("B done!") end
)

-- Run two functions concurrently until BOTH finish
parallel.waitForAll(
    function() while true do print("Tick") sleep(1) end end,
    function() sleep(10) print("Timer done!") end
)
```

A practical event loop

```
-- watchdog.lua - monitor a redstone signal, print when it changes
local lastState = rs.getInput("back")
print("Monitoring redstone on back. Ctrl+T to stop.")

while true do
    local event = os.pullEvent("redstone")
    local current = rs.getInput("back")
    if current ~= lastState then
        print("Redstone changed: " .. (current and "ON" or "OFF"))
        lastState = current
    end
end
```

`sleep(n)` is shorthand for `os.startTimer(n); os.pullEvent("timer")`.

Peripherals & Wrapping

Peripherals are external blocks connected to your computer via **Wired Modems**. They let your computer interact with inventories, tanks, and machines.

Connecting a peripheral

1. Craft a **Wired Modem** (stone + redstone, vanilla recipe)
2. Place it on the side of a block (e.g., a chest, a Mekanism tank)
3. Place your computer adjacent to the modem
4. Right-click the modem — you'll see "Peripheral: connected"

Wrapping in code

```
-- Find a modem on a specific side
local modem = peripheral.wrap("right")

-- Find ANY peripheral of a given type
local chest = peripheral.find("minecraft:chest")

-- List all connected peripherals
local names = peripheral.getNames()
for _, name in ipairs(names) do
    print(name, peripheral.getType(name))
end
```

Working with inventories

```
local chest = peripheral.wrap("right")

-- List contents
local items = chest.list()
for slot, item in pairs(items) do
    print("Slot " .. slot .. ": " .. item.name .. " x" .. item.count)
end

-- Move items
chest.pushItems("left", 1)      -- push slot 1 to left inventory
chest.pullItems("bottom", 1, 2) -- pull slots 1-2 from bottom

-- Check item count
local count = chest.getItemDetail(1)
if count then
    print("Slot 1 has " .. count.count .. " " .. count.name)
end
```

Working with fluid tanks

```
local tank = peripheral.wrap("back")
local tanks = tank.tanks()

for _, t in ipairs(tanks) do
    print(t.name .. ": " .. t.amount .. "/" .. t.capacity .. " mB")
end
```

```
-- Push fluid to adjacent tank
tank.pushFluid("left")
```

Turtles

Turtles are mobile computers — programmable robots that can move, mine, build, farm, and fight.

Crafting a turtle

You need a Basic or Advanced Computer, a Chest, and **7 TFMG Steel Ingots** (gated — see Recipe Gates above). Place them in a crafting table:

```
SSS
SCS
SXS
```

- S = TFMG Steel Ingot
- C = Basic/Advanced Computer
- X = Chest

Tool turtles

Right-click a turtle with a tool to equip it:

Tool	Turtle type	What it does
Diamond Pickaxe	Mining Turtle	Mines blocks, drops items
Diamond Axe	Felling Turtle	Chops wood faster
Diamond Shovel	Digging Turtle	Digs soft blocks faster
Diamond Hoe	Farming Turtle	Tills soil
Diamond Sword	Melee Turtle	Attacks mobs
(nothing)	Normal Turtle	Places blocks, moves inventory

Fuel

Turtles need fuel to move. Refuel by placing fuel items in the selected slot and running `turtle.refuel()`:

```
turtle.select(1)      -- select slot 1 (where your coal is)
turtle.refuel()        -- consume 1 fuel item
print(turtle.getFuelLevel()) -- fuel remaining (each coal = 80 moves)
```

Fuel values: Coal/Charcoal (80), Block of Coal (800), Lava Bucket (1000), Blaze Rod (120).

Movement

```
turtle.forward()    -- move forward 1 block
turtle.back()       -- move backward 1 block
turtle.up()         -- move up 1 block
turtle.down()       -- move down 1 block
turtle.turnLeft()   -- rotate 90° left
turtle.turnRight()  -- rotate 90° right
```

Each movement returns `true` if successful, `false` if blocked.

Inventory

Turtles have 16 inventory slots. Access them with:

```
turtle.select(3)      -- switch to slot 3
turtle.suck()          -- pick up items from front
turtle.drop()          -- drop selected slot in front
turtle.getItemCount(3) -- count items in slot 3
turtle.getItemSpace(3) -- free space in slot 3
turtle.transferTo(4)   -- move from selected to slot 4
```

Block interaction

```
turtle.dig()          -- mine block in front
turtle.digUp()         -- mine block above
turtle.digDown()       -- mine block below
turtle.place()         -- place block from selected slot in front
turtle.placeUp()       -- place block above
turtle.placeDown()     -- place block below
turtle.inspect()       -- get info about block in front
turtle.detect()        -- is there a block in front? (true/false)
turtle.compare()       -- does block in front match selected slot?
```

Example: Simple strip miner

Save this as `miner`:

```
-- 3-wide strip miner, 64 blocks deep
local depth = 64
local width = 3

-- Refuel first
turtle.select(1)
turtle.refuel()
print("Fuel: " .. turtle.getFuelLevel())

-- Check if we have a chest behind us to deposit into
print("Place a chest behind the turtle, then press Enter")
read()
```

```

for d = 1, depth do
    for w = 1, width do
        while turtle.detect() do
            turtle.dig()
            sleep(0.5)
        end
        turtle.forward()

        -- Deposit if inventory is getting full
        if turtle.getItemCount(16) > 0 then
            -- Save position and return to chest
            for _ = 1, d do turtle.down() end
            for _ = 1, w do turtle.back() end
            -- Deposit all
            for slot = 1, 16 do
                turtle.select(slot)
                turtle.drop()
            end
            -- Return to mining position
            turtle.select(1)
            turtle.refuel()
            for _ = 1, w do turtle.forward() end
            for _ = 1, d do turtle.up() end
        end
    end
end

-- Move to next row: back to start, down 1
for _ = 1, width do turtle.back() end
turtle.digDown()
turtle.down()
end

print("Mining complete! " .. depth * width .. " blocks mined.")

```

Run with `miner`. Place a chest behind the turtle, place fuel in slot 1.

Redstone

Computers can read and output redstone signals on all six sides:

```

-- Output (power a side)
rs.setOutput("left", true)      -- full power on left
rs.setOutput("right", false)    -- off
rs.setAnalogOutput("top", 7)    -- signal strength 7 (0-15)
rs.setOutput("bottom", true)    -- full power

-- Input (read a side)
local signal = rs.getInput("back") -- true if powered, false if not
local strength = rs.getAnalogInput("back") -- 0-15 signal strength

```

```
-- Bundled output (colored cables, 16 channels per side)
rs.setBundledOutput("left", colors.red + colors.green)
```

Practical redstone pattern: Create clutch control

```
-- toggle.lua - toggle a Create clutch on keypress
local powered = false
rs.setOutput("back", false)

print("Press Space to toggle clutch. Ctrl+T to stop.")
while true do
    local event, key = os.pullEvent("key")
    if key == keys.space then
        powered = not powered
        rs.setOutput("back", powered)
        print("Clutch: " .. (powered and "ENGAGED" or "DISENGAGED"))
    end
end
end
```

Place the computer so its “back” faces a Create Clutch. Run `toggle`, press Space to engage/disengage.

Wireless Networking

Wireless Modems let computers communicate across any distance — even across dimensions.

Crafting a Wireless Modem

The recipe is gated behind Create Brass (see Recipe Gates). Craft one, place it on a computer, right-click to open.

Sending and receiving

Computer A (sender):

```
rednet.open("left") -- open modem on left side
local targetID = 5 -- Computer B's ID (use `id` command)
rednet.send(targetID, "Hello, base!")
print("Message sent!")
rednet.close("left")
```

Computer B (receiver):

```
rednet.open("left")
local senderID, message = rednet.receive() -- wait for any message
print("Message from #" .. senderID .. ": " .. message)
rednet.close("left")
```

Broadcast to all computers

```
rednet.open("left")
rednet.broadcast("Fission reactor: CRITICAL")
rednet.close("left")
```

Practical intercom pattern

```
-- chat.lua — simple two-way chat between computers
rednet.open("left")
print("Chat open. Type messages. Ctrl+T to quit.")

parallel.waitForAny(
  -- Sender: type messages
  function()
    while true do
      local msg = read()
      rednet.broadcast(msg)
    end
  end,
  -- Receiver: print incoming messages
  function()
    while true do
      local id, msg = rednet.receive()
      print("#" .. id .. " " .. msg)
    end
  end
)

rednet.close("left")
```

Finding a computer's ID: Open the shell and type `id`. The computer's ID appears on the label or in the shell output.

Integration Recipes — Copy-Paste Scripts

AE2 Level Monitor

```
-- ae2monitor.lua — display ME drive capacity on a monitor
-- Attach a wired modem to an ME Interface or ME Drive

local monitor = peripheral.wrap("top") -- monitor on top
local me = peripheral.find("ae2:me_interface") or
peripheral.find("ae2:me_drive")

if not me then
  print("No AE2 peripheral found!")
  return
end
```

```

monitor.clear()
monitor.setTextScale(2)

while true do
    local items = me.listItems() or {}
    local total = 0
    for _, item in pairs(items) do
        total = total + item.count
    end

    monitor.setCursorPos(1, 1)
    monitor.write(" AE2 Storage")
    monitor.setCursorPos(1, 3)
    monitor.write(" Items: " .. total)
    monitor.setCursorPos(1, 5)
    monitor.write(" Types: " .. #items)

    -- Redstone warning if getting full
    if total > 100000 then
        rs.setOutput("back", true)
        monitor.setCursorPos(1, 7)
        monitor.write(" WARNING: NEAR CAPACITY")
    else
        rs.setOutput("back", false)
    end

    sleep(5)
end

```

Mekanism Fission Watchdog

```

-- reactor-watchdog.lua - scram Mekanism fission reactor if temperature
> 1200K
-- Place computer adjacent to reactor logic port (comparator reads
temperature)
-- Wire redstone output to Scram button or redstone-controlled safety
switch

local temp = 0
local scrambled = false

print("Reactor Watchdog active. Ctrl+T to stop.")

while true do
    temp = rs.getAnalogInput("back") * 100 -- signal 0-15 → 0-1500K
    approx

    if temp > 1200 and not scrambled then
        print("!!! SCRAM: Temperature " .. temp .. "K exceeds
1200K !!!")
    end
end

```

```

    rs.setOutput("left", true)      -- trigger scram
    rednet.broadcast("REACTOR SCRAM - temp " .. temp .. "K")
    scrambled = true
    sleep(1)
    rs.setOutput("left", false)     -- pulse complete
end

sleep(1)
end

```

Note: The signal-to-temperature mapping depends on your reactor's configuration. Calibrate by checking in-game reactor temperature against the analog signal value.

Turtle Strip-Miner

```

-- stripmine.lua - 3-wide tunnel, 64 deep, auto-deposit, self-refuel
-- Place turtle facing the wall to mine. Fuel in slot 1. Chest behind.

local DEPTH = 64
local WIDTH = 3

-- Refuel
turtle.select(1)
turtle.refuel()
local startFuel = turtle.getFuelLevel()
print("Starting fuel: " .. startFuel)

for d = 1, DEPTH do
    for w = 1, WIDTH do
        -- Dig until clear
        while turtle.detect() do
            turtle.dig()
            sleep(0.3)
        end
        turtle.forward()

        -- Auto-deposit when slot 16 fills
        if turtle.getItemCount(16) > 0 then
            print("Depositing at depth " .. d .. "...")
            local currentDepth = d
            for _ = 1, currentDepth do turtle.down() end
            for _ = 1, w do turtle.back() end
            -- Drop everything except slot 1 (fuel)
            for slot = 2, 16 do
                turtle.select(slot)
                turtle.drop()
            end
        end
    end
end

```

```

        -- Refuel and return
        turtle.select(1)
        turtle.refuel()
        for _ = 1, w do turtle.forward() end
        for _ = 1, currentDepth do turtle.up() end
    end
end

-- Return to row start, dig down for next row
for _ = 1, WIDTH do turtle.back() end
turtle.digDown()
turtle.down()
turtle.select(1)
turtle.refuel()
end

print("Done! Mined " .. (DEPTH * WIDTH) .. " blocks.")
print("Fuel used: " .. (startFuel - turtle.getFuelLevel()))

```

Stellaris Launch Sequencer

```

-- launch.lua - countdown + redstone pulse for Stellaris rocket
-- Redstone output (left) → Rocket Station
-- Monitor (top) for countdown display

local monitor = peripheral.wrap("top")
if monitor then
    monitor.clear()
    monitor.setTextScale(3)
end

local function countdown(n)
    for i = n, 1, -1 do
        if monitor then
            monitor.setCursorPos(1, 1)
            monitor.clearLine()
            monitor.write(" " .. i)
        end
        print("T-" .. i)
        sleep(1)
    end
end

print("Press Enter to initiate launch sequence")
read()

if monitor then
    monitor.clear()
    monitor.setTextScale(2)
    monitor.setCursorPos(1, 1)
end

```

```

    monitor.write(" IGNITION")
    monitor.setCursorPos(1, 3)
    monitor.write(" SEQUENCE")
end

countdown(10)

-- Fire!
rs.setOutput("left", true)
if monitor then
    monitor.setCursorPos(1, 5)
    monitor.write(" LIFTOFF!")
end
print("LIFTOFF!")
sleep(2)
rs.setOutput("left", false)
print("Sequence complete.")

```

Base Status Dashboard

```

-- dashboard.lua – cycles through power, storage, time displays
-- Monitors on multiple sides

local monitors = {}
for _, side in ipairs({"top", "left", "right", "front"}) do
    local m = peripheral.wrap(side)
    if m then
        m.clear()
        m.setTextScale(1)
        table.insert(monitors, m)
    end
end

local page = 1

while true do
    local m = monitors[page]
    if m then
        m.clear()
        m.setCursorPos(1, 1)

        if page == 1 then
            m.write("=== POWER ===")
            m.setCursorPos(1, 2)
            m.write("Mekanism Induction:")
            m.setCursorPos(1, 3)
            m.write(" Check comparator")
        elseif page == 2 then
            m.write("=== STORAGE ===")
            m.setCursorPos(1, 2)

```

```

        m.write("AE2 ME System:")
        m.setCursorPos(1, 3)
        m.write("  Check ME Drive")
    elseif page == 3 then
        m.write("=== WORLD ===")
        m.setCursorPos(1, 2)
        m.write("Day: " .. os.day())
        m.setCursorPos(1, 3)
        m.write("Time: " .. textutils.formatTime(os.time()))
    end
end

page = page + 1
if page > #monitors then page = 1 end
sleep(5)
end

```

Interdependency Notes

- **CC:Tweaked × Create:** Redstone output → Create clutch/gearshift for sequenced automation. Wired modems on Create basins for recipe monitoring. Computers can pulse gearshifts, toggle deployers, and read tank levels.
- **CC:Tweaked × Mekanism:** Reactor monitoring via redstone comparators on logic ports. Tank monitoring via wired modems on dynamic tanks. Enriched Alloy gates the Advanced Computer recipe.
- **CC:Tweaked × AE2:** ME Interface peripheral for inventory listing. Redstone comparator on ME Drive for capacity monitoring. Turtle inventory automation for bulk crafting requests. Applied Mekanistics (already in pack) bridges AE2 chemical storage with potential peripheral access.
- **CC:Tweaked × Stellaris:** Launch sequencer — countdown + redstone trigger to Rocket Station. Turtles can automate rocket component loading and fuel transfer.
- **CC:Tweaked × FTB Quests:** A dedicated “Programmable Computers” quest chapter (6 quests) guides progression from first computer to cross-mod automation.
- **CC:Tweaked × MineColonies:** Turtles can supply builder hut inventories. Monitors can display colony stats if accessible via peripheral API.

Progression Placement

- **Phase 2 (hours 40–200):** After basic Mekanism processing (Metallurgic Infuser for Enriched Alloy), Create Brass (Mixer), and TFMG steel production (Blast Furnace).
- The Basic Computer is craftable in Phase 1 for shell tinkering — harmless without peripherals or turtles.

- Turtles and wireless networking are Phase 2 proper, requiring steel and brass respectively.

Session Plan

Session	Duration	Activity	Unlocks
1 — Hello World	30 min	Craft Basic Computer. Boot it. Explore shell commands. Write <code>hello</code> program. Use <code>pastebin get</code> to download scripts.	Shell basics, editor, running programs
2 — Lua Basics	1 hour	Write small programs: number guessing game, timer display, Minecraft day counter. Experiment with <code>os.pullEvent</code> and the event loop.	Lua syntax, event-driven programming
3 — Turtles	1.5 hours	Craft Mining Turtle (needs TFMG steel). Write 3-wide strip miner from the integration recipes. Fuel it with coal. Watch it dig. Write return-to-home script.	Turtle API, autonomous mining
4 — Redstone Control	30 min	Computer + redstone output → Create clutch. Toggle factory on/off from terminal. Timer-based automation pulses. Read Mekanism tank level via comparator.	Redstone API, Create cross-mod control
5 — Peripherals & Monitoring	1 hour	Attach wired modems to Mekanism tanks + AE2 ME Drive. Print	Peripheral API, modded block integration

Session		Duration	Activity		Unlocks	
			fuel levels and item counts to monitors. Build the base status dashboard script.			
6 — Networking		1 hour	Craft Wireless Modem (needs Create brass). Two computers chatting via rednet . Remote base monitor at mining outpost. Alert system for automated alerts.		Wireless networking, multi-computer systems	
7 — Integration Deep Dive		2 hours	Write Mekanism fission watchdog. Stellaris launch sequencer. AE2 capacity monitor. Complete FTB Quests chapter. Experiment with custom automation ideas.		Cross-mod automation scripts, full ecosystem	

Keybindings Reference

Keybind		Suggested		Mod		Wave
View Recipes		R		EMI		0
View Uses		U		EMI		0
Open Fullscreen Map		J		JourneyMap		0
Toggle Minimap		Check default		JourneyMap		0
Open Waystone Menu		Ctrl+Shift+W		Waystones		0
Open Backpack		B		Sophisticated Backpacks		0
Zoom		C		Sodium		0
Toggle Dynamic Lights		Ctrl+L		LambDynamicLights		0
Open Curios Inventory		Y		Curios API		0
Open Skill Tree		P		Skill Tree		3

Keybind		Suggested		Mod		Wave
View Weapon Ability		Ctrl (hold)		Simply Swords		3
Open Rune Pouch		Set manually		Runes		3
Open Create Ponder		W (hold over block)		Create		1
Open Town Hall GUI		Right-click		MineColonies		1.5

Pro tip: Open **Options** → **Controls** → **Key Binds** and browse categories at the bottom. Every mod's binds are grouped under its name. Set the ones above first, then explore others as you install new waves.

Progression Through All Waves

Phase	What you'll be doing
Phase 2 (hours 40–200)	Full Create factory (conveyors, deployers, sequenced crafters, trains). Mekanism 3x–5x ore processing. AE2 ME network with autocrafting. Explore Twilight Forest bosses. Apotheosis Rare/Epic gear. Mutant Monsters. First Cataclysm boss. TFMG oil refining and diesel engines. Build with Rechiseled, Macaw's, Handcrafted. Small Ships for ocean travel. Immersive Aircraft for aerial exploration. Complete quest chapters. CC:Tweaked turtles and wireless automation. Stellaris Moon and early planets
Phase 3 (hours 200+)	Mekanism fusion reactor + MekaSuit. AE2 full autocrafting CPU chains. Endgame Apotheosis Mythic gear with perfect gem sockets. Twilight Forest final bosses. Cataclysm endgame bosses. End dimension exploration. Mars and Glacio space bases. CC:Tweaked cross-mod automation networks. Monumental building projects. Create mega-factories. Perpetual optimization

General Tips

Finding Help

- **EMI (R/U)** — always your first stop. Hover any item, press R for recipe, U for uses
- **Ponder (W over Create blocks)** — interactive animated tutorials. Better than any wiki
- **Mod Menu / Configured** — every installed mod's config screen
- Most mods link to their wiki or Discord from Mod Menu entry

Losing Progress

You won't. GraveStone Mod preserves your items on death. No creeper grief data pack prevents base damage. The pack is designed so you can walk away for a month and come back to exactly the world you left.

Java & Performance

Minecraft 1.21.1 requires **Java 21**. See [Wave -1](#) for JDK installation and JVM arguments.

Performance tips:

- ZGC garbage collector with generational mode (`-XX:+UseZGC -XX:+ZGenerational`)
- Distant Horizons handles far rendering — keep vanilla render distance at 10
- ModernFix `dynamic_resources = true` for fastest startup
- If you see missing textures, disable `dynamic_resources` temporarily

Cross-Wave Integration Cheat Sheet

If you want to...	Combine these mods
Auto-build a blueprint	Create Schematicannon + Blueprint and Quill
Power Create from Mekanism	Create Crafts & Additions electric motor/ alternator
Bridge TFMG oil to AE2	TFMG Energy Converter + Mekanism universal cable
Store colony resources digitally	AE2 Storage Bus on MineColonies warehouse
Feed colony kitchen automatically	Create conveyor → builder's chest + Farmer's Delight meals
Autocraft Mekanism chemicals	Applied Mekanistics + AE2 Pattern Provider

If you want to...	Combine these mods
Cook from your ME network	AE2 Export Bus on Farmer's Delight cooking pot
Get combat buffs before bosses	Farmer's Delight meal + Brewin' And Chewin' drink (effects stack)
Find the best loot	Explorer's Compass → YUNG's structure → loot → Apotheosis Rare+ gear
Build a train network	Steam 'n' Rails tracks → Train Utilities stations → Railways Navigator boards
Fly between Aether islands	Immersive Aircraft airship + Aether portal
Defend colony perimeter	Create Big Cannons on walls + Guard Towers
Explore oceans efficiently	Small Ships cog → Ocean's Delight ingredients → YUNG's Ocean Monuments
Process space ores	Stellaris desh/ostranium → Mekanism 5x line → AE2 storage
Build a megabase	Building Wands (structure) + Rechiseled blocks (aesthetic) + Handcrafted furniture (interior)
Program your base	CC:Tweaked computer + Create clutch + AE2 ME Drive → custom Lua automation scripts